

# Causal State Binding: An Intervention-Based Evaluation Framework for Language Agents

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## Supplementary overview

This supplement documents controlled action responses, execution following agent-written state and software-localization prediction. Methods Tables M1–M4 define sampling units, intervention access, predictor features and execution inference; results tables retain their S-numbered identifiers. The opening sections describe memory-content verification, directional target scoring, independent prediction, baseline replication and the linear prediction comparison. Notes 1–15 give detailed analyses, annotation procedures and acquisition conditions. Figures S1–S14 show task-, corpus- and model-level results. Sections S1–S9 provide the source experiments, control tables and metric definitions. Protocol and export identifiers connect these results to their source records. Tables round reported values; accompanying CSV and JSON files retain full precision. Entropy, bootstrap and agreement statistics use the units and definitions specified below (Shannon, 1948; Efron and Tibshirani, 1993; Cohen, 1960).

## Reading guide

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Page numbers link to the corresponding material. PDF bookmarks provide individual section and note headings; figure references lead to the artwork. Longer panel descriptions are headed by the same figure number.

The figures present execution effects, prediction diagnostics, finite-action responses, output coupling, entropy and annotation agreement. Source data and input hashes accompany every figure. Task-level means use four continuations in the core execution experiment and two in the content verification; cross-validation folds are parts of a single grouped estimate. The supplementary specificity intervals and the content-separation experiment use their separately stated resampling and acquisition procedures, without changing the fixed localization predictors. The paired Brier histograms for the linear prediction analysis (Figure S13) use symmetric geometric bins from  $10^{-4}$  to 1, including zero, with a symmetric-log threshold of 0.01. Captions specify the remaining transformations. Notes 1 and 13–15 distinguish reference-dependent prediction from the reference-free analyses.

## Study structure and supplementary methods

The studies examine controlled action responses, execution following agent-written state and software-localization prediction. The tables specify the sampling units, information available to each intervention and the complete predictor features. Primary execution effects, supplementary contrasts and predictive comparisons have distinct analysis units and inferential roles.

**Methods Table M1. Evidence roles and analysis units**

| Evidence                         | Sample and unit                                                            | Role and dependence                                                                                                                         |
|----------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Finite-action assay              | 57,816 scored rows; 803 unique corpus/event identifiers                    | Seven source corpora with shared prompts. Corpus contrasts summarize this design; rows are not independent replications.                    |
| Open-weight controls             | Four models; 126,000 rows each; 60 scaffold–task–family cells per model    | Three scaffolds, 20 families; 108,000 generation and 18,000 offline-policy rows per model. Strict Qwen-32B deletion is an additional slice. |
| Agent-written execution          | 160 tasks per core model; 80 memory and 80 goal; four continuations/branch | Eight mechanisms, ten tasks per mechanism/component; 32 redundant tasks. The task is the paired unit. Astra uses the same 40-task subset.   |
| Content validity controls        | 40 tasks per core model, 20/component                                      | Paraphrase, replacement and history compensation. Undefined replacement targets are separately reported.                                    |
| Reference-dependent localization | 1,800 model–issue rows; 300 issues; six provider identifiers               | Reference-dependent augmented features; grouped reanalysis is retrospective.                                                                |
| Reference-free bridge            | 1,200 model–issue rows; 300 issues; 297 issue/template groups              | Four consistent core configurations; all copies of a group remain within one fold. Astra is a fixed 60-issue extension.                     |
| Reused holdout                   | 400 model–issue rows; 100 issues                                           | Disjoint from bridge fitting, but outcomes were known before the service-driven model replacement. Bootstrap conditions on frozen fits.     |
| Independent prediction           | 3,200 model–issue records; 800 new issues from 12 repositories             | Frozen predictors; outcome scoring released after all predictions. Paired issue resampling carries the four configurations together.        |

| Evidence                    | Sample and unit                                                             | Role and dependence                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline replication        | 74,352 planned responses; 3,098 entries / 3,073 unique IDs                  | Seven source corpora; eight conditions and three repetitions. Membership, raw-field coverage and rule-completed compatibility output are provided separately. |
| Human output evaluation     | 1,200 items; three annotators; seven binary dimensions                      | 25,200 individual scores; 8,400 item-dimension combinations. Agreement uses individuals, not majority labels.                                                 |
| Memory-content development  | 16 tasks per API configuration; six branches and two continuations          | Two tasks per mechanism, shared by Sol and DeepSeek; 384 planned continuations. Development observations informed the subsequent fixed verification.          |
| Memory-content verification | 128 new tasks per API configuration; eight conditions and two continuations | Sixteen tasks per mechanism; 4,096 planned continuations. The paired unit is the task; task identities and analysis were fixed before acquisition.            |
| Supplementary specificity   | Same 640 core model-task records                                            | Result-informed contrast of irrelevant perturbation versus targeted deletion; 5,000 stratified draws/cell and a separate eight-test Holm family.              |

**Methods Table M2. State interventions and information access**

| Branch              | Changed material                                                          | Fixed material and scoring                                                                                                     |
|---------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Sham                | No saved component changed                                                | Agent-written memory, goal and unrelated note; same environment and handoff.                                                   |
| Memory lesion       | Entire saved memory removed                                               | Goal, unrelated note, tool schema and current request retained. Identifiers in memory may also be removed.                     |
| Goal lesion         | Entire saved goal removed                                                 | Memory, unrelated note, tool schema and current request retained. No evaluator-optimal policy inserted.                        |
| Irrelevant deletion | Unrelated note shortened toward the targeted serialized-character removal | Memory and goal retained. The available note can be shorter than the target; character matching does not imply token matching. |
| Rescue              | Original agent-written component restored                                 | Restore its actual content, including mistakes. Independent continuations, not replay of the same random draw.                 |
| Paraphrase          | Semantically preserving re-expression of stored material                  | Same task, handoff and scorer. Twenty-task component subsets.                                                                  |
| Replacement         | One registered task-relevant saved field changed                          | Original environment scorer retained; replacement-target scorer computed separately only when definable. Targets can overlap.  |

| Branch                    | Changed material                             | Fixed material and scoring                                                                                               |
|---------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| History compensa-<br>tion | Allowed earlier history supplied with lesion | The same earlier information that preceded the agent’s write; no hidden answer added.                                    |
| Redundancy stratum        | Relevant information also present at handoff | Sixteen tasks/component; stratum was fixed before acquisition. This is a task-design stratum, not an extra model branch. |

Identity-related memory fields are account (all mechanisms), participants (calendar), sku (inventory), customer code and currency (invoice), collection (merge), and report code (aggregation). The content-separation experiment partitions written memory using this classification. The original whole-memory lesion also removes an account value in 266 of 320 memory-task records. This records availability of the field; attribution of an execution failure to that field requires the separate intervention.

### Methods Table M3. Linear reference-free feature dictionary

Let  $B_0, B_1, B_2$  be normalized valid base-call file sets and  $E_1, \dots, E_L$  the extra-call sets. Invalid outputs become empty sets.  $J$  is Jaccard similarity, assigned zero when a set is empty. The consensus  $C$  is the three highest-voted paths in the base calls, with the recorded deterministic tie order. Confidence is valid only for numeric values in  $[0, 1]$ ; a missing confidence remains missing until training-fold imputation. Path-vote entropy uses natural logarithms and is zero when there are no votes. The normalized margin is the top path’s vote count minus the second path’s count, divided by the number of calls (zero when no votes exist).

| Block              | Feature                    | Exact definition                                                              |
|--------------------|----------------------------|-------------------------------------------------------------------------------|
| Shared numeric     | Issue length               | Character count of visible problem statement.                                 |
|                    | Candidate count            | Number of retrieved candidate implementation paths.                           |
|                    | Base agreement             | Mean $J(B_i, B_j)$ over the three call pairs.                                 |
|                    | Base vote margin           | Normalized top-minus-second path-vote count over three base calls.            |
|                    | Base entropy               | Entropy of the base path-vote distribution, nats.                             |
|                    | Base invalid fraction      | Number of invalid base outputs divided by three.                              |
|                    | Base confidence            | Mean valid base-call confidence; missing if none valid.                       |
| Shared categorical | Model, repository          | One-hot encoded within the training fold; unseen categories ignored.          |
| State augmentation | 1. Current agreement       | $J(P_c, C)$ .                                                                 |
|                    | 2. Minimal-state agreement | $J(P_c, P_m)$ .                                                               |
|                    | 3. Removal difference      | $1 - J(P_c, P_{\text{removed}})$ when both outputs are valid; otherwise zero. |
|                    | 4. Irrelevant agreement    | $J(P_c, P_i)$ .                                                               |
|                    | 5. Mismatch difference     | $1 - J(P_c, P_x)$ when both outputs are valid; otherwise zero.                |
|                    | 6–7. Validity/confidence   | Invalid fraction and mean valid confidence over the five state probes.        |

| Block                | Feature                  | Exact definition                                                  |
|----------------------|--------------------------|-------------------------------------------------------------------|
| Generic augmentation | 1. Consensus agreement   | Mean $J(E_i, C)$ over extra calls.                                |
|                      | 2. Extra agreement       | Mean pairwise $J(E_i, E_j)$ .                                     |
|                      | 3. Extra vote margin     | Normalized top-minus-second path-vote count over $L$ extra calls. |
|                      | 4. Extra entropy         | Entropy of the extra path-vote distribution, nats.                |
|                      | 5. Cross agreement       | Mean $J(B_i, E_j)$ over every base/extra pair.                    |
|                      | 6–7. Validity/confidence | Invalid fraction and mean valid confidence over extra calls.      |
|                      |                          |                                                                   |

The state sets  $P_c, P_m, P_{\text{removed}}, P_i, P_x$  correspond to current, minimal, removed, irrelevant and mismatched saved working state. Mismatch candidates come from the same visible issue-specific pool and exclude the first output’s paths; no gold path or cross-fold donor is used. Both augmentations contain seven numeric features and use the same shared features, categories and estimator. Fixed generic diagnostics use five calls. Token funding can extend this to ten, governed by recorded usage alone.

Path normalization inspects at most the first three list entries, keeps strings, strips surrounding whitespace, replaces backslashes by forward slashes and removes a leading `./`. Empty paths, absolute paths, paths with a `..` component and duplicate normalized paths are discarded. A valid output additionally requires a list of at most three supplied files, a nonempty normalized set and membership of every retained path in the visible implementation-file inventory. These checks do not consult reference patch paths. For example, two valid sets sharing one of three total paths have  $J = 1/3$  regardless of whether that path is correct; the scorer-only outcome is evaluated separately.

## Supplementary specificity and replacement classification

The specificity estimator uses paired task differences, then equal mechanism means, as defined in the main paper. Each of eight core model/component cells contains 80 tasks. The bootstrap resamples ten tasks with replacement within each of eight mechanisms, recomputing the entire contrast for each of 5,000 draws. The 2.5th and 97.5th percentiles form a two-sided interval. Tests flip each task difference independently with probability one half, compare the absolute mean with its sign-flip reference and apply the plus-one correction over 50,000 draws. The eight supplementary tests share a separate Holm correction from the registered primary family. The all-zero Mistral memory sample produces a degenerate empirical interval and does not establish population equivalence.

Every one of the 640 replacement continuations is assigned exactly once. The replacement target is first checked for definability. Defined cases are partitioned by the original and replacement scorer: original-only, replacement-only, both, neither with a valid transaction, or neither with an invalid transaction. Thus overlapping targets do not count as two independent successes. Both targets are scored against the same saved environment and continuation action.

## Memory content and transaction identifiers

The content-separation experiment specified 16 new memory tasks across eight mechanisms, shared by Sol and DeepSeek, before model acquisition. Its six conditions were full memory, identity fields only, content fields only, empty memory, semantic paraphrase and irrelevant-note perturbation. Each used two fresh continuations. Identity fields follow Supplementary Methods Table M2; the partition uses the agent’s actual memory without inserting missing or corrected values. The persistent goal, tool schema, execution context and initial environment are shared across conditions. This exploratory experiment

was informed by the execution results; its complete cohort and analysis were fixed before acquisition, with failures retained.

All 32 model-task records and 384 continuations were collected. Valid state writes numbered 15/16 for Sol and 16/16 for DeepSeek; the invalid write remains in the primary denominator. Sham success was 27/32 and 23/32. With identifiers retained but other memory content removed, success fell to 6/32 and 3/32. The paired content-loss contrasts were 0.656 [0.531, 0.781] and 0.625 [0.500, 0.750]. These descriptive 95% intervals use 5,000 paired-task bootstrap draws within each mechanism, seed 2026092602. With two tasks per mechanism and two continuations per branch, they describe variation within the tested set of mechanisms.

Removing identifiers while retaining content and removing all memory both yielded zero primary successes in each configuration. Thus execution depends on identity information as well as content. Sol’s paraphrase-minus-sham contrast was  $-0.094$  [ $-0.156$ ,  $-0.031$ ], while DeepSeek’s was  $0.000$  [ $-0.063$ ,  $0.063$ ], indicating different responses to paraphrase in the two configurations. Irrelevant-minus-sham contrasts were  $-0.063$  [ $-0.188$ ,  $0.063$ ] and  $0.000$  [ $-0.063$ ,  $0.063$ ]. The table below gives the six development-condition counts. The independent content verification used 128 new tasks and eight conditions; its results appear in main Figure 5 and in Verification of memory-content dependence (p. 8).

| Branch                  | Sol success/planned | DeepSeek success/planned |
|-------------------------|---------------------|--------------------------|
| Sham                    | 27/32               | 23/32                    |
| Identity fields only    | 6/32                | 3/32                     |
| Content fields only     | 0/32                | 0/32                     |
| Empty memory            | 0/32                | 0/32                     |
| Paraphrase              | 24/32               | 23/32                    |
| Irrelevant perturbation | 25/32               | 23/32                    |

**Service and resource accounting.** The first credential’s development batch returned 416 terminal insufficient-quota refusals and no valid model answer. That batch is retained as a service-access failure, outside the behavioral cohort. A separately registered second-credential batch used the same fixed development specification and completed 828 logical calls in 834 transport attempts. Final responses comprised 827 HTTP 200 and one terminal HTTP 404; six additional failed attempts and that terminal response lacked token usage. The saved reported attempt total was 1,194,825 tokens, with seven unmetered attempts. Two separate availability probes reported 4,445 tokens. Missing billing is not assigned zero. These records are separate from the earlier administrative upper bound of 42 potentially unlogged attempts. All model responses in this experiment were obtained through the API service.

**Restoration from accessible history.** A source-record baseline uses the same accessible structured history available to every candidate policy. On 192 fixed interface-validation cases (32 tasks and six state conditions), it exactly restored the declared memory and goal fields, without changing the 32 initially correct states or making a model call. Its implementation receives no task or evaluator argument. This establishes restoration feasibility at the interface, before evaluating downstream execution. The structured history supplies every declared field in these cases, making direct restoration the appropriate reference policy for this setting. The finite-action guard and software guard in the main text retain their measured outcomes and tradeoff.

**Methods Table M4. Execution inference and reproducibility**

| Analysis               | Resampling / test                                                | Multiplicity and seed                              |
|------------------------|------------------------------------------------------------------|----------------------------------------------------|
| Primary execution      | 5,000 within-mechanism task draws;<br>50,000 sign flips          | Eight primary contrasts; shared<br>seed 2026092411 |
| Mechanism sensitivity  | 5,000 draws of eight mechanism means;<br>leave one mechanism out | Descriptive; same ordered stream                   |
| Specificity supplement | 5,000 task draws; 50,000 sign flips                              | Separate eight-test family; seed<br>2026092601     |
| Content development    | 5,000 stratified task draws                                      | Descriptive; seed 2026092602                       |
| Content verification   | 10,000 stratified task draws; 50,000<br>sign flips               | Two primary contrasts; seed<br>2026092701          |

For each task, the four continuation outcomes are averaged within branch. The primary statistic is the mean of the eight mechanism-specific mean sham-minus-lesion differences. Five thousand bootstrap draws resample tasks within each mechanism. A separate sensitivity analysis resamples the eight mechanism means 5,000 times and reports leave-one-mechanism-out estimates. Two-sided sign-flip tests independently multiply each task difference by  $-1$  or  $1$  with probability one half, then recompute the mechanism-equal mean. The reference assumes symmetry of task differences under the null. The  $p$  value is  $(1 + \#\{|T^*| \geq |T| - 10^{-12}\})/50,001$ . Holm’s monotone step-down procedure adjusts the eight primary model-component tests.

The analysis initializes NumPy’s default random generator once with seed 2026092411. Its ordered traversal is Sol, DeepSeek, Qwen, Mistral and Astra, with memory before goal; task files and mechanism keys are sorted. Bootstrap, mechanism sensitivity and sign-flip draws consume this shared stream in the archived analysis order, including secondary controls. Thus the seed reproduces the complete analysis sequence, not an independently restarted stream for each table cell. The Astra subset is outside the eight-test primary family. The separate specificity analysis uses seed 2026092601, 5,000 stratified draws and 50,000 sign flips; content development uses seed 2026092602 and 5,000 descriptive draws. The 128-task content verification uses seed 2026092701 and its separately fixed 10,000-draw analysis and two-test Holm family, described below.

### Checkpoint encoding and tool access

Preparation exposes the previous request, remembered facts, persistent commitment and unrelated archived material through `inspect_history`. A subsequent `write_state` request must contain exactly three keys, `memory`, `goal` and `unrelated`, whose values are dictionaries. This validates the state-write structure; agreement with the observed facts is recorded separately. Two preparation calls are allowed, each with a maximum completion length of 1,536 tokens. A failed preparation retains its planned continuations and receives primary success zero.

The checkpoint stores the task, realized state, preparation calls, handoff messages and initial environment. Each branch deep-copies that environment. A lesion changes the selected dictionary to `{}` without removing its key; restoration reinstates its original content, including any error. Irrelevant editing shortens string values in the unrelated dictionary in sorted-key order, toward the number of serialized characters removed by the lesion. Continuation starts with fresh system and handoff messages, not the preparation conversation. It includes the stored state, tool schema and complementary component information; the preassigned redundant stratum also restates the targeted information. Only history compensation supplies the earlier history.

Each continuation allows two calls (maximum 1,024 completion tokens each): `inspect_environment`, then the designated transaction. The first call must have empty arguments. The transaction requires the exact argument-key set, registered outer value types and mechanism-specific feasibility checks; a failed call stops that continuation. The evaluator then

compares the realized transaction with the valid task solutions. The original 160-task experiment shuffles branch-repeat pairs within task using a Python random generator seeded by the first twelve hexadecimal digits of the canonical request-prefix hash. Environment reset fixes the starting state while repeats obtain new model draws.

## Shared repository retrieval

At the pinned base commit, candidate implementation paths have extensions `.py`, `.pyx`, `.c`, `.cpp`, `.h`, `.js` or `.ts`. Paths are excluded when a lower-cased component equals `test`, `tests`, `testing`, `doc` or `docs`, or starts with `test_`. Camel-case boundaries are split before lower-casing; retrieval tokens match `[a-z][a-z0-9_]{2,}`. The visible problem statement supplies query counts  $q_t$ . For path-token frequency  $d_{pt}$ , document frequency  $df_t$  and  $N$  eligible paths, the score is

$$r(p) = \sum_{t \in p \cap q} \min(q_t, 3) d_{pt} \log \left( 1 + \frac{N + 1}{df_t + 1} \right).$$

Paths are sorted by descending score and then lexicographic path; the first fifty are supplied. For each of the first eight paths, the source line maximizing the sum of  $\min(q_t, 2)$  over its distinct matching tokens anchors an excerpt; ties select the earliest line. The excerpt begins up to eight lines before that anchor, includes at most 32 numbered lines and is truncated to 1,500 characters. The model-visible issue is truncated to 12,000 characters. All diagnostic arms use the same cached context. Feature tokenization is defined separately in *Reference-free response representations* and includes English stop-word removal; retrieval tokenization does not.

## Verification of memory-content dependence

The content study used 128 newly generated tasks per API configuration: 16 instances of each of the eight transaction mechanisms. The tasks, identity-field partitions, conditions, repeats, scoring rule and analysis were fixed before acquisition. Their instance identifiers were disjoint from the 16-task development set. The agent first inspected the earlier interaction and wrote its own memory. Each saved checkpoint was then continued twice under each of eight conditions: the complete record, identity fields only, non-identity content only, empty memory, deterministic rewording, exact JSON text, irrelevant editing and restoration. This yielded 256 model-task checkpoints and 4,096 continuations. The same realized record was used across conditions; inaccurate agent-written values were retained.

Identity fields were account identifiers in allocation, routing and dependency tasks; account and participant identifiers in calendar tasks; account and item identifiers in inventory; account, customer and currency identifiers in invoice tasks; account and collection identifiers in record merging; and account and report identifiers in aggregation. Identity-only and content-only conditions formed disjoint partitions of the actual memory dictionary. The goal was held fixed. The exact-text condition used a lossless canonical JSON serialization of the dictionary. The deterministic rewording retained every original field value and used the same transformation as the development study. Comparing these two text conditions separated rewording from the change from a structured object to a string. Condition and repeat order were randomized within checkpoint using deterministic request-derived seeds.

The primary outcome required a valid state write and a successful final environment transaction. Raw environment success was also retained. The independent analysis unit was the task; its two continuation outcomes were averaged before comparison. Tasks received equal weight within each mechanism, and the eight mechanisms received equal weight. The two primary full-memory-minus-identity-only contrasts were evaluated with 10,000 stratified paired-task bootstrap draws and 50,000 two-sided sign-flip draws, with Holm adjustment across the two API configurations. The analysis seed was 2026092701. All registered secondary contrasts were retained. Reworded text versus exact JSON text

additionally had a prespecified, descriptive two-one-sided-test analysis with a  $\pm 0.05$  probability margin. Its standard error respected the eight equal-weight strata, with Welch–Satterthwaite degrees of freedom. This secondary equivalence analysis was not part of the family-adjusted superiority claim. Mechanism-resolved intervals in the figure are descriptive within-mechanism resamples with seed 2026092705.

Sol produced a valid state write for all 128 tasks, and DeepSeek for 127. Every planned continuation remained in the primary denominator. Sol used 4,352 logical requests and 4,353 recorded transport attempts; DeepSeek used 4,320 logical requests and attempts. Reported attempt usage was 12,204,784 and 3,746,841 tokens, respectively. One Sol transport failure and 30 DeepSeek HTTP 404 responses lacked token metering; their unknown costs were not assigned zero. No terminal scientific failure was resampled.

After the registered analysis, a separate service-failure sensitivity assigned each primary outcome affected by a terminal non-200 preparation or continuation request an arbitrary value in  $[0, 1]$ . A retry failure followed by a valid final response remained an observed outcome; malformed model content returned with HTTP 200 remained an observed failure. The resulting deterministic bounds were  $[0.656, 0.656]$  for Sol and  $[0.543, 0.582]$  for DeepSeek. These are worst-case missing-outcome bounds, not confidence intervals, and do not replace the registered all-planned estimates. The complete task records, intervention states, transactions, individual outcomes, attempts, source hashes and analysis outputs are retained in the research archive.

## Direction of action under content replacement

The saved replacement experiments allow action direction to be distinguished from an undirected change in performance. For each task, we evaluate the four original-state continuations and the four replacement-state continuations against both the original and alternative targets. Let  $S_{ij}$  be the mean success of actions produced with state  $i$ , evaluated against target  $j$ , where 0 denotes the original and 1 the replacement. We summarize target alignment by

$$D = \frac{1}{2}\{(S_{00} - S_{01}) + (S_{11} - S_{10})\}. \quad (1)$$

The contrast ranges from  $-1$  to  $1$ . A positive value indicates that actions under each state align more closely with that state’s corresponding target. Both target scores are retained when their accepted actions overlap. The evaluator uses the saved environment and continuation actions; no model response is generated during cross-scoring.

The analysis includes all 160 model–task records in the fixed replacement subset. An alternative target is defined for 130 records, supplying 1,040 cross-scored actions. Each table entry first averages the four continuations within task and then gives equal weight to the defined tasks. Mechanism-equal summaries and the redundant-information strata accompany the task-level source data. The alternative target is defined for all 20 memory and 20 goal tasks in each API model, 19 memory and 20 goal tasks in Qwen, and three memory and eight goal tasks in Mistral.

This supplementary cross-scoring was specified after inspecting the existing replacement results. It provides an explanatory comparison on those tasks. The original and replacement continuations were independently sampled; they are paired by task, without an assumption of common random numbers. Undefined targets are excluded from the conditional scores and retained in the planned denominator. Assigning their unknown task contrasts any value in  $[-1, 1]$  yields all-planned bounds of  $[-0.025, 0.075]$  for Qwen memory,  $[-0.850, 0.850]$  for Mistral memory and  $[-0.600, 0.600]$  for Mistral goals. These are sensitivity bounds, not confidence intervals.

## Independent prediction of localization success

The independent study fixed 800 new SWE-bench test issues from twelve repositories after excluding the 423-identifier exclusion set, repository/commit duplicates and registered text/patch near-duplicates.

Table S4: Crossed scoring of original and replacement actions. The “Tasks” column reports defined/planned counts; each defined task contributes four actions per state. Target scores can overlap. Qwen and Mistral refer to the fixed Qwen2.5-14B and Mistral-7B configurations.

| Model    | State  | Tasks | $S_{00}$ | $S_{01}$ | $S_{10}$ | $S_{11}$ | $D$   |
|----------|--------|-------|----------|----------|----------|----------|-------|
| Sol      | Memory | 20/20 | 0.938    | 0.050    | 0.238    | 0.763    | 0.706 |
| Sol      | Goal   | 20/20 | 0.938    | 0.050    | 0.188    | 0.738    | 0.719 |
| DeepSeek | Memory | 20/20 | 0.713    | 0.038    | 0.088    | 0.700    | 0.644 |
| DeepSeek | Goal   | 20/20 | 0.800    | 0.050    | 0.088    | 0.613    | 0.638 |
| Qwen     | Memory | 19/20 | 0.105    | 0.026    | 0.079    | 0.053    | 0.026 |
| Qwen     | Goal   | 20/20 | 0.163    | 0.163    | 0.125    | 0.125    | 0.000 |
| Mistral  | Memory | 3/20  | 0.000    | 0.000    | 0.000    | 0.000    | 0.000 |
| Mistral  | Goal   | 8/20  | 0.000    | 0.000    | 0.000    | 0.000    | 0.000 |

The exclusion set is the disjoint union of 300 SWE-bench Lite test issues used for development, 100 SWE-bench Verified test issues used for reassessment, and the full 23-issue Lite development split. Twenty of those 23 development issues were used for implementation checks; all 23 were excluded to separate the development split from the independent sample. Issue-level membership is provided with the source data. There were 1,754 eligible unused issues. Selection was stratified by repository before outcomes, and the same issues were assigned to all four configurations. The fixed 800-issue size was a finite precision and acquisition choice: a development-conditional approximation required 658 groups for an AUC difference of 0.005 at 80% power, or 1,027 with 1.25-fold standard-deviation inflation. Sampling was not extended according to the new results. Membership, base commits and context hashes accompany the data.

The endpoint is success when at least two of three shared base calls each identify a reference implementation file among at most three returned paths. Gold paths are absent from generation and feature extraction. Three further current-state reevaluations, including the current-state probe and two ordinary repeats, combine with state removal and state mismatch to form five hybrid diagnostic calls. Pure state diagnosis uses the five original intervention conditions; ordinary self-consistency uses at least five calls. Token-funded generic sampling continues to the larger recorded state/hybrid budget, at most ten generic calls, based only on usage. Responses can be reused across strategies in joint acquisition, but deployment cost assigns every strategy its required calls.

The feature map uses model/repository identity, issue length and candidate count, base-answer agreement, validity, vote margin, entropy and confidence, then within-response confidence/validity, normalized retrieval ranks, lexical query agreement, state-answer agreement, pairwise file-set overlap and contrasts between query positions. Set overlap is Jaccard similarity with the frozen empty-set and invalid-output rules. Each arm has access to the original vector, its augmentation with ordered and aggregate features, its augmentation with aggregate features alone, and a compact relation-focused augmentation. Numeric inputs are represented consistently as float32 before the frozen transforms. Model/repository encoding ignores unseen categories. Training-fold median imputation retains entirely missing columns with zero fill and adds indicators for columns with missing training values; numeric features and indicators are standardized using training-fold statistics. Estimator random seeds are 17.

Development uses five outer issue/template-group folds and three inner folds. The common search has 60 configurations per arm: logistic  $C \in \{0.1, 1, 10\}$ ; histogram gradient boosting with maximum leaves 7 or 15 and minimum leaf size 10 or 30; extra trees with depth 4 or 8 and minimum leaf size 5 or 15; and CatBoost with depth 4 or 6 and L2 penalty 3 or 10, crossed with the four representations. Histogram boosting uses 150 iterations, learning rate 0.05, L2 penalty 5 and no early stopping. Extra trees uses 160 trees and all features; CatBoost uses 160 iterations and learning rate 0.05. Logistic regression uses liblinear and at most 3,000 iterations. For each outer validation fold, inner AUC selects

one member of each learner family. Final configurations are then selected within each family by pooled five-fold out-of-fold AUC over the development cohort. Both selections use log loss and fixed candidate order to break ties. The four probabilities receive equal weight. The selected models are refitted on all development data and frozen before new acquisition; the 800 new issues do not enter either selection.

Table S5: Frozen main candidate and comparator configurations. Column counts precede one-hot expansion; each ensemble member has weight 0.25. Common settings are stated above.

| Arm                     | Learner        | Representation      | Columns | Selected settings  |
|-------------------------|----------------|---------------------|---------|--------------------|
| Repetition-intervention | CatBoost       | Aggregate           | 130     | depth=4, l2=3      |
| Repetition-intervention | Extra trees    | Aggregate           | 130     | depth=8, leaf=5    |
| Repetition-intervention | Hist. boosting | Aggregate           | 130     | leaves=15, leaf=10 |
| Repetition-intervention | Logistic       | Aggregate           | 130     | C=0.1              |
| State interventions     | CatBoost       | Aggregate           | 130     | depth=6, l2=3      |
| State interventions     | Extra trees    | Aggregate           | 130     | depth=8, leaf=5    |
| State interventions     | Hist. boosting | Aggregate           | 130     | leaves=15, leaf=10 |
| State interventions     | Logistic       | Aggregate           | 130     | C=0.1              |
| Five generic queries    | CatBoost       | Aggregate           | 130     | depth=6, l2=3      |
| Five generic queries    | Extra trees    | Ordered + aggregate | 279     | depth=8, leaf=5    |
| Five generic queries    | Hist. boosting | Aggregate           | 130     | leaves=15, leaf=10 |
| Five generic queries    | Logistic       | Aggregate           | 130     | C=0.1              |
| Token-funded generic    | CatBoost       | Aggregate           | 130     | depth=6, l2=10     |
| Token-funded generic    | Extra trees    | Ordered + aggregate | 279     | depth=8, leaf=5    |
| Token-funded generic    | Hist. boosting | Aggregate           | 130     | leaves=15, leaf=10 |
| Token-funded generic    | Logistic       | Aggregate           | 130     | C=0.1              |

## Reference-free response representations

The five ordered diagnostic positions are fixed as follows. Superscripts index diagnostic responses, and subscripts index the three common base answers.

| Position | Pure state diagnostic | Hybrid diagnostic | Generic diagnostic |
|----------|-----------------------|-------------------|--------------------|
| 0        | Current record        | Current record    | Ordinary repeat 1  |
| 1        | Minimal record        | Ordinary repeat 1 | Ordinary repeat 2  |
| 2        | Removed record        | Removed record    | Ordinary repeat 3  |
| 3        | Irrelevant edit       | Ordinary repeat 2 | Ordinary repeat 4  |
| 4        | Mismatched record     | Mismatched record | Ordinary repeat 5  |

Current, removed and mismatched responses are shared between pure-state and hybrid acquisition; the hybrid’s ordinary repeats reuse the first two generic responses. Deployment costs charge each strategy for all five required responses, even when joint acquisition reuses them. A funded generic arm can add positions beyond four. Let  $P^{(i)}$  be the normalized ordered paths from diagnostic response  $i$ ,  $B_j$  those from base response  $j \in \{0, 1, 2\}$ , and  $C$  the three most frequent paths across the base responses. Let  $T_i$ ,  $U_i$  and  $T_0$  denote the input, returned and original state’s candidate-file sets. Jaccard agreement is  $J(A, B) = |A \cap B| / |A \cup B|$  when both sets are nonempty and zero otherwise. Path normalization, repository-membership validation and the at-most-three-path rule are identical for every diagnostic

arm. A response is parsed when a nonempty JSON object is recovered; validity additionally requires the registered file-output schema and valid implementation paths.

Each response supplies 21 quantities: confidence, validity, parsing success, path count, valid-path fraction;  $J(P^{(i)}, C)$ ,  $J(P^{(i)}, B_0)$ , rank-weighted base votes and top-path agreement with  $C$ ;  $J(P^{(i)}, T_i)$ ,  $J(P^{(i)}, U_i)$ ,  $J(P^{(i)}, T_0)$ , the following shift  $J(P^{(i)}, T_i) - J(B_0, T_i)$  and input-state distance  $1 - J(T_i, T_0)$ ; retrieval coverage and best/mean retrieval rank; mean path–query lexical overlap, returned-state–query lexical overlap, serialized state length and a nonempty-returned-state indicator. An empty input state gives missing input-state distance. Rank-weighted base votes equal  $\sum_{k=0}^{|P^{(i)}|-1} (3-k)v(P_k^{(i)})/18$ , where  $v$  counts the three base outputs. Retrieval ranks are zero-based among the fifty visible candidates, with absent paths assigned rank 50; best and mean ranks are divided by 50, and both default to one for an empty output. Fractions use denominator  $\max(|P^{(i)}|, 1)$ . Lexical overlap applies  $J$  to lower-cased identifier tokens of more than two characters after English stop-word removal; returned state is serialized as JSON for this calculation. No reference implementation path enters these quantities.

The ordered representation records the first five response vectors, ten pairwise overlaps and their valid-output versions. The latter are missing unless both responses are valid. For positions  $i = 1, \dots, 4$ , it also records differences from position zero in confidence, base-consensus agreement, original-state agreement, following shift and input-state agreement. Its additional relative-disagreement term is

$$\{1 - J(P^{(0)}, P^{(i)})\} - \{1 - J(P^{(0)}, P^{(3)})\}.$$

For  $i = 3$ , this relative-disagreement term is identically zero; it is retained in the frozen ordered column list. It does not introduce an additional contrast. The aggregate representation summarizes every response quantity by finite-value mean, population standard deviation, minimum, maximum and missing fraction; it also summarizes all pairwise overlaps, the proportion of unique file sets, file-vote entropy in nats, the largest file-vote count divided by the number of calls, and that call count. Aggregates use all funded generic calls when more than five are available. These fields supplement the common base features and the seven original diagnostic features. Every arm can select the original feature vector, the original plus ordered and aggregate fields, the original plus aggregate fields, or a compact subset containing validity/parsing, state correspondence, following shifts, differences and pairs. The frozen column lists specify the compact representation exactly.

The auxiliary noise-relative representation compares altered responses with ordinary repetition. Neutral positions are  $N = \{0, 1, 3\}$  and altered positions  $A = \{2, 4\}$ ; in the hybrid arm these contain the current-state response and two ordinary repeats, then removal and mismatch. The corresponding positions are treated identically in the generic arm. The representation records the mean, population standard deviation and minimum overlap among neutral pairs. For each of confidence, base-consensus agreement, input/returned/original-state agreement, following shift, validity, parsing, best retrieval rank and retrieval coverage, it retains the neutral mean and standard deviation, plus each altered response’s difference from the neutral mean and that difference divided by  $(SD + 0.05)$  and clipped to  $[-10, 10]$ . It also retains altered-to-neutral mean agreement, excess disagreement relative to neutral pairs, the difference between minimum neutral-pair agreement and maximum altered-to-neutral agreement, relative instability  $(1 - \bar{J}_{i,N})/(1 - \bar{J}_N + 0.05)$ , overlap between altered responses, and the mean and difference of their excess disagreements. These numeric contrasts are computed after the fixed float32 conversion. They receive their own equally specified candidate and generic fits and do not replace the frozen primary representation.

Independent fixed-weight inference used the pinned weights and llama.cpp runtime; the final Mistral configuration used four concurrent slots. Backend settings were checked on development inputs before use on the fixed evaluation cohort. Each logical request allowed at most three attempts, with completed responses retained and terminal failures included in the planned denominator.

The prespecified noise-contrast variant is an auxiliary candidate developed separately, with corresponding generic models receiving the same representation and search opportunities. Its comparison is

a prespecified secondary member of the three-comparison family.

All four 800-issue collections and their 3,200 predictions were completed before the separate gold file was released. The registered analysis retains failures and resamples issues 10,000 times within repository, carrying all four models together and conditioning on fixed predictors. Marginal intervals use the 2.5th and 97.5th percentiles. The three AUC comparisons use 0.8333rd and 99.1667th percentiles for Bonferroni family coverage. Brier score, log loss, ten fixed calibration bins and failure risk at 25/50/75/100% retained fractions are secondary measures; score ties are resolved by issue and model identifiers. Risk intervals condition on that observed selection. Calibration functions and retention rules remain fixed during evaluation.

Table S6: Prespecified independent AUC comparisons against the corresponding funded generic predictor, with marginal 95% and family-adjusted 98.333% intervals.

| Candidate               | Difference | Marginal 95%        | Family 98.333%      |
|-------------------------|------------|---------------------|---------------------|
| Repetition–intervention | 0.00382    | [0.00023, 0.00743]  | [-0.00055, 0.00821] |
| State interventions     | 0.00168    | [-0.00247, 0.00589] | [-0.00345, 0.00677] |
| Noise contrast          | 0.00194    | [-0.00214, 0.00608] | [-0.00305, 0.00697] |

Table S7: Model-resolved independent discrimination. Each configuration has 800 issues; subgroup intervals are marginal and descriptive.

| Model    | Positive | Negative | Hybrid AUC | Generic AUC | $\Delta$ AUC [95% interval]  |
|----------|----------|----------|------------|-------------|------------------------------|
| Sol      | 698      | 102      | 0.8343     | 0.8415      | -0.00721 [-0.02631, 0.01151] |
| DeepSeek | 564      | 236      | 0.9325     | 0.9282      | 0.00430 [-0.00401, 0.01287]  |
| Qwen     | 531      | 269      | 0.7722     | 0.7352      | 0.03699 [0.01318, 0.06103]   |
| Mistral  | 20       | 780      | 0.9879     | 0.9886      | -0.00071 [-0.00496, 0.00327] |

Reported joint acquisition usage was 241,102,491 tokens. There were 3,121 fully metered and sufficiently funded records out of 3,200; 81 attempts lacked metering. The full cohort is primary. On the complete-and-funded subset, the primary AUC difference is 0.00403 [0.00041, 0.00766]; the same-call-count comparison on all records is 0.00392 [0.00044, 0.00739]. Both are additional marginal descriptions. Costs include transport attempts and indivisible-call overshoot. The resource measure is provider-reported tokens; unmetered attempts form a separate accounting category. Requested and returned identifiers specify each API configuration.

The research archive includes the analysis specification, response files, frozen predictions, scoring-release record and bootstrap draws. Model-service interruptions in the development acquisitions are accounted for separately in Supplementary Notes 13–15.

## Development ablations of diagnostic information

The 300-issue development cohort supports a targeted comparison of information blocks. All arms retain the same common base features. Quality fields describe confidence, parsing, path validity, retrieval compatibility and lexical/output properties. Response-relation fields describe agreement and contrasts across generated answers. State-correspondence fields describe input, returned and original state agreement, following shifts and input-state distance. We removed blocks and repeated the same nested search and learner-balanced ensemble construction within each diagnostic arm. Thus an ablation changes the available information while giving the reduced predictor the same opportunity to adapt.

The full hybrid ensemble has development AUC 0.94803, compared with 0.94331 using quality fields, 0.94870 with quality and answer relations, and 0.94254 with quality and state correspondence. The

corresponding pure-state values are 0.94756, 0.94071, 0.94690 and 0.94087. Relations among answers account for more of the observed increment than state correspondence alone. The auxiliary noise-relative hybrid attains 0.94469, below the original full hybrid, and remains a secondary candidate. The primary candidate and independent test population remained fixed throughout the ablation analysis.

Tables S8 and S9 report all ensembles and paired comparisons in the ablation analysis. Intervals use 10,000 paired issue/template-group resamples of the saved out-of-fold predictions (1,200 model–issue records in 297 groups), conditional on those fits and preceding development choices. These explanatory intervals resample predictions with the estimators held fixed; the independent evaluation is reported separately.

Table S8: Development performance for all retained information blocks. Hybrid combines repeated observations with state interventions; the two generic arms use five calls or the funded total.

| Representation                         | AUC     | Conditional 95%    | Brier   |
|----------------------------------------|---------|--------------------|---------|
| State / Full                           | 0.94756 | [0.93447, 0.95997] | 0.08499 |
| State / Quality                        | 0.94071 | [0.92726, 0.95338] | 0.09262 |
| State / Quality + relations            | 0.94690 | [0.93351, 0.95942] | 0.08547 |
| State / Quality + state                | 0.94087 | [0.92724, 0.95379] | 0.09248 |
| Generic (five) / Full                  | 0.94264 | [0.92908, 0.95530] | 0.09120 |
| Generic (five) / Quality               | 0.93989 | [0.92616, 0.95265] | 0.09457 |
| Generic (five) / Quality + relations   | 0.94233 | [0.92882, 0.95488] | 0.09200 |
| Generic (five) / Quality + state       | 0.94018 | [0.92671, 0.95273] | 0.09456 |
| Generic (funded) / Full                | 0.94334 | [0.93008, 0.95563] | 0.09128 |
| Generic (funded) / Quality             | 0.93922 | [0.92540, 0.95198] | 0.09534 |
| Generic (funded) / Quality + relations | 0.94243 | [0.92876, 0.95525] | 0.09113 |
| Generic (funded) / Quality + state     | 0.94022 | [0.92680, 0.95282] | 0.09442 |
| Hybrid / Full                          | 0.94803 | [0.93490, 0.96042] | 0.08433 |
| Hybrid / Quality                       | 0.94331 | [0.93024, 0.95556] | 0.09114 |
| Hybrid / Quality + relations           | 0.94870 | [0.93578, 0.96076] | 0.08458 |
| Hybrid / Quality + state               | 0.94254 | [0.92922, 0.95503] | 0.09112 |
| Hybrid / Noise-relative                | 0.94469 | [0.93113, 0.95737] | 0.08668 |
| Generic (five) / Noise-relative        | 0.94063 | [0.92709, 0.95346] | 0.09314 |
| Generic (funded) / Noise-relative      | 0.94041 | [0.92661, 0.95340] | 0.09325 |

Table S9: Paired development AUC comparisons. Intervals resample issue groups in the saved out-of-fold predictions, conditional on the fitted estimators and development choices.

| First representation    | Second representation                  | Difference | Conditional 95%     |
|-------------------------|----------------------------------------|------------|---------------------|
| State / Full            | State / Quality                        | 0.00684    | [0.00384, 0.01002]  |
| State / Full            | State / Quality + relations            | 0.00065    | [−0.00076, 0.00212] |
| State / Full            | State / Quality + state                | 0.00669    | [0.00364, 0.00993]  |
| Generic (five) / Full   | Generic (five) / Quality               | 0.00275    | [−0.00013, 0.00571] |
| Generic (five) / Full   | Generic (five) / Quality + relations   | 0.00031    | [−0.00100, 0.00161] |
| Generic (five) / Full   | Generic (five) / Quality + state       | 0.00246    | [−0.00052, 0.00537] |
| Generic (funded) / Full | Generic (funded) / Quality             | 0.00412    | [0.00132, 0.00700]  |
| Generic (funded) / Full | Generic (funded) / Quality + relations | 0.00092    | [−0.00052, 0.00245] |
| Generic (funded) / Full | Generic (funded) / Quality + state     | 0.00313    | [0.00001, 0.00639]  |

| First representation         | Second representation                  | Difference | Conditional 95%      |
|------------------------------|----------------------------------------|------------|----------------------|
| Hybrid / Full                | Hybrid / Quality                       | 0.00472    | [0.00091, 0.00872]   |
| Hybrid / Full                | Hybrid / Quality + relations           | -0.00067   | [-0.00246, 0.00101]  |
| Hybrid / Full                | Hybrid / Quality + state               | 0.00549    | [0.00205, 0.00911]   |
| State / Full                 | Generic (funded) / Full                | 0.00421    | [-0.00124, 0.00964]  |
| Hybrid / Full                | Generic (funded) / Full                | 0.00469    | [-0.00053, 0.00983]  |
| State / Quality              | Generic (funded) / Quality             | 0.00149    | [-0.00375, 0.00686]  |
| Hybrid / Quality             | Generic (funded) / Quality             | 0.00409    | [-0.00046, 0.00871]  |
| State / Quality + relations  | Generic (funded) / Quality + relations | 0.00448    | [-0.00104, 0.01011]  |
| Hybrid / Quality + relations | Generic (funded) / Quality + relations | 0.00627    | [0.00107, 0.01154]   |
| State / Quality + state      | Generic (funded) / Quality + state     | 0.00065    | [-0.00448, 0.00579]  |
| Hybrid / Quality + state     | Generic (funded) / Quality + state     | 0.00232    | [-0.00209, 0.00674]  |
| Hybrid / Noise-relative      | Generic (funded) / Noise-relative      | 0.00428    | [-0.00110, 0.00966]  |
| Hybrid / Noise-relative      | Hybrid / Full                          | -0.00334   | [-0.00643, -0.00035] |

### Frozen auxiliary predictor settings

The auxiliary noise-relative ensemble uses the same four learner families and parameter grid. Its four representation choices are the original features, original plus noise contrasts, original/quality plus noise contrasts, and original plus the complete ordered/aggregate and contrast fields. The principal hybrid and funded-generic configurations used in the prespecified auxiliary comparison are listed below. Each member has probability weight 0.25; all other learner settings match the main ensemble.

Table S10: Frozen noise-relative candidate and corresponding funded-generic configurations.

| Arm              | Learner        | Representation   | Settings           |
|------------------|----------------|------------------|--------------------|
| Hybrid           | Logistic       | Full + contrasts | C=0.1              |
| Hybrid           | Hist. boosting | Full + contrasts | leaves=15, leaf=10 |
| Hybrid           | Extra trees    | Full + contrasts | depth=8, leaf=5    |
| Hybrid           | CatBoost       | Full + contrasts | depth=4, l2=3      |
| Generic (funded) | Logistic       | Original         | C=0.1              |
| Generic (funded) | Hist. boosting | Full + contrasts | leaves=7, leaf=10  |
| Generic (funded) | Extra trees    | Original         | depth=8, leaf=5    |
| Generic (funded) | CatBoost       | Full + contrasts | depth=4, l2=3      |

### Structured-state responses across source corpora

The seven-corpus structured-state experiment used a versioned sample drawn from the retained event pools. Sampling selected up to 500 entries per corpus; the available narrative pool contributed 98 entries. The resulting sample contained 3,098 entries and 3,073 unique corpus–event identifiers. Each entry was evaluated under the eight specified conditions in three repeated acquisitions, yielding all 74,352 planned records. The request route was `gpt-5.4-mini`, with a recorded output limit of 1,536 tokens, the specified condition-specific temperatures and memory initialization at each 50-entry block. Repetition indices do not imply service-side seed determinism.

The complete membership table identifies each source file, row, corpus, event and episode. Overlap

with the source-intake, comparison and finite-action samples is reported by event identifier. The two baseline acquisitions share a design and source pools; their response records are retained separately.

Table S11: Membership of the complete baseline acquisition. Duplicate event identifiers refer to distinct retained source rows and remain visible in the membership file.

| Corpus     | Available rows | Sample entries | Unique events |
|------------|----------------|----------------|---------------|
| Narratives | 98             | 98             | 84            |
| ds000210   | 3,720          | 500            | 500           |
| ds001618   | 3,636          | 500            | 500           |
| ds001882   | 4,737          | 500            | 500           |
| ds000212   | 3,584          | 500            | 500           |
| ds004917   | 38,195         | 500            | 500           |
| ds004042   | 4,017          | 500            | 489           |
| Total      |                | 3,098          | 3,073         |

Table S12: Event overlap between the complete baseline sample and three related samples. Shared IDs are unique corpus–event identifiers; counts refer to the complete baseline sample.

| Related sample        | Entries | Unique IDs | Shared IDs |
|-----------------------|---------|------------|------------|
| Initial source intake | 224     | 224        | 81         |
| Comparison pool       | 497     | 493        | 251        |
| Finite-action sample  | 818     | 803        | 501        |

Raw model fields and outputs completed by the rule-based implementation were analyzed separately. The raw reason-response index is defined when a final action, an initial impulse and a reason graph with a node list are present; it equals one when the graph is nonempty and the final action differs from the initial impulse. Among paired scorable records, the structured condition exceeded reason removal in all seven corpora. The equal-corpus mean difference was 0.679, with a conditional 95% episode-resampling interval [0.657, 0.708]. These estimates describe the paired scorable subset. Coverage is retained in Table S13; unsupplied raw fields are not assigned zero.

Table S13: Structured-minus-reason-removal contrasts on paired raw fields. Planned pairs count entry–repetition pairs. Episodes are the source episode identifiers represented in the scorable subset.

| Corpus     | Planned pairs | Reason pairs | Episodes | Difference |
|------------|---------------|--------------|----------|------------|
| Narratives | 294           | 60           | 4        | 0.217      |
| ds000210   | 1,500         | 387          | 4        | 0.765      |
| ds001618   | 1,500         | 394          | 3        | 0.640      |
| ds001882   | 1,500         | 371          | 8        | 0.571      |
| ds000212   | 1,500         | 189          | 14       | 0.884      |
| ds004917   | 1,500         | 508          | 1        | 0.927      |
| ds004042   | 1,500         | 201          | 2        | 0.751      |

The primary field definitions also require numeric identity and continuity weights for the self-state index and a Boolean application field for the veto index. No paired self-state records satisfied those requirements. Only five pairs supplied the required veto fields on both sides, all with a zero contrast. The complete raw-field coverage table accompanies the source data. Outputs from the original field-completion rules are retained in a separate compatibility table, identifying their computational provenance.

The high-randomness condition had higher empirical final-action-string entropy than the structured condition in every corpus. Entropy is computed in bits on the same paired nonmissing actions within each corpus; it describes variation in the emitted strings, including their wording. It does not measure an open-ended action ontology or establish an entropy-matched comparison. Paired effects and entropy differences use the registered 2,000 resamples of source episode identifiers, carrying each selected episode’s conditions and repeats together. Corpora remain fixed in the equal-corpus summary. Where a corpus contributes one episode, resampling cannot estimate variation between episodes; the point estimate is retained without an inferential interval.

Table S14: Empirical action-string entropy in bits on matched nonmissing records. The paired denominator is shown for each corpus.

| Corpus     | Paired actions | Structured | Random | Difference |
|------------|----------------|------------|--------|------------|
| Narratives | 233/294        | 7.090      | 7.769  | 0.678      |
| ds000210   | 871/1500       | 9.173      | 9.415  | 0.242      |
| ds001618   | 896/1500       | 7.973      | 9.038  | 1.064      |
| ds001882   | 1015/1500      | 4.022      | 4.110  | 0.088      |
| ds000212   | 363/1500       | 6.875      | 7.809  | 0.934      |
| ds004917   | 1183/1500      | 7.625      | 9.846  | 2.221      |
| ds004042   | 1444/1500      | 10.172     | 10.390 | 0.218      |

The acquisition saved 74,352 terminal logical requests and 74,353 attempts. Reported usage totalled 31,851,562 tokens; 3 attempts lacked usage. Those attempts and all terminal failures remain in the acquisition accounting. Missing usage contributes an unknown amount to the total cost.

## Linear localization prediction

The six-model reference-dependent localization study found a substantial association between state diagnostics and correctness. Across 300 issues, 18,000 condition/repeat observations aggregate to 1,800 model–issue records, of which 1,425 meet the two-of-three localization criterion. Issue-grouped analysis gives AUC 0.737 for the ordinary baseline and 0.924 after adding the state features, a difference of 0.187 [0.127, 0.268] from 2,000 full-refit issue-bootstrap draws. Several augmented features use reference implementation files during scoring. The association therefore concerns reference-dependent diagnostics, even though reference files were excluded from generation prompts and the ordinary baseline.

The reference-free experiment tests predictive utility under information available before scoring. Sol, DeepSeek, Qwen and Mistral each evaluated the same 300 bridge issues. The common baseline was augmented with state probes, five generic probes, or generic acquisition funded to at least the state-probe token total when sufficient metering and calls were available. The resulting predictors use identical preprocessing and model complexity (Table S15).

Table S15: Reference-free localization reliability. Bridge estimates are grouped out-of-fold. Reused-holdout estimates condition on the bridge fits; these disjoint test outcomes had been viewed before the service-driven model amendment.

| Feature set             | Bridge AUC | Brier | Reused AUC | Brier |
|-------------------------|------------|-------|------------|-------|
| Shared baseline         | 0.920      | 0.108 | 0.950      | 0.074 |
| Shared + state probes   | 0.923      | 0.106 | 0.954      | 0.072 |
| Shared + five generic   | 0.924      | 0.105 | 0.957      | 0.072 |
| Shared + funded generic | 0.924      | 0.105 | 0.957      | 0.072 |

State-probe AUC was 0.923 on the bridge and 0.954 on the 100-issue reassessment. The relevant

incremental comparison is against the funded generic diagnostic: the paired AUC differences were  $-0.000972$   $[-0.013172, 0.004733]$  and  $-0.003221$   $[-0.010227, 0.002950]$ , respectively. The reassessment is separate from model fitting, but its outcomes were observed before the amendment that replaced an unavailable API service; it is a conditional held-out evaluation.

Pooled discrimination combines within-model and between-model outcome variation. The bridge contains 1,200 model-issue records in 297 issue/template groups, with 678 positive outcomes. Within-model CSB/funded AUCs were 0.826/0.858 for Sol, 0.892/0.899 for DeepSeek, 0.674/0.651 for Qwen and 0.891/0.889 for Mistral. The reassessment contains 400 records and 258 positives; corresponding AUCs were 0.828/0.846, 0.916/0.924, 0.711/0.746 and 0.973/0.976. Sol has only four negative test outcomes and Mistral only three positives, making those estimates particularly sensitive to individual observations. Figure S12 presents the pooled and model-specific estimates with their support.

All four held-model point comparisons, and all four model-specific reassessment comparisons, favored funded generic diagnostics. Repository-held-out and token-prefix results were mixed. The conservative prior-interruption subset contains 693 bridge records and gives a saved out-of-fold AUC difference of 0.000495; its changed composition makes it a sensitivity analysis.

Recorded usage was complete and generic acquisition sufficiently funded in 1,151/1,200 bridge pairs; 68 attempts lacked usage. Figure S14 reports these accounting categories, retrospective budget comparisons and the upper bound of 42 potentially unlogged administrative attempts. Figure S13 complements discrimination with observed calibration, paired probability differences and selective localization risk. The figures summarize fixed predictions and the corresponding resampling distributions; model-resolved calibration is given in Supplementary Figure S11.

## Supplementary analyses and research records

These notes detail the samples, controls and statistical analyses for controlled action responses, agent-written state and software localization, specifying the sampling units, available information and estimators.

### Supplementary Note 1. Reference-dependent localization analysis

The localization endpoint requires at least two of three base calls each to retrieve a reference implementation file within their top three predictions. The 1,800 model-issue records contain 1,425 positive and 375 negative outcomes under this definition. The ordinary baseline uses reference-free features; the augmented predictor includes diagnostic components calculated with reference files.

Issue-grouped five-fold cross-validation with training-only preprocessing gives AUC 0.737216 for the baseline and 0.924486 after adding the reference-dependent CSB features. The difference is 0.187269, with a two-sided 95% full-refit issue-bootstrap interval  $[0.126941, 0.267774]$ ; all 2,000 draws are estimable. Copies of a bootstrap-sampled issue share a fold. Numerical scaling is fitted separately from categorical encoding.

For held-group evaluation, each model or repository is excluded from training and predicted directly from the remaining groups. The archive also contains cross-validation on the retained groups as a separate sensitivity calculation. Direct held-group tests retain single-class groups as non-estimable for AUC. A held-model evaluation can include other models' records for the same issues in training, so its target is transfer across models conditional on those tasks.

## Supplementary Note 2. Samples and generation accounting

| Finite-action accounting layer                           | Count  |
|----------------------------------------------------------|--------|
| Frozen intake: 32 source events in each of seven corpora | 224    |
| Sampled event entries for the 120-per-corpus request     | 818    |
| Planned calls: $818 \times 6 \times 4 \times 3$          | 58,896 |
| Acquisition log: recorded calls                          | 58,836 |
| Acquisition log: remote generations                      | 58,835 |
| Acquisition log: unrecovered generations                 | 1      |
| Merged scored source rows                                | 61,056 |
| Duplicate scored keys removed                            | 3,240  |
| Verified deduplicated scored rows                        | 57,816 |
| Distinct corpus/base-event pairs in scored rows          | 803    |

The accounting distinguishes planned calls, generation logs, merged exports and deduplicated scored observations. The difference  $58,896 - 58,835$  spans these layers and is not a generation-failure count. The intake contains 32 source events per corpus; the finite-action evaluation requested 120 entries per corpus. All 803 unique identifiers in the finite-action list occur in the scored behavior set. The two lists share 73 identifiers, with 151 intake events absent from the finite-action sample and 730 finite-action events absent from the intake. This overlap specifies their shared source material.

The baseline lesion matrix uses a separate sample from the same seven-corpus pool. Its saved configuration sets a maximum of 500 events per corpus, eight conditions and three random seeds. The observed sample size depends on the available events in each corpus. The saved run-level metrics contain all 168 corpus-condition-seed cells, and the run summary records 74,352 calls, 74,336 remote generations and 16 unrecovered generations. These are run-level baseline totals, reported separately from the finite-action sample; the complete baseline replication provides entry-level membership in Tables S11–S12.

| Sample                  | Observed entries<br>or configured cap | Rows | Unique<br>IDs | Overlap<br>with 32 | Overlap<br>with 120 |
|-------------------------|---------------------------------------|------|---------------|--------------------|---------------------|
| Frozen intake           | 32                                    | 224  | 224           | —                  | 73                  |
| Finite-action<br>sample | Six corpora: 120;<br>Narratives: 98   | 818  | 803           | 73                 | —                   |
| Comparison snapshot     | 71                                    | 497  | 493           | 74                 | 237                 |
| Baseline lesion matrix  | Configured cap:<br>500/corpus         | —    | —             | —                  | —                   |

*Table note.* Counts and intersections are computed directly from the retained Parquet files; unique IDs are corpus/event pairs. The 71-per-corpus comparison snapshot contains 497 entries and is distinct from the baseline execution cohort, which uses a cap of 500 per corpus. The 818-entry sample is stored in `full_sample_manifest.parquet`. The baseline row specifies its configured cap; entry-level membership and overlap are reported for the enumerated replication in Tables S11–S12. Dashes distinguish aggregate-only fields from observed sample counts. In the finite-action list, `ds004042` contributes 119 unique IDs among 120 entries and `Narratives` contributes 84 among 98; the other five corpora each contribute 120 unique IDs. Verified memberships and source hashes accompany the analysis records.

For each of the four controlled open-weight models, the  $3 \times 20 \times 100 \times 7 \times 3 = 126,000$  scored rows comprise 108,000 open-weight generation records and 18,000 offline-policy records, with no duplicate scoring keys. Recorded error rows are 689 (Qwen 14B), 1,523 (Qwen 7B), 212 (Qwen 32B) and 3,942 (Mistral 7B). The core seven variants include post-hoc/scrambled controls; strict decisive-field removal is a separate slice. Frozen plans identify the offline policy as the expected-action marginal within selected events, not a fitted empirical structured-output marginal.

### Supplementary Note 3. Explicit rules and visible-output consistency

The deterministic baseline reconstructs the visible finite-action rule template and module settings without a hidden-target argument. It matches the target in all 14,454 structured-condition records. The result identifies explicit rule following as sufficient for performance in that experiment.

The output-only consistency rule excludes condition and provenance information and can score at least one dimension in 1.9075% of 26,946 traces. An exploratory semantic assessment uses 252 fixed-hash outputs, four per variant/corpus cell, with two judges and 504 logged calls. Sampling metadata is withheld. Each accepted judgment requires a quotation occurring exactly in the supplied text; absent or ambiguous evidence is unscorable. Coverage and agreement are reported separately because conditional agreement describes only the scorable subset. Equal-weight and four leave-one-component-out AFCI summaries assess sensitivity to the score definition; their weights were fixed without optimizing the contrasts.

| Component                           | Jointly scorable / 252 | Three-category $\kappa$ | Agreement within jointly scorable |
|-------------------------------------|------------------------|-------------------------|-----------------------------------|
| Reason                              | 186                    | 0.410                   | 0.984                             |
| Memory                              | 64                     | 0.810                   | 0.969                             |
| Persistent goal (judge field: self) | 52                     | 0.759                   | 0.981                             |
| Veto                                | 7                      | 0.403                   | 0.857                             |

The three categories are unscorable, inconsistent and consistent; failed responses enter the unscorable category. Five of 504 terminal judgments returned HTTP 403, and the logs retain 506 total attempts. Coverage stratified by each corpus and variant is exported separately. Accepted quotations had to occur exactly in the supplied output; all were within the requested 160-character limit.

The equal-weight composites, including AFCI and the finite-action response summaries, are operational score definitions. Equal weights give each registered term the same arithmetic contribution while preserving its separate operational definition. Component-level values and the fixed leave-one-component-out analysis show how differences in coverage and numerical range affect the composite.

### Supplementary Note 4. Patch application and repair outcomes

The official harness received 100 model-issue submissions from 50 issues and two models; 44 completed, one resolved and 56 returned harness errors. The application-and-repair experiment generated 100 nonempty initial patches and made 93 repair attempts. Eight final patches applied locally, spanning seven issues; all eight completed the official harness and none resolved its issue. Final apply categories are: 48 format errors, 25 context/hunk mismatches, 13 recorded empty repair patches, two missing-path errors, four other recorded failures and eight applicable patches. The 92 zero-length final exports represent filtered or rejected patches after initially nonempty generation.

### Supplementary Note 5. Human evaluation and uncertainty reporting

**Sample construction.** The human-evaluation pool contains 532,800 candidate records from Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-32B-Instruct and Mistral-7B-Instruct-v0.3 experiments and their accompanying controls. Stratified sampling selects 150 records from each of eight conditions: structured, full-context stochastic, state-field removal, context scrambling, goal deletion, post-hoc explanation/scrambling, distribution-matched and strict goal deletion. Within each condition, quotas are allocated across source blocks, with grouped sampling by prompt scaffold, task family and probe component. Both sampling and display-order randomization use seed 20260509. The resulting 1,200 items cover three prompt scaffolds and 20 task families: 1,050 are open-weight experiment records and 150 are offline-policy control records. Sampling and scoring programs are retained with the source records.

**Masking and presentation.** Each item receives a unique anonymous identifier. Model identity, condition, task source and reference target are stored in a separate hidden manifest. The rating form displays only the initial action tendency, candidate actions, reason graph, self-state, memory, veto status, final action and its rationale, in addition to the item identifier and display index. A predefined lexicon replaces model-, scaffold- and condition-related strings in the visible text with uniform placeholders; 223 records contain such replacements. All three annotators use the same item list and randomized display order.

**Annotators and scoring instructions.** Three graduate students in relevant university disciplines, including doctoral and master’s students, perform the evaluation using a common codebook and written instructions. They participate voluntarily, having been informed that the ratings will be used for this research. The seven dimensions assess consistency between the final action and stated reasons; adherence to relevant memory or prior commitments; appropriateness of veto; whether veto changes the initial action tendency; post-hoc characteristics of explanations; verbosity or templated expression without corresponding behavioral control; and inappropriate action changes accompanying irrelevant cues. The codebook defines each dimension, its binary judgment criteria and the handling of unjudgeable cases. These common criteria and written procedures standardize scoring across annotators. Instructions require independent judgments based only on visible item information, no discussion of individual items, no access to the hidden manifest, and no changes to item identifiers or output text.

**Scores and aggregation.** Each dimension is binary: 1 means its judgment statement holds and 0 means it does not. Because the statements have different directions, higher scores do not uniformly indicate better performance. Unjudgeable dimensions are marked missing and accompanied by an unscorable-item flag; forms also record three-level confidence, a metadata-exposure flag and optional notes. Each annotator completes all 1,200 items, with individual returns preserved separately. At each item and dimension, when all three scores are valid, the label agreed on by at least two annotators is the majority-vote result.

**Agreement.** Agreement is computed on individual scores before majority aggregation, for each of three annotator pairs and seven dimensions. Pairwise jointly valid observations are used if values are missing. All retained core ratings are valid binary values, yielding 25,200 individual scores. Of 8,400 item–dimension combinations, 8,250 are unanimous and 150 have a 2:1 split, affecting 68 items. The 21 Cohen’s  $\kappa$  values have mean 0.961, minimum individual pair-by-dimension value 0.860 and minimum dimension mean 0.897. The two-annotator evaluation used a separate 300-item sample.

In the control-summary tables, the column named “lower 95% bound” is the 2.5th percentile of the bootstrap distribution, corresponding to the lower endpoint of a two-sided 95% percentile interval. The controlled comparisons use seven corpus units; the descriptive Student  $t$  intervals in main Figure 2b and Supplementary Figure S4b are distinct from these percentile-bootstrap bounds. In the execution study, the task is the paired unit: the primary analysis resamples tasks within each of eight fixed mechanism strata and gives each mechanism equal weight. Resampling the eight mechanism clusters and leaving one mechanism out assess sensitivity to mechanism composition. Additional continuations improve measurement within a task without increasing the task count. Main Methods 5.1–5.2 specify these distinct uncertainty summaries. Methods 5.2 also collects the model and generation settings; Methods 5.3 defines the linear and richer predictive representations, budget comparisons and cohort roles.

## Supplementary Note 6. Cohort membership and evaluation roles

The execution study contains 160 formal instances across eight mechanisms, including 32 predetermined redundancy cases and a 40-instance control subset; development instances are excluded. The linear

localization comparison separates 20 implementation-development issues, 300 bridge issues and 100 deduplicated reassessment issues. Its state features use no reference files. The 100-issue outcomes had been observed before the service-driven substitution of DeepSeek for Grok, so this cohort supports a conditional reassessment of the refitted four-model predictors. The independent prediction sample comprises 800 unused issues acquired after predictor selection, with all predictions frozen before outcome scoring. Supplementary Note 14 documents the model-service cohorts, their acquisition order and the service failures. Request records distinguish shared calls, diagnostic queries, retries and tokens, separating joint acquisition cost from the cost of each strategy.

## Supplementary Note 7. Comparisons on shared model subsets

All five configurations are compared on the same 40 execution tasks and 60 bridge issues. The analysis uses DeepSeek together with the saved Sol, Qwen, Mistral and Astra observations. Supplementary Note 14 specifies the cohort composition and service-driven model substitution.

Table S16: Execution effects on the shared 40-task subset.

| Model          | State  | Tasks | Effect | 95% interval   |
|----------------|--------|-------|--------|----------------|
| Sol            | memory | 20    | 0.911  | [0.854, 0.984] |
| Sol            | goal   | 20    | 0.510  | [0.344, 0.667] |
| DeepSeek route | memory | 20    | 0.703  | [0.635, 0.776] |
| DeepSeek route | goal   | 20    | 0.328  | [0.208, 0.448] |
| Qwen 14B       | memory | 20    | 0.120  | [0.042, 0.198] |
| Qwen 14B       | goal   | 20    | 0.062  | [0.010, 0.115] |
| Mistral 7B     | memory | 20    | 0.000  | [0.000, 0.000] |
| Mistral 7B     | goal   | 20    | 0.000  | [0.000, 0.000] |
| Astra          | memory | 20    | 0.984  | [0.969, 1.000] |
| Astra          | goal   | 20    | 0.604  | [0.458, 0.760] |

Table S17: Prediction on the shared 60-issue subset.

| Model          | CSB AUC | Funded AUC | $\Delta$ AUC [conditional 95%] | Estimable |
|----------------|---------|------------|--------------------------------|-----------|
| DeepSeek route | 0.854   | 0.881      | -0.02736 [-0.07446, 0.01563]   | 2000/2000 |
| Sol            | 0.933   | 0.925      | 0.00721 [-0.03366, 0.04575]    | 2000/2000 |
| Astra          | 0.973   | 0.964      | 0.00893 [0.00000, 0.03571]     | 1971/2000 |
| Mistral 7B     | 0.977   | 0.953      | 0.02339 [0.00000, 0.10169]     | 1912/2000 |
| Qwen 14B       | 0.619   | 0.620      | -0.00118 [-0.10296, 0.09933]   | 2000/2000 |

## Supplementary Note 8. Execution controls and sensitivity

Estimates use the planned task denominator and equal mechanism weights. Each cell reports a signed difference and a two-sided 95% within-mechanism task-bootstrap interval. The three core controls use all registered tasks: 80 per component for each core model and 20 per component for Astra. These intervals describe secondary contrasts. The eight-comparison multiplicity correction applies to the primary effects; no equivalence margin was specified for the controls.

Table S18: Restoration and irrelevant-deletion controls.

| Model          | State  | Tasks | Rescue – lesion      | Sham – irrelevant     | Sham – rescue          |
|----------------|--------|-------|----------------------|-----------------------|------------------------|
| Sol            | memory | 80    | 0.753 [0.662, 0.834] | 0.006 [-0.016, 0.028] | -0.003 [-0.019, 0.013] |
| Sol            | goal   | 80    | 0.366 [0.272, 0.462] | 0.006 [-0.031, 0.044] | 0.000 [-0.016, 0.016]  |
| DeepSeek route | memory | 80    | 0.519 [0.431, 0.600] | 0.003 [-0.028, 0.034] | 0.022 [-0.016, 0.059]  |

Table S18: Restoration and irrelevant-deletion controls. (continued)

| Model          | State  | Tasks | Rescue – lesion      | Sham – irrelevant      | Sham – rescue          |
|----------------|--------|-------|----------------------|------------------------|------------------------|
| DeepSeek route | goal   | 80    | 0.238 [0.156, 0.319] | -0.013 [-0.047, 0.022] | 0.022 [-0.013, 0.059]  |
| Qwen 14B       | memory | 80    | 0.072 [0.025, 0.122] | 0.003 [-0.037, 0.044]  | 0.009 [-0.009, 0.031]  |
| Qwen 14B       | goal   | 80    | 0.041 [0.013, 0.075] | 0.003 [-0.022, 0.031]  | 0.003 [-0.009, 0.016]  |
| Mistral 7B     | memory | 80    | 0.000 [0.000, 0.000] | 0.003 [0.000, 0.009]   | 0.003 [0.000, 0.009]   |
| Mistral 7B     | goal   | 80    | 0.016 [0.000, 0.037] | 0.000 [-0.009, 0.009]  | -0.009 [-0.022, 0.000] |
| Astra          | memory | 20    | 1.000 [1.000, 1.000] | -0.016 [-0.031, 0.000] | -0.016 [-0.031, 0.000] |
| Astra          | goal   | 20    | 0.604 [0.458, 0.760] | 0.000 [0.000, 0.000]   | 0.000 [0.000, 0.000]   |

The following validity contrasts use the fixed 40-task subset in each core model (20 per component). Fact replacement changes the stored fact or policy while leaving the environment unchanged. This contrast measures success against the original task target; following the replacement target is evaluated separately. Astra did not receive these additional branches. Supplementary Note 12 reports the corresponding replacement-target following rates.

Table S19: Fixed-subset paraphrase, replacement and history-compensation contrasts.

| Model          | State  | Paraphrase – sham      | Replacement – sham      | Compensation – lesion |
|----------------|--------|------------------------|-------------------------|-----------------------|
| Sol            | memory | -0.042 [-0.094, 0.000] | -0.693 [-0.844, -0.542] | 0.563 [0.333, 0.771]  |
| Sol            | goal   | 0.016 [-0.031, 0.062]  | -0.745 [-0.875, -0.615] | 0.198 [0.062, 0.333]  |
| DeepSeek route | memory | -0.016 [-0.078, 0.042] | -0.615 [-0.719, -0.495] | 0.438 [0.250, 0.620]  |
| DeepSeek route | goal   | -0.021 [-0.104, 0.047] | -0.719 [-0.797, -0.641] | 0.250 [0.125, 0.380]  |
| Qwen 14B       | memory | 0.000 [0.000, 0.000]   | -0.021 [-0.104, 0.042]  | 0.010 [0.000, 0.031]  |
| Qwen 14B       | goal   | 0.021 [-0.062, 0.104]  | -0.031 [-0.094, 0.021]  | 0.000 [-0.042, 0.031] |
| Mistral 7B     | memory | 0.000 [0.000, 0.000]   | 0.000 [0.000, 0.000]    | 0.000 [0.000, 0.000]  |
| Mistral 7B     | goal   | 0.000 [0.000, 0.000]   | 0.000 [0.000, 0.000]    | 0.000 [0.000, 0.000]  |

The mechanism sensitivity resamples the eight tested mechanisms, measuring dependence on their composition. Its precision is limited by those eight clusters. Holm adjustment is confined to the eight registered core primary contrasts.

Table S20: Mechanism-bootstrap sensitivity and adjusted primary tests.

| Model          | State  | Primary effect | Mechanism-bootstrap 95% CI | Holm-adjusted $p$ |
|----------------|--------|----------------|----------------------------|-------------------|
| Sol            | memory | 0.750          | [0.656, 0.800]             | < 0.001           |
| Sol            | goal   | 0.366          | [0.269, 0.491]             | < 0.001           |
| DeepSeek route | memory | 0.541          | [0.344, 0.712]             | < 0.001           |
| DeepSeek route | goal   | 0.259          | [0.147, 0.378]             | < 0.001           |
| Qwen 14B       | memory | 0.081          | [0.016, 0.162]             | 0.031             |
| Qwen 14B       | goal   | 0.044          | [0.009, 0.081]             | 0.137             |
| Mistral 7B     | memory | 0.003          | [0.000, 0.009]             | 1.000             |
| Mistral 7B     | goal   | 0.006          | [0.000, 0.019]             | 1.000             |

Redundancy strata and the analysis restricted to a valid stored state are secondary. A valid write satisfies the tool/schema contract and need not contain correct facts. Selection on write validity changes the estimand; it does not replace the planned denominator. Each cell gives its actual task count.

Table S21: Redundancy and valid-write selected sensitivity.

| Model          | State  | No redundant source             | Redundant source                  | Valid writes only               |
|----------------|--------|---------------------------------|-----------------------------------|---------------------------------|
| Sol            | memory | $n = 64$ ; 0.938 [0.898, 0.973] | $n = 16$ ; 0.000 [0.000, 0.000]   | $n = 80$ ; 0.750 [0.659, 0.838] |
| Sol            | goal   | $n = 64$ ; 0.465 [0.359, 0.570] | $n = 16$ ; -0.031 [-0.062, 0.000] | $n = 80$ ; 0.366 [0.272, 0.459] |
| DeepSeek route | memory | $n = 64$ ; 0.680 [0.625, 0.730] | $n = 16$ ; -0.016 [-0.031, 0.000] | $n = 79$ ; 0.550 [0.465, 0.628] |
| DeepSeek route | goal   | $n = 64$ ; 0.312 [0.230, 0.402] | $n = 16$ ; 0.047 [-0.016, 0.109]  | $n = 78$ ; 0.264 [0.188, 0.345] |
| Qwen 14B       | memory | $n = 64$ ; 0.090 [0.035, 0.148] | $n = 16$ ; 0.047 [0.000, 0.094]   | $n = 76$ ; 0.081 [0.034, 0.134] |
| Qwen 14B       | goal   | $n = 64$ ; 0.055 [0.016, 0.102] | $n = 16$ ; 0.000 [-0.031, 0.031]  | $n = 70$ ; 0.046 [0.009, 0.090] |
| Mistral 7B     | memory | $n = 64$ ; 0.004 [0.000, 0.012] | $n = 16$ ; 0.000 [0.000, 0.000]   | $n = 31$ ; 0.008 [0.000, 0.025] |
| Mistral 7B     | goal   | $n = 64$ ; 0.008 [0.000, 0.023] | $n = 16$ ; 0.000 [0.000, 0.000]   | $n = 29$ ; 0.042 [0.000, 0.083] |
| Astra          | memory | $n = 20$ ; 0.984 [0.969, 1.000] | Not estimable                     | $n = 20$ ; 0.984 [0.969, 1.000] |
| Astra          | goal   | $n = 20$ ; 0.604 [0.458, 0.750] | Not estimable                     | $n = 20$ ; 0.604 [0.458, 0.750] |

### Supplementary Note 9. Preparation, failures and execution cost

Valid/exact counts are tasks. The two sham columns are successful continuations over all planned sham continuations. “Raw” removes only the preparation-validity gate and is secondary. “Chars” is the number of tasks with equal serialized character counts after targeted and irrelevant deletion; this is not token or semantic equivalence.

Table S22: State preparation and sham-success denominators.

| Model          | State  | $N$ | Valid | Exact | Primary sham | Raw sham | Chars |
|----------------|--------|-----|-------|-------|--------------|----------|-------|
| Sol            | memory | 80  | 80    | 80    | 296/320      | 296/320  | 80/80 |
| Sol            | goal   | 80  | 80    | 70    | 300/320      | 300/320  | 80/80 |
| DeepSeek route | memory | 80  | 79    | 79    | 216/320      | 216/320  | 80/80 |
| DeepSeek route | goal   | 80  | 78    | 78    | 218/320      | 218/320  | 80/80 |
| Qwen 14B       | memory | 80  | 76    | 76    | 34/320       | 34/320   | 80/80 |
| Qwen 14B       | goal   | 80  | 70    | 69    | 40/320       | 40/320   | 79/80 |
| Mistral 7B     | memory | 80  | 31    | 31    | 1/320        | 13/320   | 80/80 |
| Mistral 7B     | goal   | 80  | 29    | 29    | 4/320        | 20/320   | 80/80 |
| Astra          | memory | 20  | 20    | 20    | 79/80        | 79/80    | 20/20 |
| Astra          | goal   | 20  | 20    | 18    | 80/80        | 80/80    | 20/20 |

The branch failure counts pool the two task components. Invalid/missing transactions and policy failures count continuations; invalid JSON and non-200 terminal HTTP responses count logical calls. Categories can overlap. The primary analysis includes all of these failed observations.

Table S23: Execution branch failures and retained calls.

| Model | Branch            | Contin. | Calls | Invalid transaction | Policy failure | Invalid JSON | HTTP failure |
|-------|-------------------|---------|-------|---------------------|----------------|--------------|--------------|
| Sol   | fact replace-ment | 160     | 320   | 0                   | 126            | 0            | 0            |

Table S23: Execution branch failures and retained calls. (continued)

| Model          | Branch     |          | Contin. | Calls | Invalid transaction | Policy failure | Invalid JSON | HTTP failure |
|----------------|------------|----------|---------|-------|---------------------|----------------|--------------|--------------|
| Sol            | history    | com-     | 160     | 320   | 0                   | 14             | 0            | 0            |
|                | pensation  |          |         |       |                     |                |              |              |
| Sol            | irrelevant |          | 640     | 1280  | 1                   | 47             | 0            | 0            |
| Sol            | lesion     |          | 640     | 1280  | 200                 | 201            | 0            | 0            |
| Sol            | paraphrase |          | 160     | 320   | 0                   | 13             | 0            | 0            |
| Sol            | rescue     |          | 640     | 1280  | 0                   | 43             | 0            | 0            |
| Sol            | sham       |          | 640     | 1280  | 1                   | 43             | 0            | 0            |
| DeepSeek route | fact       | replace- | 160     | 318   | 40                  | 106            | 40           | 1            |
|                | ment       |          |         |       |                     |                |              |              |
| DeepSeek route | history    | com-     | 160     | 319   | 31                  | 12             | 31           | 3            |
|                | pensation  |          |         |       |                     |                |              |              |
| DeepSeek route | irrelevant |          | 640     | 1270  | 173                 | 30             | 172          | 7            |
| DeepSeek route | lesion     |          | 640     | 1274  | 312                 | 150            | 311          | 11           |
| DeepSeek route | paraphrase |          | 160     | 318   | 29                  | 14             | 29           | 4            |
| DeepSeek route | rescue     |          | 640     | 1274  | 175                 | 45             | 171          | 9            |
| DeepSeek route | sham       |          | 640     | 1271  | 173                 | 33             | 171          | 12           |
| Qwen 14B       | fact       | replace- | 160     | 320   | 39                  | 105            | 0            | 0            |
|                | ment       |          |         |       |                     |                |              |              |
| Qwen 14B       | history    | com-     | 160     | 320   | 22                  | 122            | 0            | 0            |
|                | pensation  |          |         |       |                     |                |              |              |
| Qwen 14B       | irrelevant |          | 640     | 1280  | 98                  | 470            | 0            | 0            |
| Qwen 14B       | lesion     |          | 640     | 1280  | 180                 | 426            | 0            | 0            |
| Qwen 14B       | paraphrase |          | 160     | 320   | 31                  | 106            | 0            | 0            |
| Qwen 14B       | rescue     |          | 640     | 1280  | 97                  | 473            | 0            | 0            |
| Qwen 14B       | sham       |          | 640     | 1280  | 99                  | 467            | 0            | 0            |
| Mistral 7B     | fact       | replace- | 160     | 304   | 51                  | 98             | 0            | 0            |
|                | ment       |          |         |       |                     |                |              |              |
| Mistral 7B     | history    | com-     | 160     | 303   | 64                  | 88             | 0            | 0            |
|                | pensation  |          |         |       |                     |                |              |              |
| Mistral 7B     | irrelevant |          | 640     | 1220  | 175                 | 432            | 0            | 0            |
| Mistral 7B     | lesion     |          | 640     | 1239  | 189                 | 425            | 0            | 0            |
| Mistral 7B     | paraphrase |          | 160     | 301   | 47                  | 102            | 0            | 0            |
| Mistral 7B     | rescue     |          | 640     | 1241  | 162                 | 448            | 1            | 0            |
| Mistral 7B     | sham       |          | 640     | 1241  | 165                 | 442            | 0            | 0            |
| Astra          | irrelevant |          | 160     | 320   | 0                   | 0              | 0            | 0            |
| Astra          | lesion     |          | 160     | 320   | 101                 | 29             | 0            | 0            |
| Astra          | rescue     |          | 160     | 320   | 0                   | 0              | 0            | 0            |
| Astra          | sham       |          | 160     | 320   | 0                   | 1              | 0            | 0            |

Execution acquisition charges preparation once and includes all continuation branches and transport attempts. The totals quantify reported usage, with unmetered attempts counted separately. A failed first tool call terminates the continuation, reducing its logical call count. API and local token totals follow their respective tokenizers.

Table S24: Execution acquisition and missing token metering.

| Model          | Logical calls | Attempts | Reported tokens | Unmetered attempts |
|----------------|---------------|----------|-----------------|--------------------|
| Sol            | 6400          | 6403     | 2,795,583       | 3                  |
| DeepSeek route | 6363          | 6367     | 5,384,953       | 54                 |
| Qwen 14B       | 6400          | 6400     | 2,760,862       | 0                  |
| Mistral 7B     | 6071          | 6071     | 2,787,200       | 0                  |
| Astra          | 1360          | 1360     | 555,915         | 0                  |

## Supplementary Note 10. Prediction by model and cohort

Each row compares the state probes with the token-funded generic comparator using the same issues. Bridge predictions are pooled-model, grouped out-of-fold estimates; reused-holdout predictions use the four-model frozen bridge fit. Counts are model–issue rows and are not independent across models. AP denotes average precision. Brier scores quantify probability error; the calibration plots additionally show observed success against predicted probability.

Table S25: Prediction metrics by model, feature set and data role.

| Model          | Role   | Probes  | Positive | $n$ predicted | AUC   | AP    | Brier |
|----------------|--------|---------|----------|---------------|-------|-------|-------|
| Sol            | Bridge | State   | 270/300  | 300/300       | 0.826 | 0.963 | 0.071 |
| Sol            | Bridge | Generic | 270/300  | 300/300       | 0.858 | 0.967 | 0.068 |
| DeepSeek route | Bridge | State   | 212/300  | 300/300       | 0.892 | 0.935 | 0.104 |
| DeepSeek route | Bridge | Generic | 212/300  | 300/300       | 0.899 | 0.933 | 0.098 |
| Qwen 14B       | Bridge | State   | 186/300  | 300/300       | 0.674 | 0.770 | 0.225 |
| Qwen 14B       | Bridge | Generic | 186/300  | 300/300       | 0.651 | 0.755 | 0.230 |
| Mistral 7B     | Bridge | State   | 10/300   | 300/300       | 0.891 | 0.604 | 0.023 |
| Mistral 7B     | Bridge | Generic | 10/300   | 300/300       | 0.889 | 0.529 | 0.023 |
| Sol            | Reused | State   | 96/100   | 100/100       | 0.828 | 0.989 | 0.035 |
|                | test   |         |          |               |       |       |       |
| Sol            | Reused | Generic | 96/100   | 100/100       | 0.846 | 0.990 | 0.039 |
|                | test   |         |          |               |       |       |       |
| DeepSeek route | Reused | State   | 83/100   | 100/100       | 0.916 | 0.976 | 0.063 |
|                | test   |         |          |               |       |       |       |
| DeepSeek route | Reused | Generic | 83/100   | 100/100       | 0.924 | 0.981 | 0.068 |
|                | test   |         |          |               |       |       |       |
| Qwen 14B       | Reused | State   | 76/100   | 100/100       | 0.711 | 0.852 | 0.166 |
|                | test   |         |          |               |       |       |       |
| Qwen 14B       | Reused | Generic | 76/100   | 100/100       | 0.746 | 0.878 | 0.159 |
|                | test   |         |          |               |       |       |       |
| Mistral 7B     | Reused | State   | 3/100    | 100/100       | 0.973 | 0.758 | 0.023 |
|                | test   |         |          |               |       |       |       |
| Mistral 7B     | Reused | Generic | 3/100    | 100/100       | 0.976 | 0.667 | 0.023 |
|                | test   |         |          |               |       |       |       |

The completely metered and sufficiently funded bridge subset is selected solely from recorded metering and the generic token total. It is a sensitivity analysis, not a replacement of the full planned denominator. Its estimates use predictions from the full-cohort grouped fits.

Table S26: Prediction on the complete-and-funded metering subset.

| Feature set     | Rows | Positive | AUC   | Brier |
|-----------------|------|----------|-------|-------|
| Shared baseline | 1151 | 642      | 0.919 | 0.109 |
| State probes    | 1151 | 642      | 0.923 | 0.106 |
| Five generic    | 1151 | 642      | 0.923 | 0.106 |
| Funded generic  | 1151 | 642      | 0.923 | 0.106 |

Joint bridge acquisition reports 76,484,918 tokens, counting shared calls once; 68 attempts have unreported usage. All 2,000 bridge full-refit bootstrap draws and all 5,000 fixed-predictor test-cluster draws were estimable for the primary differences. Single-class or otherwise non-estimable metrics remain NA in subgroup analyses. Repository- and model-held-out estimates and the attempt-level cost records accompany the numerical results. Held-model tests may contain the same issues under other models in training.

## Supplementary Note 11. Retrospective token-budget sensitivity

The prefixes below use realized token counts without inspecting outcome labels. These retrospective thresholds summarize costs already incurred during acquisition, including the query that exceeded each threshold. Only pairs with sufficient metering enter this sensitivity; their denominator can vary by cap.

Table S27: Retrospective token-prefix sensitivity.

| Cap    | Pairs | Probes  | AUC   | Brier | Mean calls | Mean tokens |
|--------|-------|---------|-------|-------|------------|-------------|
| 16,000 | 872   | csb     | 0.882 | 0.126 | 0.396      | 14171.331   |
| 16,000 | 872   | generic | 0.884 | 0.125 | 0.390      | 14162.151   |
| 24,000 | 1160  | csb     | 0.918 | 0.109 | 1.792      | 21617.003   |
| 24,000 | 1160  | generic | 0.918 | 0.109 | 1.778      | 21614.700   |
| 32,000 | 1162  | csb     | 0.923 | 0.105 | 3.380      | 29184.468   |
| 32,000 | 1162  | generic | 0.923 | 0.106 | 3.361      | 29156.966   |
| 48,000 | 1151  | csb     | 0.922 | 0.106 | 4.834      | 37017.755   |
| 48,000 | 1151  | generic | 0.922 | 0.106 | 4.996      | 37815.036   |
| 64,000 | 1151  | csb     | 0.923 | 0.106 | 4.986      | 38087.260   |
| 64,000 | 1151  | generic | 0.923 | 0.106 | 5.260      | 39544.712   |
| 96,000 | 1151  | csb     | 0.923 | 0.106 | 5.000      | 38211.056   |
| 96,000 | 1151  | generic | 0.923 | 0.106 | 5.288      | 39774.057   |

There were 553 excluded model–issue–cap cases, with reasons retained in the prefix exclusion ledger.

## Supplementary Note 12. Replacement targets and their denominators

Replacement-target following is defined only when the changed record yields a definable target. Undefined cases remain in the planned task-success analysis but are not counted as failures to follow a nonexistent replacement target. These are continuation counts, with four continuations per validity task. Astra did not receive this branch.

Table S28: Replacement-target following with definable and planned denominators.

| Model          | State  | Followed / definable | Undefined | Planned |
|----------------|--------|----------------------|-----------|---------|
| Sol            | memory | 61/80                | 0         | 80      |
| Sol            | goal   | 59/80                | 0         | 80      |
| DeepSeek route | memory | 56/80                | 0         | 80      |
| DeepSeek route | goal   | 49/80                | 0         | 80      |
| Qwen 14B       | memory | 4/76                 | 4         | 80      |
| Qwen 14B       | goal   | 10/80                | 0         | 80      |
| Mistral 7B     | memory | 0/12                 | 68        | 80      |
| Mistral 7B     | goal   | 0/32                 | 48        | 80      |

Main Figures 3–4 and Supplementary Figures S1–S3 show the execution effects and controls. The task estimates use four continuations per branch within eight mechanisms; the secondary-control intervals describe differences without an equivalence test.

## Supplementary Note 13. Held-group generalization and resource accounting

Each named group is excluded from training and evaluated directly. Repository tests withhold its issues across all four models. Model tests retain other models on the same issues and therefore do not simultaneously hold out tasks.

Table S29: Prediction in held-out repositories and models.

| Excluded group        | Train | Test | Positive | CSB AUC | Funded AUC | Brier | CSB / funded |
|-----------------------|-------|------|----------|---------|------------|-------|--------------|
| repo: astropy         | 1176  | 24   | 16       | 0.977   | 0.977      | 0.048 | 0.054        |
| repo: django          | 744   | 456  | 253      | 0.944   | 0.948      | 0.099 | 0.098        |
| repo: matplotlib      | 1108  | 92   | 58       | 0.944   | 0.962      | 0.073 | 0.070        |
| repo: seaborn         | 1184  | 16   | 11       | 0.964   | 0.964      | 0.100 | 0.100        |
| repo: flask           | 1188  | 12   | 9        | 1.000   | 1.000      | 0.022 | 0.021        |
| repo: requests        | 1176  | 24   | 13       | 0.881   | 0.867      | 0.135 | 0.140        |
| repo: xarray          | 1180  | 20   | 14       | 0.952   | 0.917      | 0.089 | 0.109        |
| repo: pylint          | 1176  | 24   | 11       | 0.832   | 0.790      | 0.175 | 0.182        |
| repo: pytest          | 1132  | 68   | 40       | 0.929   | 0.929      | 0.089 | 0.091        |
| repo: scikit-learn    | 1108  | 92   | 64       | 0.941   | 0.930      | 0.073 | 0.076        |
| repo: sphinx          | 1136  | 64   | 32       | 0.913   | 0.905      | 0.121 | 0.122        |
| repo: sympy           | 892   | 308  | 157      | 0.880   | 0.902      | 0.150 | 0.132        |
| model: DeepSeek route | 900   | 300  | 212      | 0.908   | 0.911      | 0.112 | 0.108        |
| model: Sol            | 900   | 300  | 270      | 0.863   | 0.901      | 0.074 | 0.072        |
| model: Mistral 7B     | 900   | 300  | 10       | 0.677   | 0.680      | 0.038 | 0.040        |
| model: Qwen 14B       | 900   | 300  | 186      | 0.624   | 0.644      | 0.257 | 0.254        |

Table S30: Bridge service failures and request-level metering.

| Bridge model   | Logical calls | Attempts | Reported tokens | Unknown | Issues with HTTP failure |
|----------------|---------------|----------|-----------------|---------|--------------------------|
| Sol            | 3900          | 3910     | 15,533,921      | 11      | 1/300                    |
| DeepSeek route | 4058          | 4061     | 20,647,973      | 37      | 33/300                   |
| Qwen 14B       | 3909          | 3909     | 17,112,923      | 0       | 0/300                    |
| Mistral 7B     | 4095          | 4109     | 23,190,101      | 20      | 3/300                    |
| Astra          | 780           | 781      | 3,102,504       | 1       | 0/60                     |

The four-model development cohort reports 76,484,918 tokens with 68 unmetered attempts. The conservative prior-interruption sensitivity retains 693 of 1,200 pairs; its CSB/funded AUC is 0.8313/0.8308. The administrative-interruption bound covers up to 42 additional attempts with indeterminate usage; it is reported separately from observed failures and token totals.

## Supplementary Note 14. Service configurations and cohort composition

The Grok service cohort contained 604 terminal HTTP failures in 4,221 logical calls, affecting 101/300 issues (92 base, 211 state and 301 generic failures). The Sol and Grok service cohorts each had 1,300 HTTP403 quota refusals and no valid model answer. Service-access refusals are classified separately from generated task responses. The Sol/Grok/Qwen/Mistral bridge CSB/funded AUCs were 0.944870/0.944280, with difference 0.000590 and full-refit interval [-0.008658, 0.005812]. Its 100-issue comparison difference was 0.005166, with conditional interval [-0.002629, 0.013293], and was confounded by the unavailable API rows.

Sol responses were recovered through a second credential on the same 100 issues. DeepSeek was initially evaluated on those issues with the first predictor’s unknown-category encoding, without DeepSeek bridge training. The Sol recovery CSB/funded AUCs were 0.833333/0.848958, a difference of -0.015625 with interval [-0.061224, 0.012249]. The DeepSeek transfer values were 0.922750/0.915663, a difference of 0.007087 with interval [-0.025514, 0.036864]. These estimates condition on the Sol/Grok/Qwen/Mistral fits; Supplementary Note 15 reports the Sol/DeepSeek/Qwen/Mistral predictors refitted on the consistent model cohort.

The amendment collected DeepSeek’s 160 execution tasks and 300 bridge issues and refitted the four reliability predictors on the consistent model cohort. Sol, Qwen, Mistral, Astra and all 100-issue responses were reused. The change followed service failure and observation of the preceding results. It consequently defines an amended analysis, with no additional independent issue cohort.

The API configurations are specified by their requested and returned third-party aliases. DeepSeek’s request configuration and second-credential route match the preceding transfer evaluation. The provider’s V4-Pro alias-migration notice documents the naming transition (DeepSeek, 2026). Eight preceding connectivity calls report 4,980 tokens and four unmetered attempts, counted once outside the formal denominators.

## Supplementary Note 15. Prediction on the reused holdout

Estimates use the same 100 held-out issues with saved Sol, DeepSeek, Qwen and Mistral responses and the refitted bridge predictors. These issues remain excluded from bridge training, while their outcomes were known before the amendment. The intervals therefore describe a post-amendment reassessment conditional on the fixed fits.

Table S31: Reused-holdout results for the consistent-model predictors.

| Model          | Features       | Positive<br>/100 | AUC    | AP     | Brier  |
|----------------|----------------|------------------|--------|--------|--------|
| Sol            | Base           | 96               | 0.8255 | 0.9894 | 0.0395 |
| Sol            | state probes   | 96               | 0.8281 | 0.9894 | 0.0349 |
| Sol            | five generic   | 96               | 0.8490 | 0.9907 | 0.0380 |
| Sol            | funded generic | 96               | 0.8464 | 0.9905 | 0.0387 |
| DeepSeek route | Base           | 83               | 0.9227 | 0.9796 | 0.0647 |
| DeepSeek route | state probes   | 83               | 0.9157 | 0.9759 | 0.0627 |
| DeepSeek route | five generic   | 83               | 0.9235 | 0.9809 | 0.0677 |
| DeepSeek route | funded generic | 83               | 0.9242 | 0.9810 | 0.0676 |
| Qwen 14B       | Base           | 76               | 0.6787 | 0.8294 | 0.1698 |
| Qwen 14B       | state probes   | 76               | 0.7111 | 0.8524 | 0.1664 |
| Qwen 14B       | five generic   | 76               | 0.7429 | 0.8778 | 0.1594 |
| Qwen 14B       | funded generic | 76               | 0.7456 | 0.8779 | 0.1585 |
| Mistral 7B     | Base           | 3                | 0.9828 | 0.7917 | 0.0225 |
| Mistral 7B     | state probes   | 3                | 0.9725 | 0.7576 | 0.0226 |
| Mistral 7B     | five generic   | 3                | 0.9759 | 0.6667 | 0.0222 |
| Mistral 7B     | funded generic | 3                | 0.9759 | 0.6667 | 0.0229 |

Table S32: Reused-holdout paired comparisons and conditional uncertainty.

| CSB minus               | AUC difference | Estimable /5000 | Conditional 95% interval |
|-------------------------|----------------|-----------------|--------------------------|
| Shared + funded generic | -0.003221      | 5000            | [-0.010227, 0.002950]    |
| Shared + five generic   | -0.003194      | 5000            | [-0.010034, 0.002680]    |
| Shared baseline         | 0.003931       | 5000            | [-0.001138, 0.009401]    |

Sol has four negative outcomes and Mistral three positives, limiting the resolution of their model-specific AUCs. Terminal and invalid responses remain in the planned denominator. The recovered Sol records contain six terminal HTTP 524 responses across five issues; the DeepSeek records contain ten terminal HTTP 404 responses across nine issues and two retried HTTP 429 responses. Original quota attempts, these failures and unreported costs are preserved. Reanalysis of the saved responses incurred no further model-query cost.

## Supplementary figures

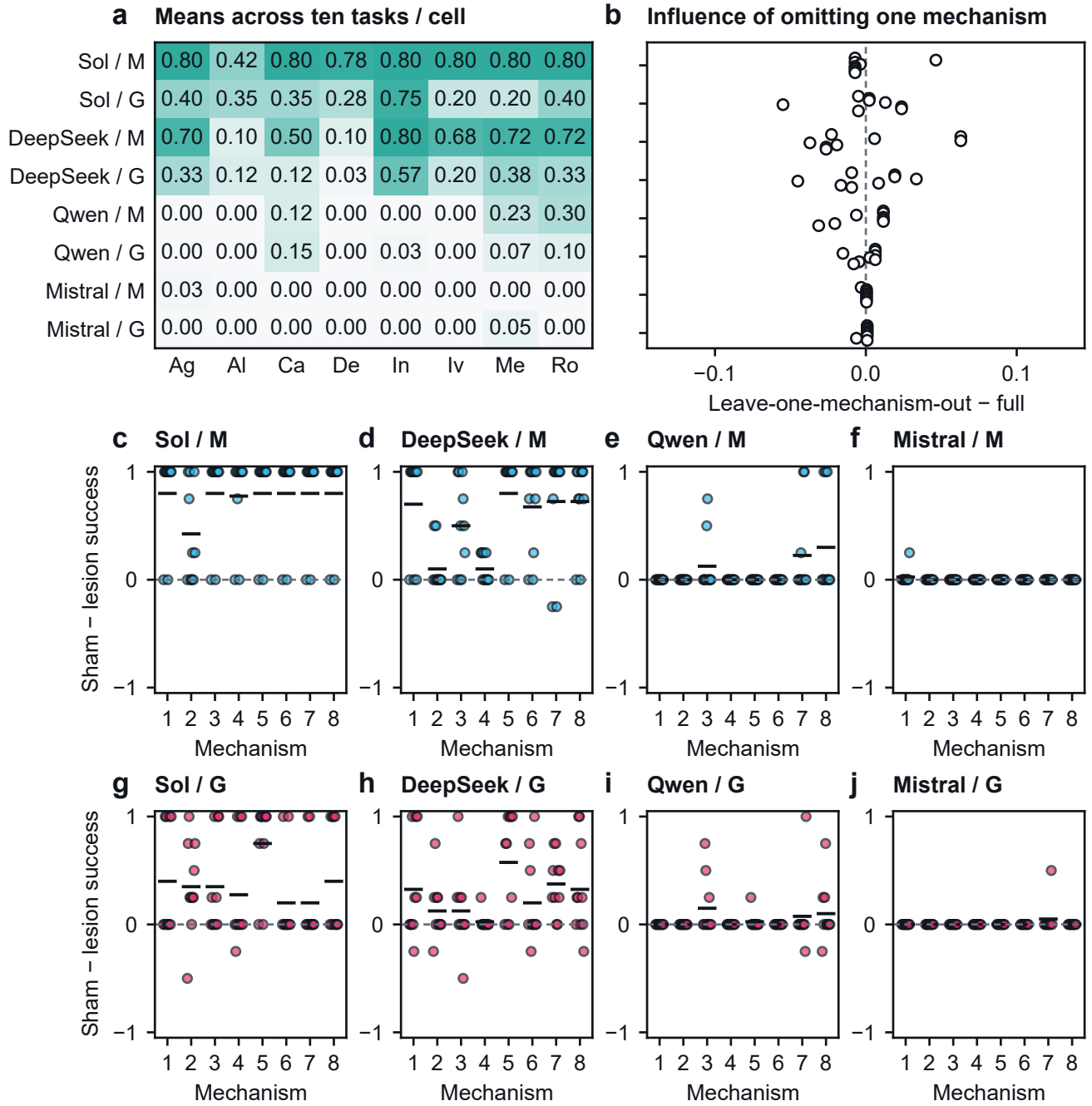


Figure S1: Mechanism heterogeneity and task variation. **a**, The 64 model/component-by-mechanism means, fixed diverging scale  $[-1, 1]$ . **b**, Each leave-one-mechanism-out estimate minus its corresponding full estimate, with the same row order as **a**; zero is no change, not a significance boundary. **c–j**, All ten task differences within each of eight mechanisms for each model/component cell, using four continuations per branch. Horizontal jitter is deterministic by task ID and is not uncertainty; black horizontal marks show each mechanism mean, with no connecting trend line between categorical mechanisms. Mechanism IDs 1–8 (Ag, Al, Ca, De, In, Iv, Me, Ro in **a**) denote aggregation, allocation, calendar, dependencies, inventory, invoice, merge and routing. The two rows are memory and goal; all 640 task differences are retained. Mechanism-bootstrap intervals are reported in Table S20 and the numerical results.

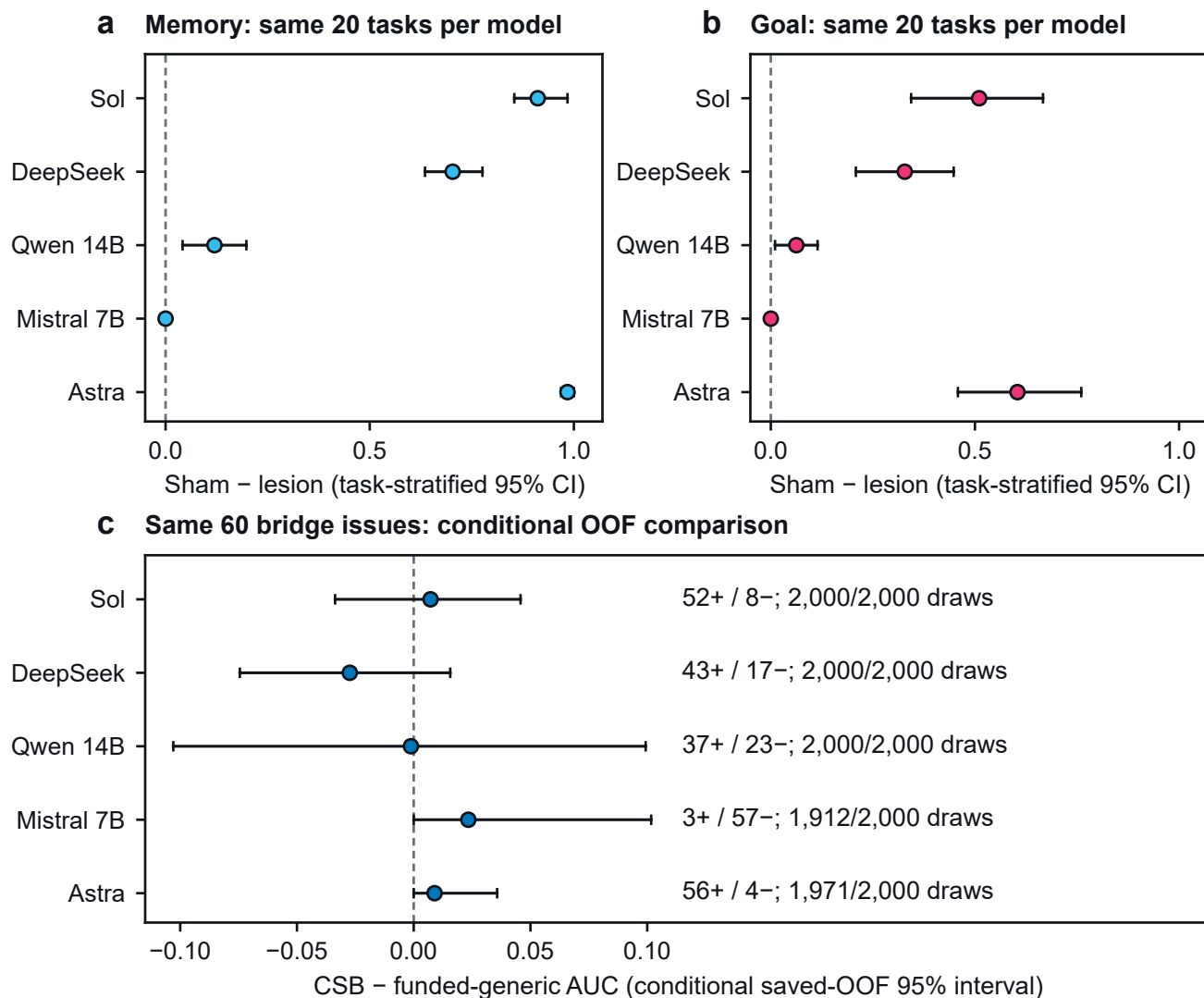


Figure S2: Fixed common subsets with Astra. **a,b**, All five models on the same 40 execution tasks, 20 per component, with equal-mechanism effects and task-stratified 95% intervals. The subset's weighting can differ from a raw continuation average. **c**, CSB minus funded-generic AUC on the same 60 bridge issues, with 95% intervals conditional on the out-of-fold (OOF) predictions. Printed positive/negative counts and estimable/2,000 draw counts retain single-class bootstrap failures. Intervals condition on the five-model subset fits and resample its issues; they do not include training variation.

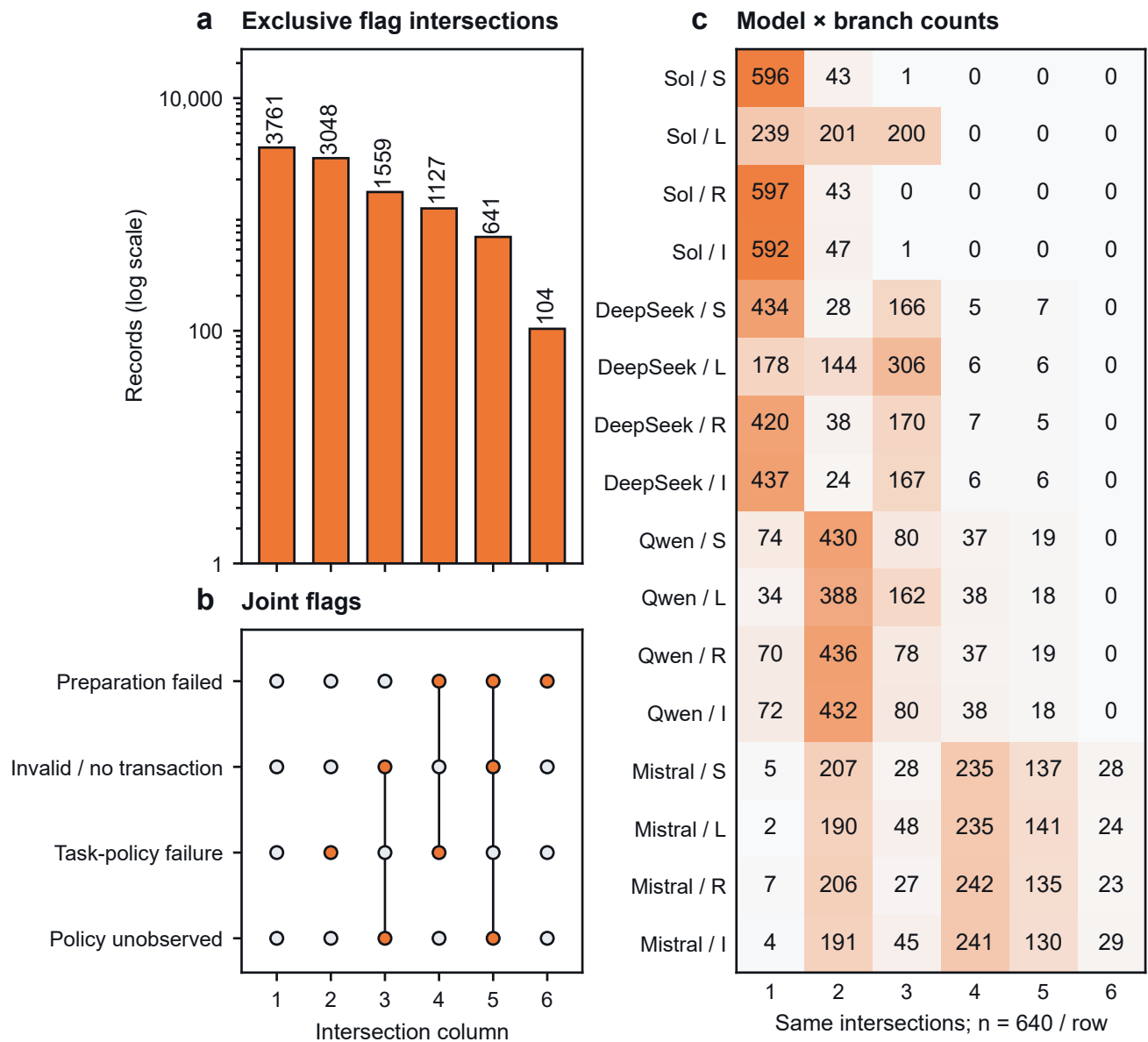


Figure S3: Failure flags without double counting. **a,b**, Mutually exclusive intersections of the three recorded failure flags plus an explicit unobserved-policy flag. Intersection 1 has no recorded failure flags. Log-count bars and incidence dots share the same intersection-column order. All 10,240 core continuations across sham, lesion, rescue and irrelevant branches are counted exactly once; 1,768 records have multiple failure flags and 2,200 lack an observed task-policy flag. **c**, The same intersections resolved by model and branch (S, sham; L, lesion; R, rescue; I, irrelevant), with exactly 640 records per row and a 0–640 sequential color scale. Unobserved policy outcomes form a separate category. The intersections describe the observed execution outcomes and response validity.

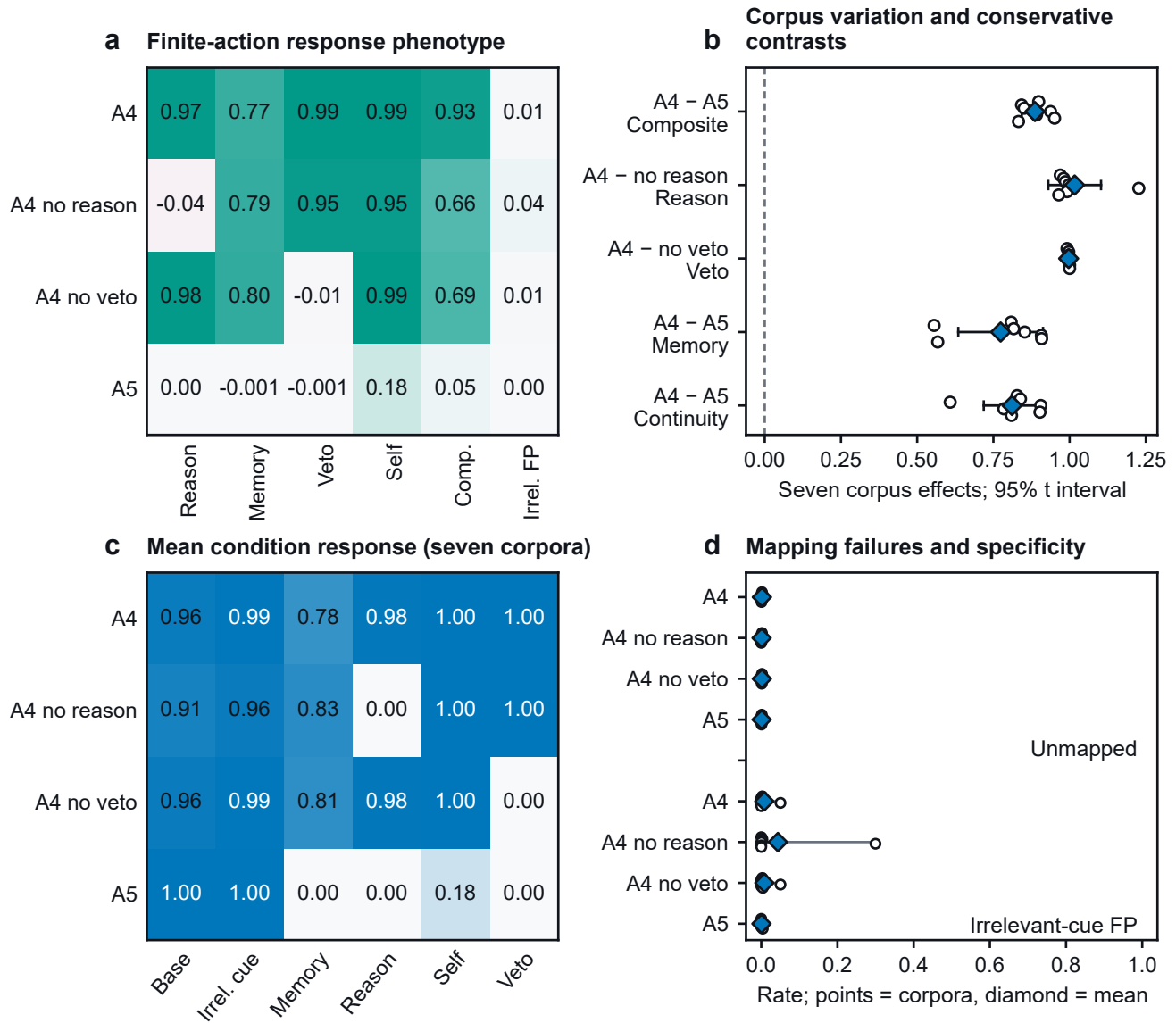


Figure S4: Explicit-rule finite-action responses. **a**, Mean behavioral response profiles, including reason, memory, veto, self, composite and irrelevant-cue false-positive measures; the symmetric diverging scale makes signed responses visible. **b**, Seven corpus effects (hollow points), means (diamonds) and descriptive two-sided 95% Student  $t$  intervals (6 degrees of freedom) for five registered contrasts. Table S45 separately retains the original percentile-bootstrap criterion. **c**, Condition-level mean response over seven corpora and three retained replicate indices; detailed corpus cells are shown in Figure S5. **d**, Unmapped-action and irrelevant-cue false-positive rates, with individual corpus observations, ranges and means. A4 denotes the structured condition, A5 the high-stochasticity comparator, and the two ablations remove reason or veto. Visible rules permit deterministic reconstruction of all 14,454 structured targets. The assay measures responses to supplied information across corpora with shared templates.

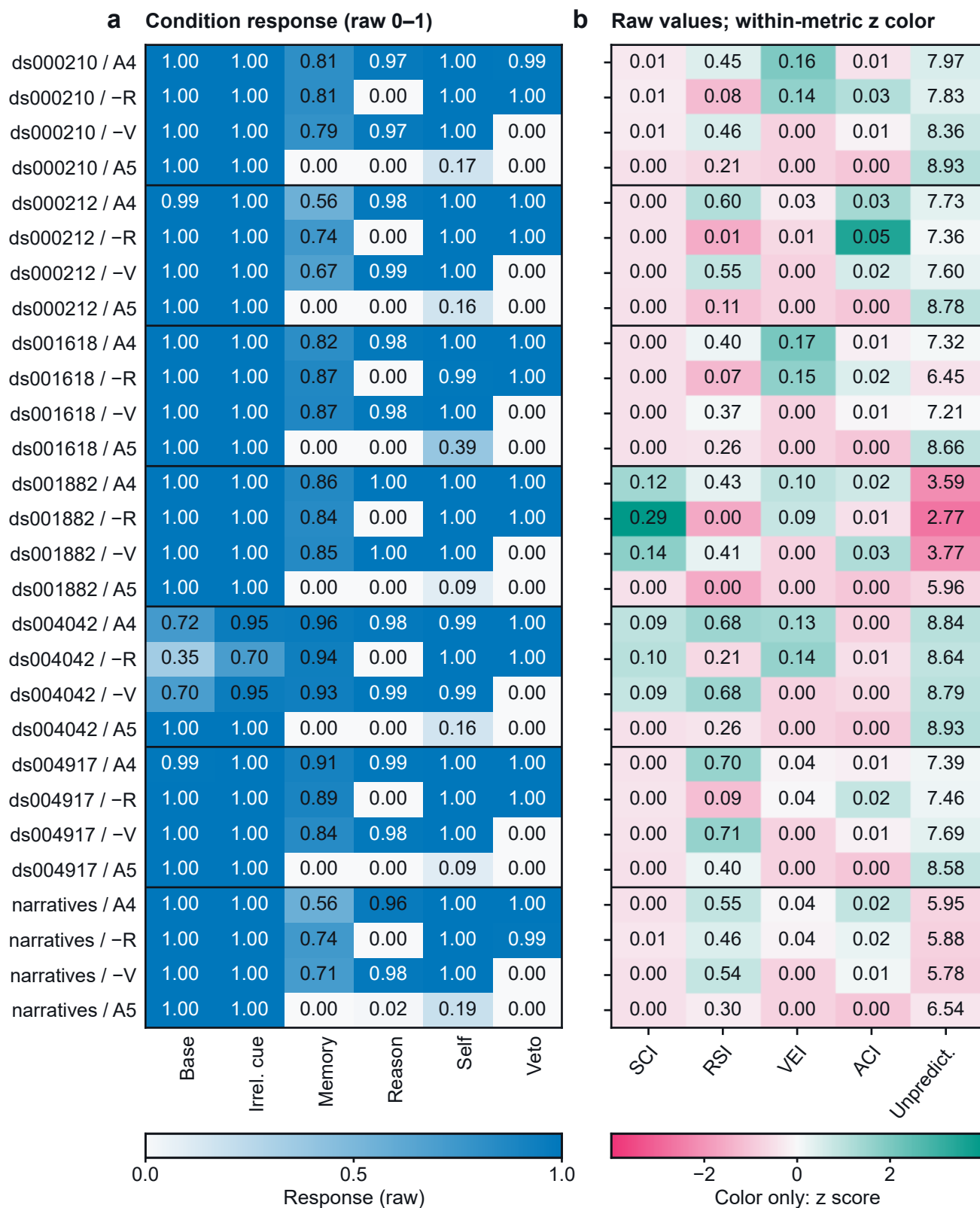


Figure S5: Corpus-resolved action and trace profiles. Cells average three replicate indices from separate finite-action (a) and structured-state trace (b) cohorts; matching corpus-variant names do not identify paired tasks. **a**, Raw condition responses (0–1). **b**, Raw metric values, colored by within-metric z scores across 28 corpus-variant means with symmetric limits. SCI, RSI, VEI and ACI denote self-continuity, reason-sensitivity, veto-effect and action-continuity indices. Rules separate corpora; -R/-V remove reason/veto. Source data retain 420 replicate metric values and 504 condition/replicate rows.

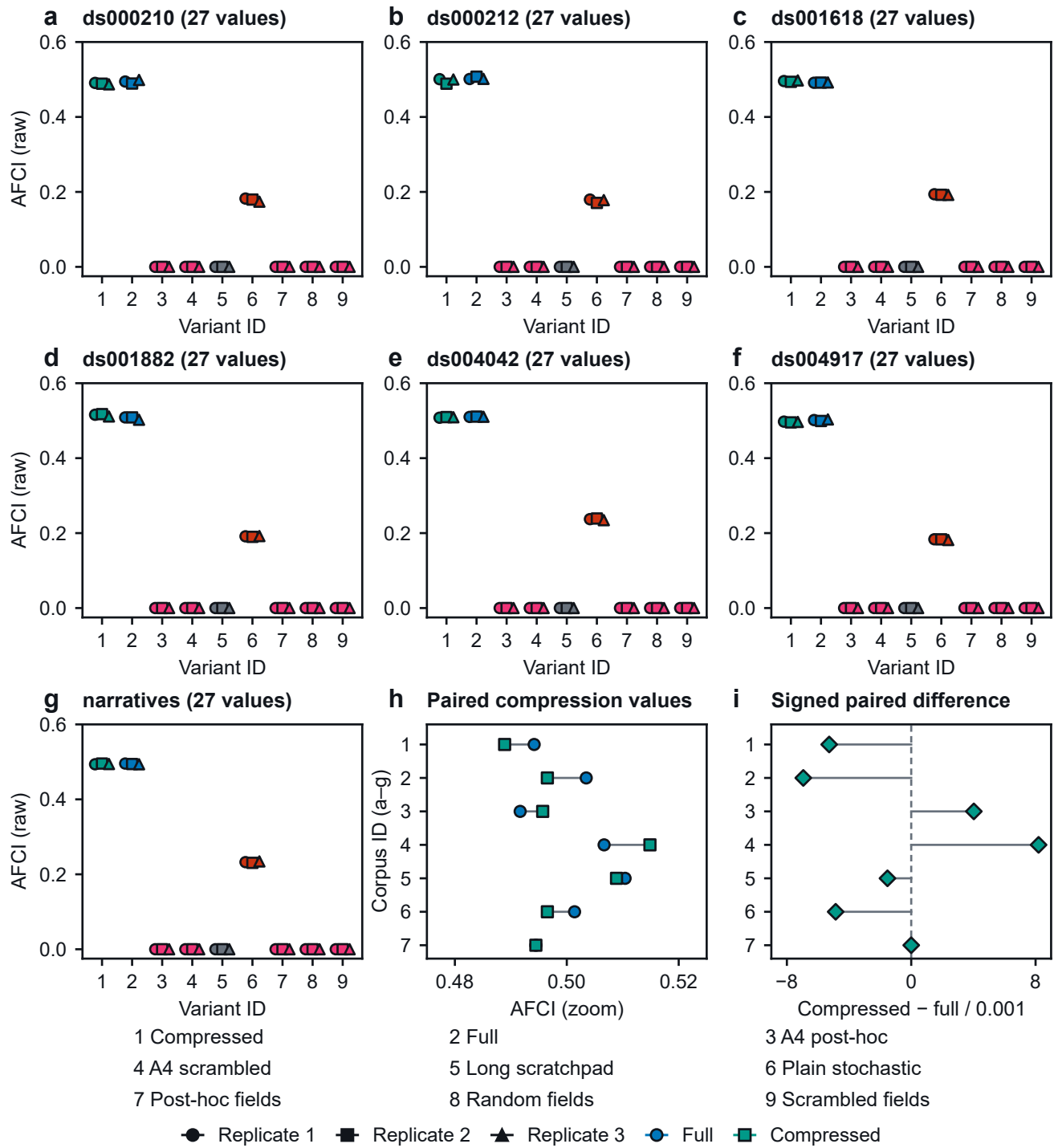


Figure S6: Action-field coupling by corpus and reporting condition. **a–g**, Seven corpus panels display all 189 original observations: nine variants times three retained replicate indices per corpus. The common displayed AFCI scale is 0–0.6 and contains all observations; the possible score range is 0–1; small horizontal offsets and circle/square/triangle shapes identify replicate indices 1/2/3, not uncertainty. The in-figure key maps variant IDs 1–9 to the original variants. Colors distinguish compressed, full, plain-stochastic, long-scratchpad and field-control families. AFCI uses both reported fields and their source information. **h**, Full and compressed structured-condition means paired within each corpus, with an explicitly magnified AFCI axis. **i**, Their signed compressed-minus-full differences in units of 0.001; **a–g** provide the full-scale context. Corpus IDs 1–7 in **h,i** follow panels **a–g**. Compression is a positive sensitivity case outside the predefined negative comparator set. Source data give both signed contrast directions; the points summarize the observed corpus means.

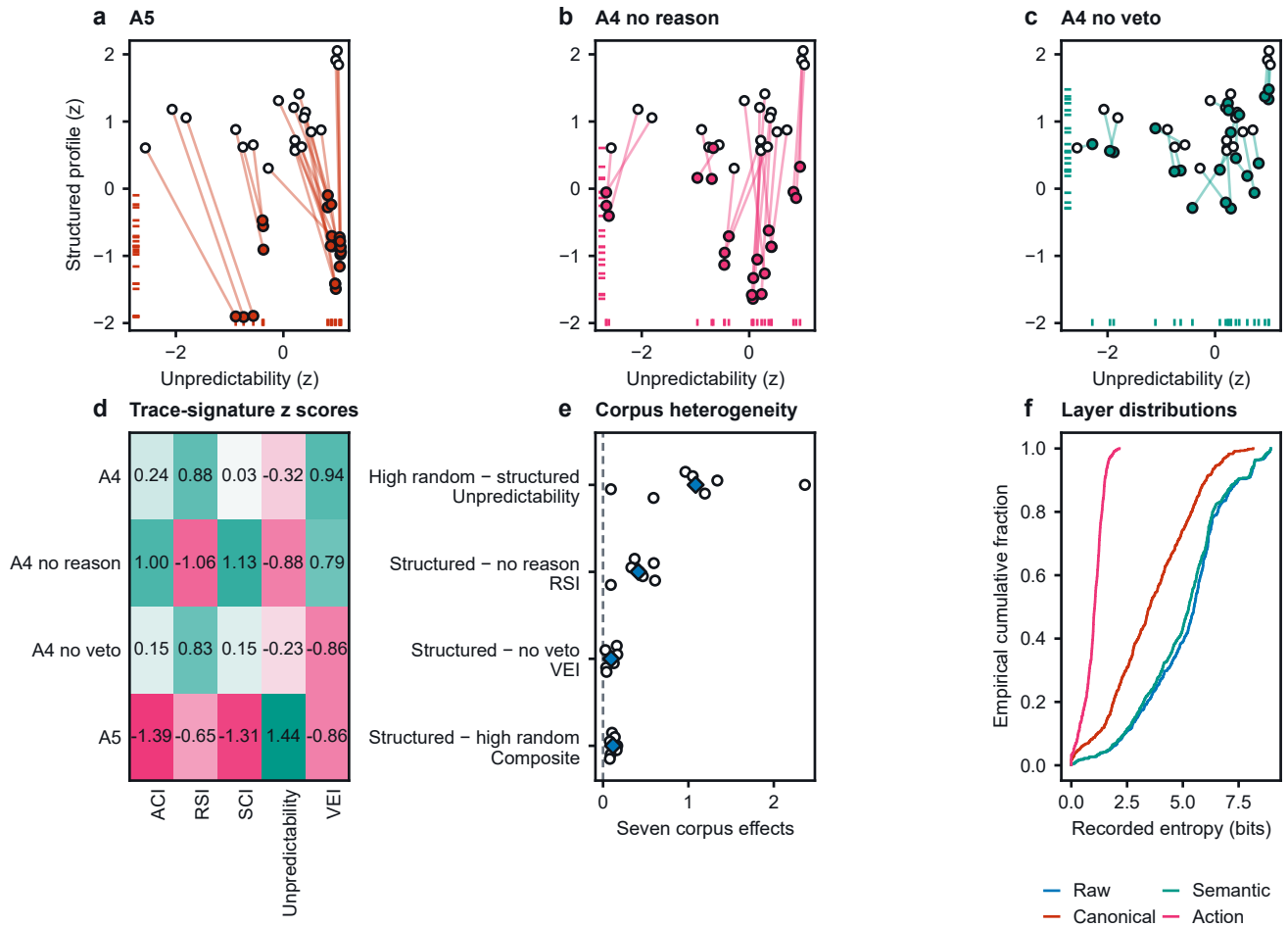


Figure S7: Paired trace geometry and entropy profiles. **a–c**, All 21 corpus–replicate displacements from A4 (open markers) to the indicated variant (colored markers), on identical standardized axes. Marginal rugs show the variant endpoints, and lines connect paired observations. **d**, Standardized trace signatures with the original metric names and exact values; symmetric color limits span the displayed data. **e**, Seven corpus effects for each indicated contrast (open points) with the corpus mean (diamond). These are descriptive contrasts. U denotes unpredictability and C the structured composite. **f**, Empirical distributions of recorded entropy by layer, retaining all rows. Entropy is in bits from the original base-2 implementation; these layers differ in representation and are not independent datasets. The standardized coordinates describe relationships among the recorded trace metrics.

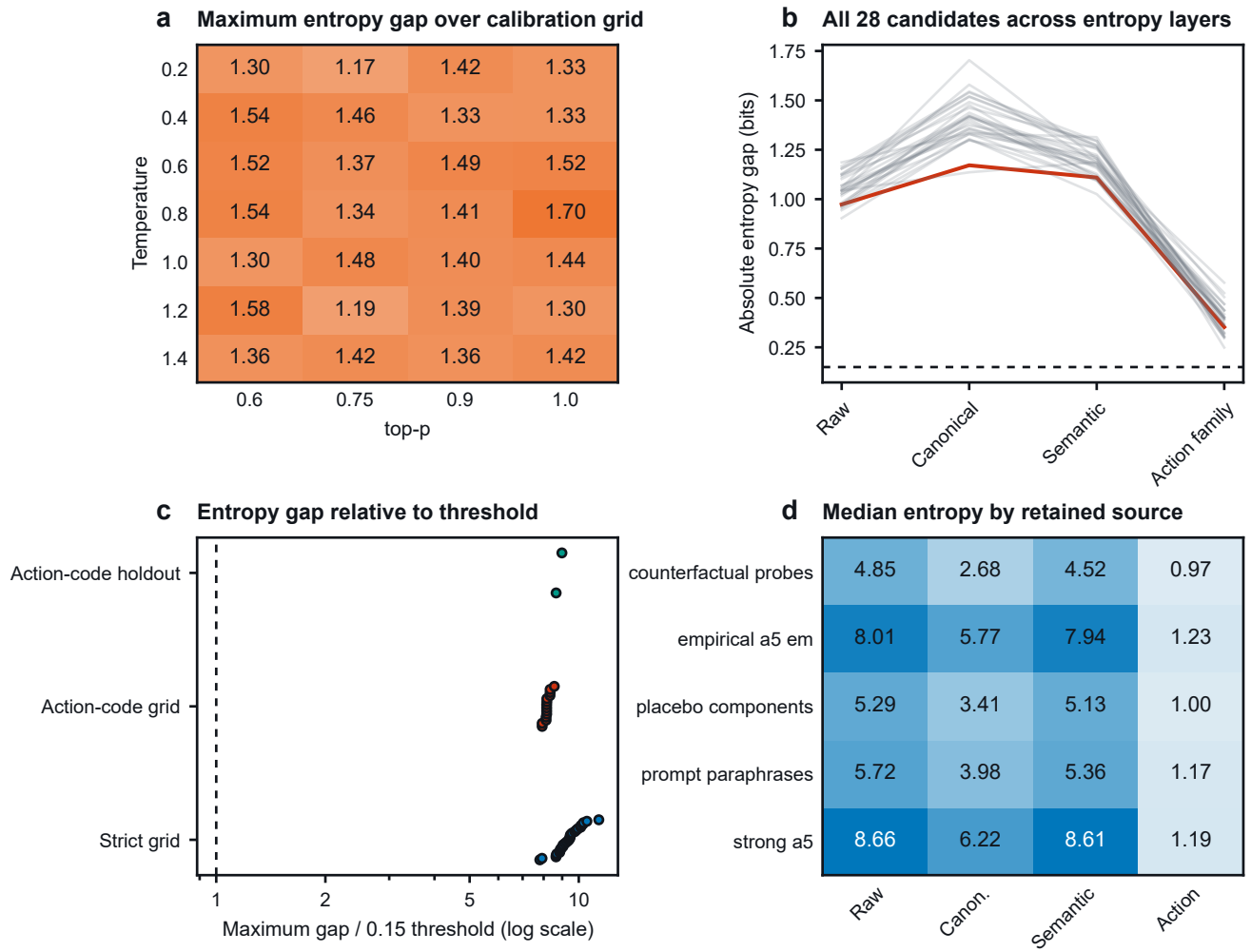


Figure S8: Entropy differences across the calibration search. **a**, Maximum entropy-layer gap for every cell of the 28-candidate strict temperature/top- $p$  grid (bits). **b**, The same candidates' four entropy gaps; the candidate with the smallest gap is red and the dashed line is the 0.15-bit threshold. **c**, All 44 retained candidates across calibration/holdout sources, shown by maximum-gap/threshold ratio on a logarithmic axis. Every ratio exceeds 1; source names identify the calibration and holdout samples; all displayed candidates belong to the documented search. **d**, Median entropy by source and layer (bits), with calibration and holdout sources shown separately. The minimum maximum gap is 1.171 bits, compared with the 0.15-bit threshold.

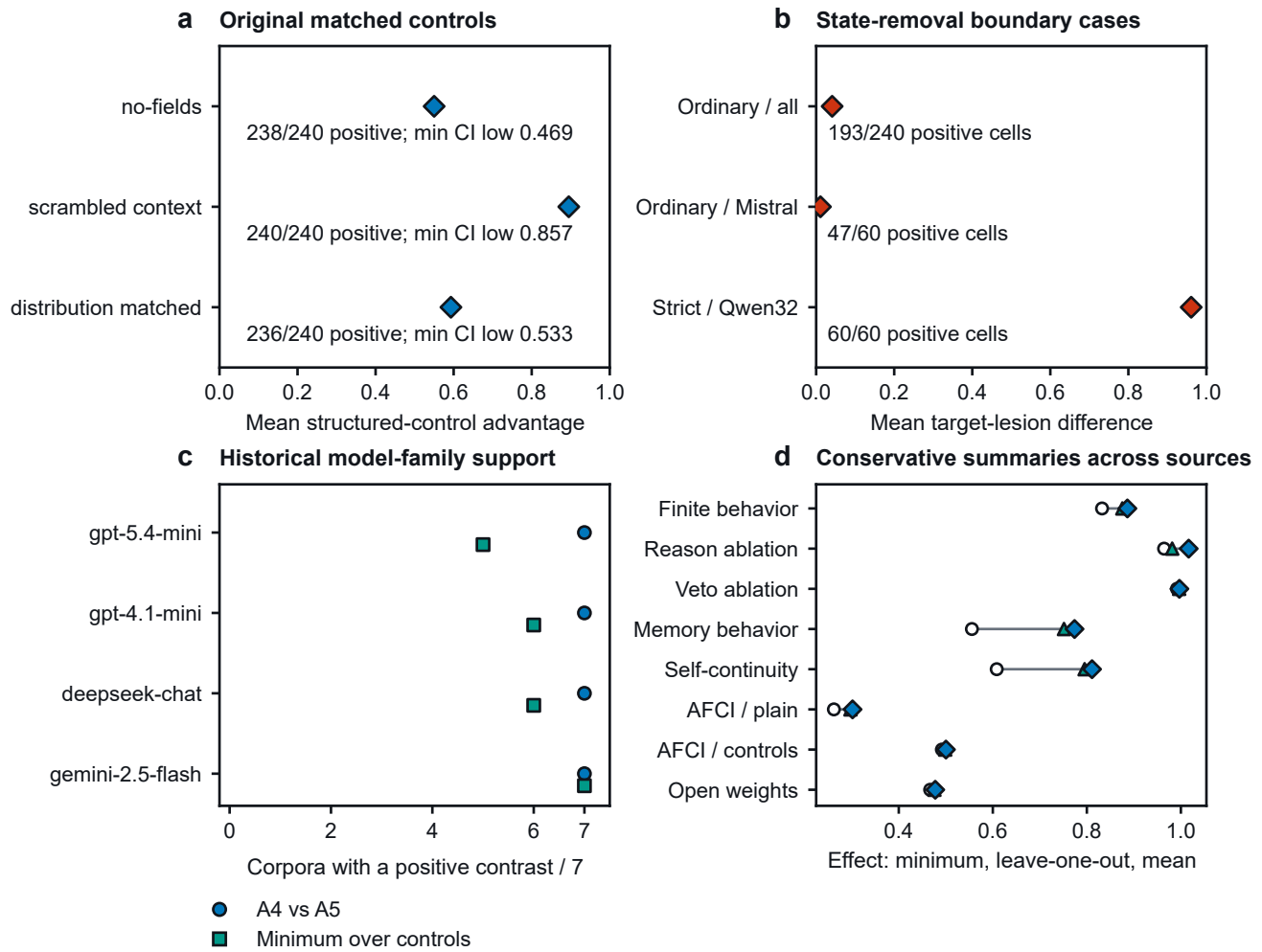


Figure S9: Open-weight control contrasts and corpus sensitivity. **a**, Three original matched controls in 240 model-scaffold-family cells, with mean accuracy differences, positive-cell counts and the minimum lower endpoint across scaffold, model, task-family and model-scaffold-family bootstrap clusters. These endpoints summarize lower limits, not complete intervals. **b**, Ordinary and stricter target-lesion contrasts with their original denominators. **c**, Provider identifiers and positive-corpus counts against A5 and the least favorable control; these are different configurations from the core model cohort. **d**, Source-specific minimum corpus effect (hollow circle), minimum leave-one-corpus-out mean (triangle) and mean (diamond). Each comparison retains its own analysis unit and score scale; values are summarized within their source study.

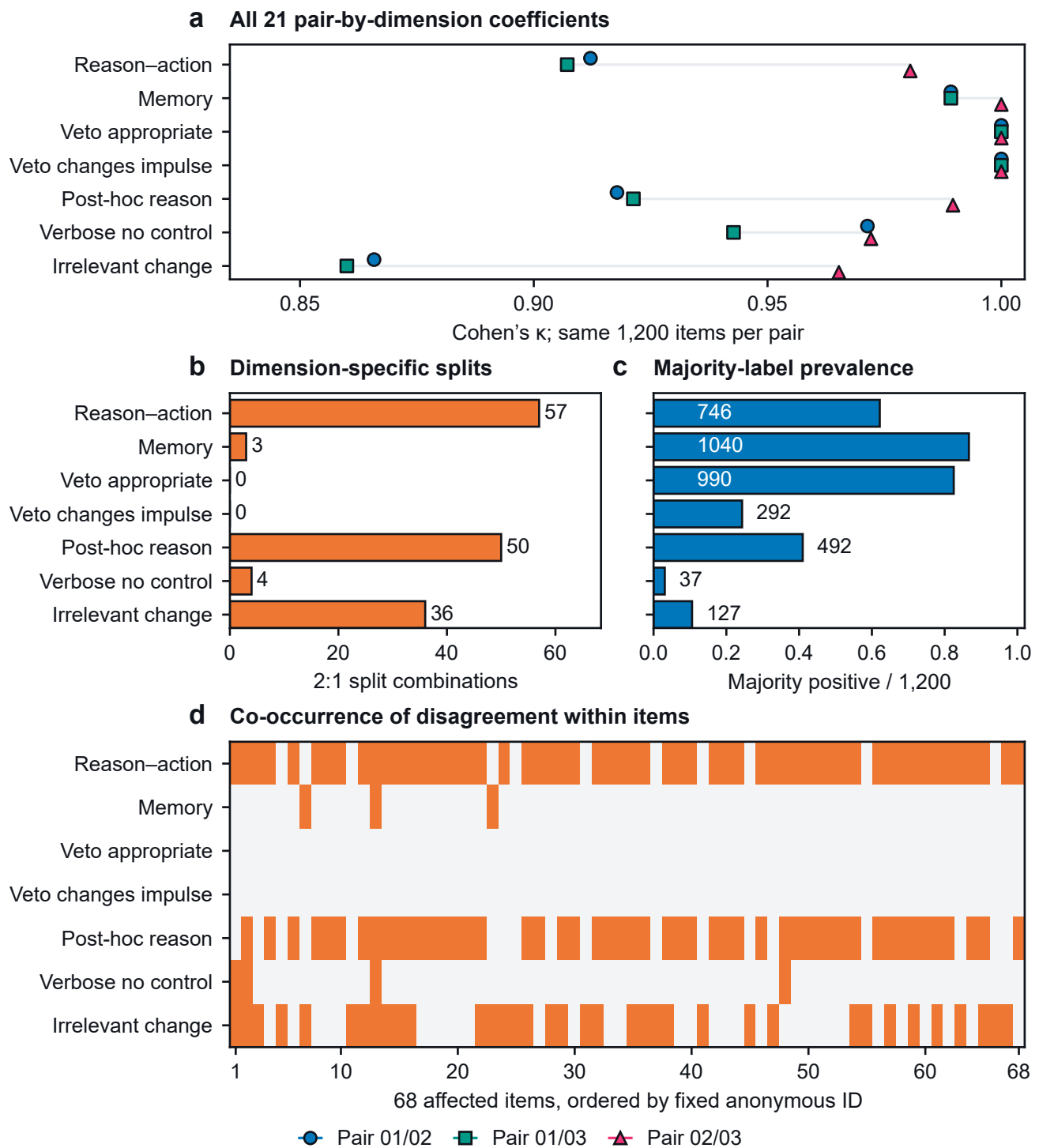


Figure S10: Annotation agreement, prevalence and item-level disagreements. **a**, All 21 pair-by-dimension Cohen's  $\kappa$  values, each based on 1,200 jointly valid binary ratings. Pairs 01/02, 01/03 and 02/03 identify the anonymous annotators; the light range segments are descriptive, not uncertainty intervals. **b**, Two-to-one split counts by dimension. **c**, Majority-positive prevalence, with actual positive counts printed and denominator 1,200. **d**, All 68 affected items in fixed anonymous-ID order; colored cells identify the 150 split combinations among 8,400 item-dimension combinations. The other 1,132 items have no disagreement and remain in the agreement and prevalence denominators. Dimensions differ in direction and are not combined into a quality score. The same annotators and items are reused across dimensions. Supplementary Note 5 specifies sampling, masking and independent scoring.

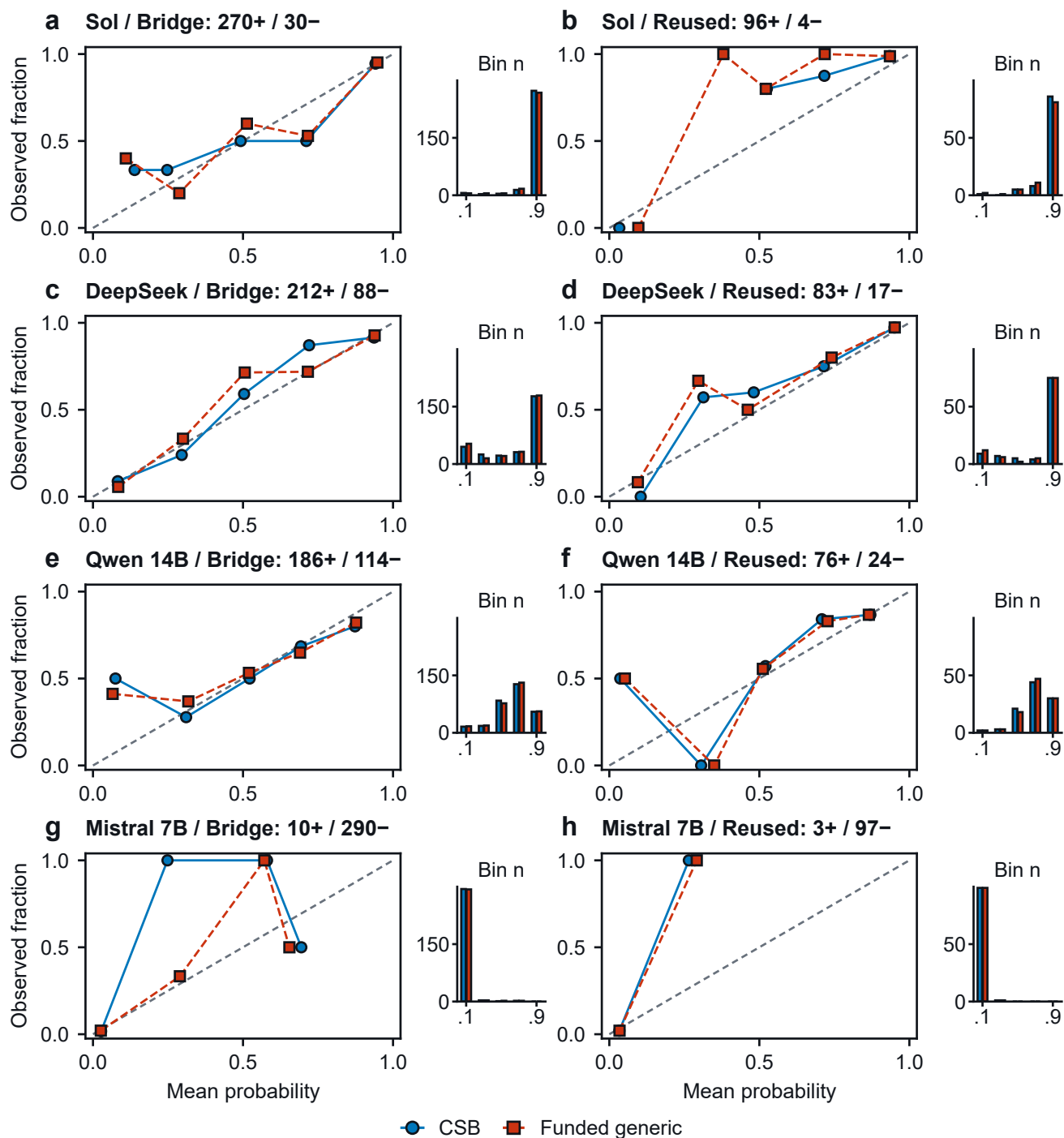


Figure S11: Calibration and support resolved by model. Rows are the four core models; columns are bridge and reused holdout. Blue circles are CSB and red squares funded generic. Calibration uses the same five fixed bins as Supplementary Figure S13; empty bins stay disconnected. The adjacent marginal bars show sample counts in all five bins, left to right from low to high probability, with separate bars per strategy and total denominators 300 or 100. Titles retain positive/negative counts, including Sol's four holdout negatives and Mistral's three positives. The source data include every bin, including empty bins. The plots are descriptive: no calibration curve was fitted, and model records sharing an issue remain statistically dependent.

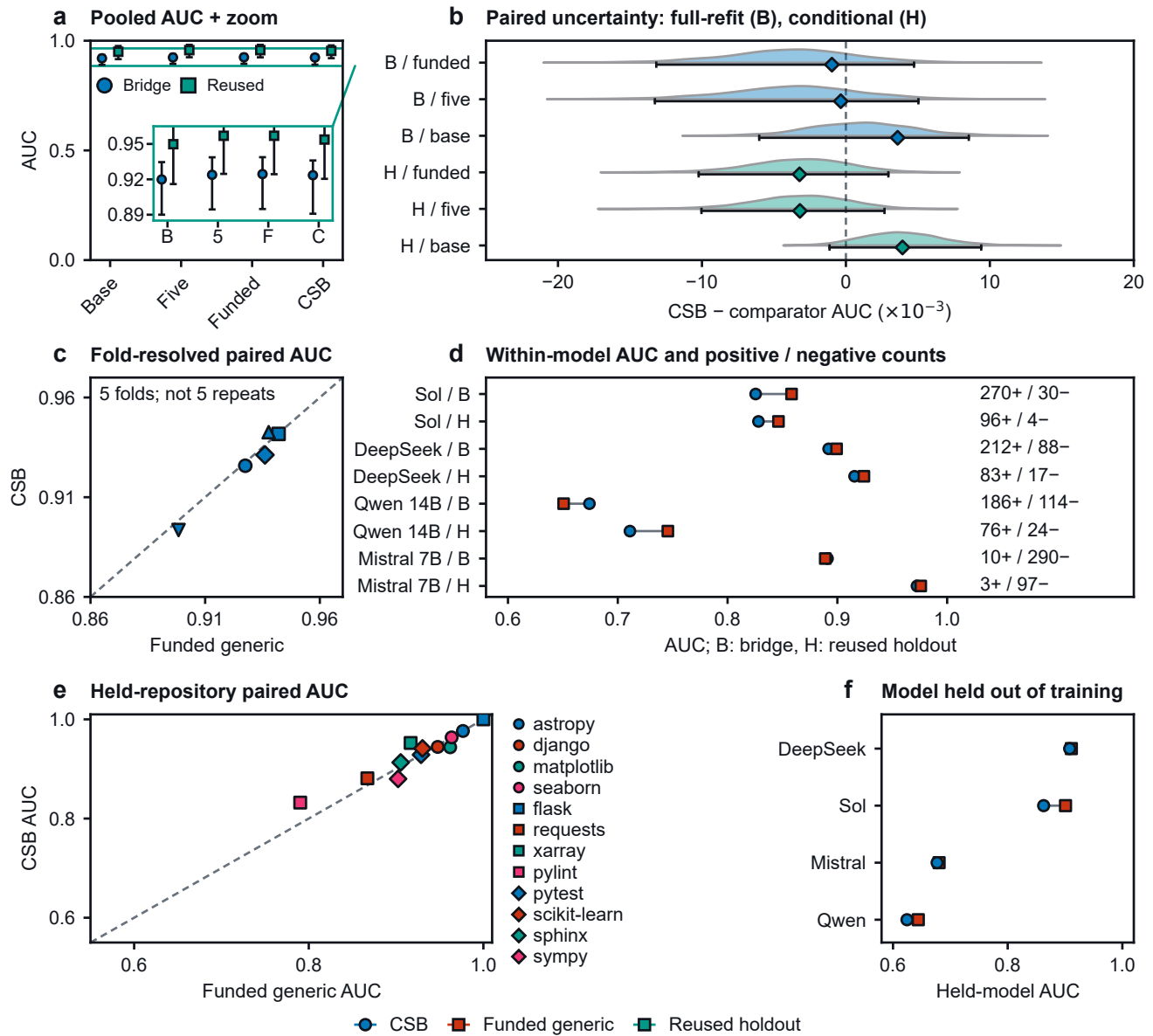


Figure S12: Predictive information from pooled performance to paired and grouped comparisons. **a**, Four strategies on a full AUC scale with an explicitly bounded magnification (B, baseline; 5, five generic probes; F, funded generic; C, CSB). Blue circles denote bridge and teal squares reused holdout in a–b. Intervals in a are marginal percentile intervals from grouped bootstrap draws. **b**, Paired CSB-minus-comparator differences: observed diamonds, two-sided 95% intervals and Scott-bandwidth half-densities of the same draws, with independently normalized heights. Bridge B uses 2,000 full-refit grouped draws and holdout H 5,000 conditional test-group draws. Axis units are 0.001 AUC. **c**, Five cross-validation (CV) fold AUC pairs; folds share training data and are not five independent replications. **d**, Model-resolved AUC and positive/negative counts for both stages. **e,f**, Repository- and model-held-out AUC pairs; in e marker size is constant and shape/color identify the held repository. Group counts are supplied in source data. Model exclusion retains other models on the same issues. In c–f blue denotes CSB and red funded generic, except the repository key in e. These are descriptive point estimates. The reused holdout is conditional on the model-cohort amendment.

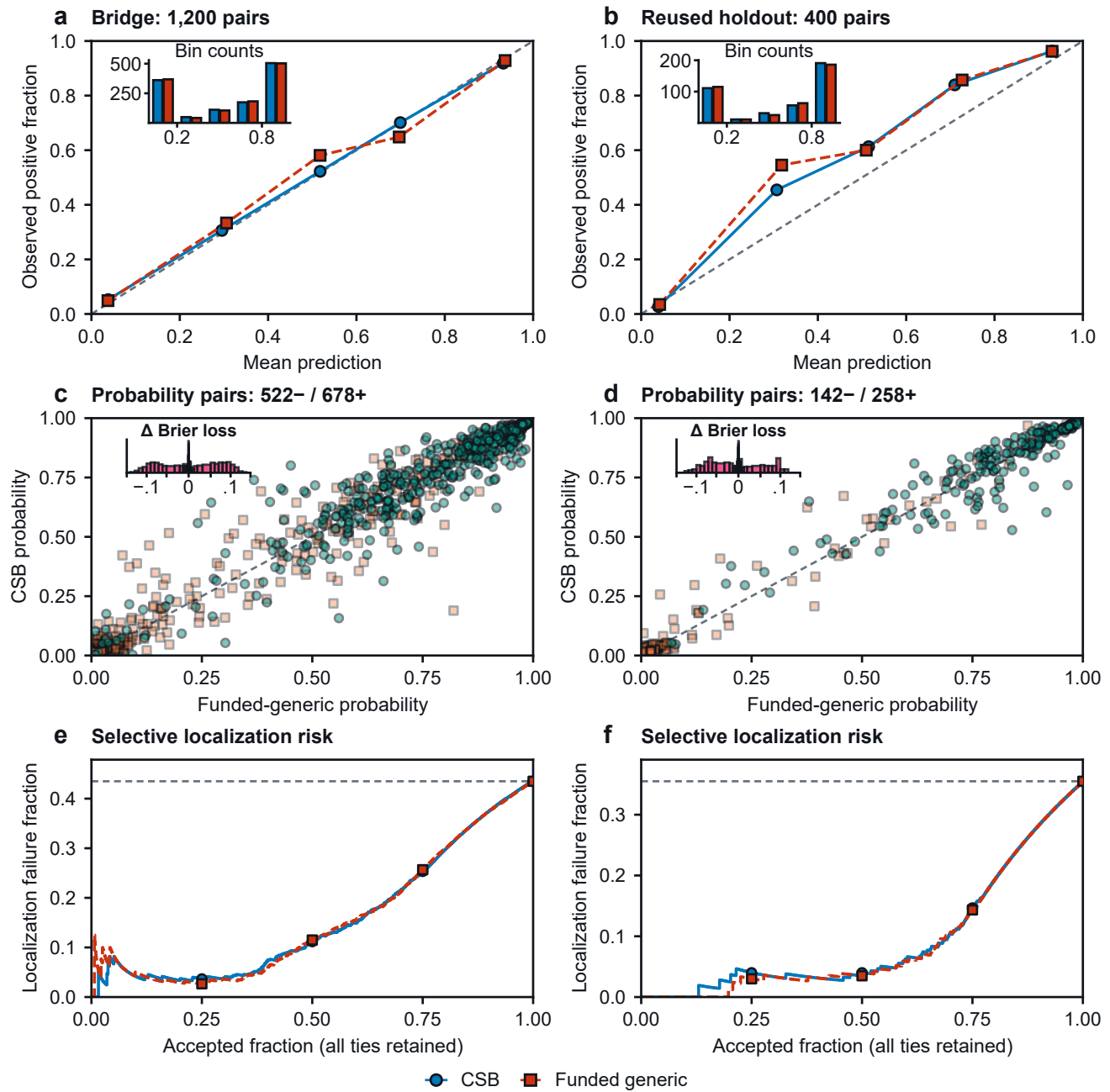


Figure S13: Probability calibration, paired prediction disagreement and selective risk. Columns are the 1,200 bridge model–issue pairs and 400 reused holdout pairs. **a,b**, Empirical positive fraction versus mean probability in five fixed bins,  $[0, 0.2), \dots, [0.8, 1]$ , with bin-count insets and ideal-calibration diagonals. **c,d**, All matched CSB and funded-generic probabilities, with scorer-side outcome classes shown as orange squares (negative) and teal circles (positive); titles report class counts. Insets show the complete distribution of paired Brier-loss differences, CSB minus funded generic, using signed geometric bins and a symmetric-log axis with linear threshold 0.01. Negative loss differences favor CSB. **e,f**, Descriptive localization failure fraction among predictions retained at every unique saved probability threshold; ties are retained together. Symbols locate the closest available 25/50/75/100% coverage values, and dashed horizontal lines mark all-pair failure fractions. The curves use fixed predictions without recalibration or confidence bands and were not used to select an operating threshold. Model records share issues. SI Figure S11 gives model-specific calibration and the observation counts supporting each bin.

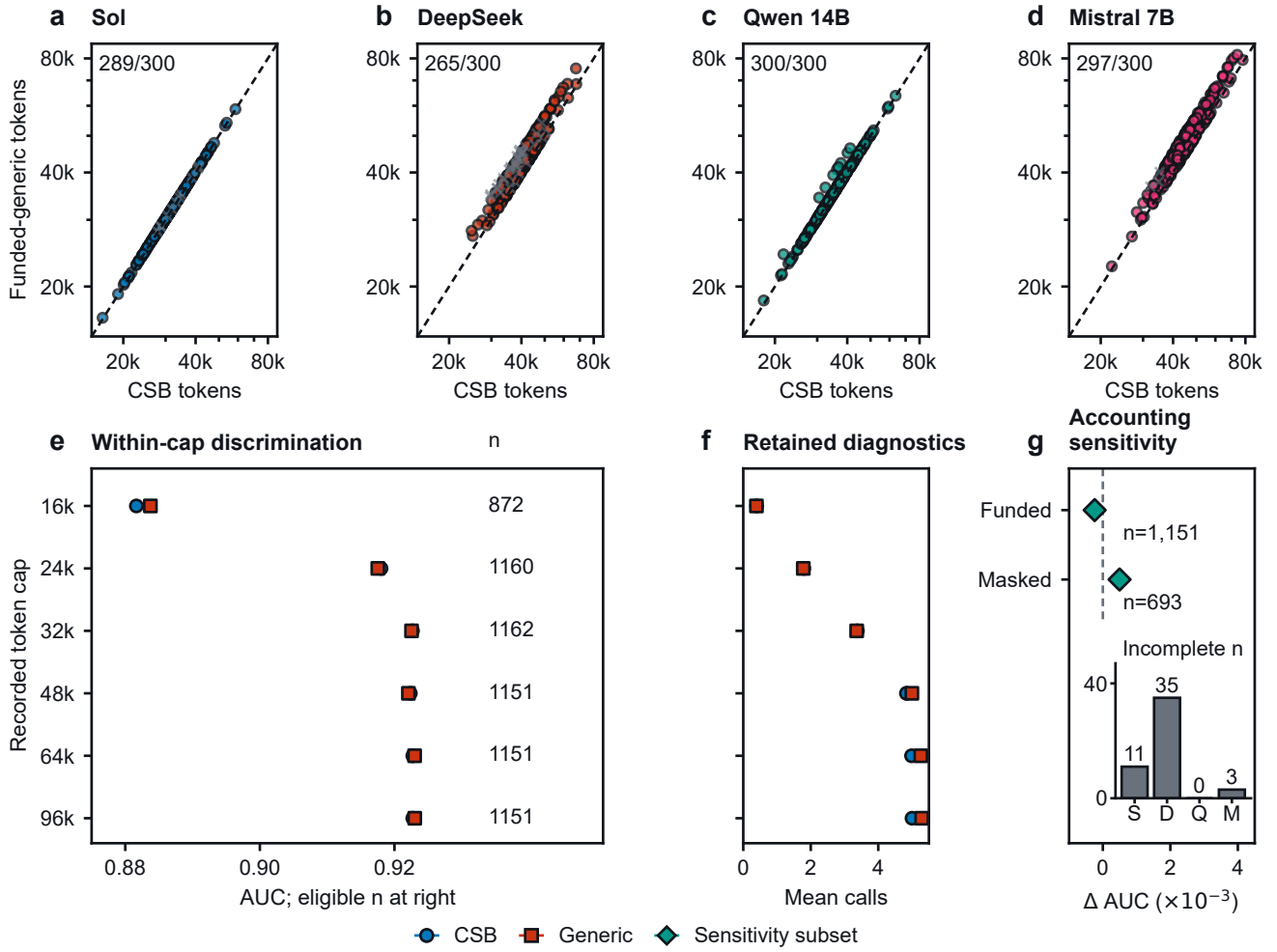


Figure S14: Reported resource use and budget comparisons. **a–d**, All 300 reported-token pairs per model on common log-log axes, with equality diagonals. Colored circles have complete recorded usage and funded-generic tokens at least equal to CSB; gray crosses retain other pairs. Printed fractions give complete-and-funded counts. All plotted pairs have positive reported totals; the flags identify pairs with incomplete metering. **e,f**, Within-cap retrospective AUC comparisons and actual mean retained diagnostic calls. Segments pair strategies within a cap only; the eligible population changes (872, 1,160, 1,162, then 1,151 pairs), so these are not common-cohort scaling curves. Actual overshoot calls remain paid and the specified missing-prefix rules apply. **g**, Selected-subset AUC differences for complete-and-funded accounting ( $n = 1,151$ ) and the conservative prior-interruption mask ( $n = 693$ ), shown descriptively without refitted intervals. Funded and Masked label the complete-and-funded and conservative prior-interruption subsets, respectively. The inset retains all 49 incomplete/unfunded pairs by model (S, Sol; D, DeepSeek; Q, Qwen; M, Mistral); these are accounting categories, not 49 model failures. Resource use is measured in provider-reported tokens. Unmetered usage and the upper bound of 42 potentially unlogged administrative attempts are accounted for separately in Supplementary Note 13.

Table S33: Variant protocol-identifier registry. Canonical labels are raw protocol identifiers retained for reproducibility; alias notes identify export names or descriptive names that appear in derived result exports.

| Canonical variant            | Used in                                                | Definition                                                                                                    | Export aliases or descriptive names                                   | Role                                                     |
|------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------|
| A4                           | baseline, behavior, strong controls                    | structured variant with branching, self-state, memory, reason graph and veto enabled                          | none                                                                  | primary structured variant                               |
| A5                           | baseline, behavior, strong controls                    | stochastic comparator with branching and high-temperature sampling while self/memory/reason/veto are disabled | none                                                                  | primary stochastic comparator                            |
| A4_no_reason                 | baseline, behavior, robustness                         | A4 with reason component removed                                                                              | none                                                                  | reason-ablation comparator                               |
| A4_no_veto                   | baseline, behavior, robustness                         | A4 with veto component removed                                                                                | none                                                                  | veto-ablation comparator                                 |
| A4_full                      | matched-interface AFCI                                 | full structured matched-interface variant                                                                     | none                                                                  | positive matched-interface variant verbosity sensitivity |
| A4_compressed                | matched-interface AFCI                                 | compressed positive sensitivity variant excluded from the predefined control-comparator set                   | none                                                                  | predefined control comparator                            |
| A5_plain                     | matched-interface AFCI                                 | plain stochastic matched-interface comparator                                                                 | none                                                                  | predefined control comparator                            |
| A5_schema_random_fields      | matched-interface AFCI                                 | A5 matched-interface fields generated as random or template fields                                            | random-field schema A5                                                | predefined control comparator                            |
| A5_schema_posthoc_fields     | matched-interface AFCI and open-weight reference model | A5 final action generated before post-hoc fields                                                              | A5_schema posthoc in post-hoc schema A5; open-weight runtime exports  | predefined control comparator                            |
| A5_schema_scrambled_fields   | matched-interface AFCI                                 | A5 fields scrambled from another event/action source                                                          | scrambled-field schema A5                                             | predefined control comparator                            |
| A4_posthoc_fields            | matched-interface AFCI                                 | A4 final action retained while fields are generated after final-action selection                              | A4 post-hoc fields                                                    | predefined control comparator                            |
| A4_scrambled_fields          | matched-interface AFCI                                 | A4 final action retained while fields are paired with wrong event/action information                          | A4 scrambled fields                                                   | predefined control comparator                            |
| A5_long_scratchpad           | matched-interface AFCI                                 | A5 stochastic comparator with extended scratchpad/field budget                                                | long-scratchpad A5                                                    | predefined control comparator                            |
| A5_entropy_attempt           | strong stochastic controls, entropy diagnostics        | lower-temperature stochastic attempt                                                                          | lower-temperature stochastic-attempt export alias                     | predefined entropy-closeness criterion unsatisfied       |
| A5_empirical_entropy_attempt | entropy-calibration diagnostic                         | selected temperature/top-p candidate from empirical calibration diagnostic                                    | calibration candidate aliases A5_attempt_t02_p07 A5_attempt_t020_p075 | predefined entropy-closeness criterion unsatisfied       |

Continued on next page

Table S33: Variant protocol-identifier registry. Canonical labels are raw protocol identifiers retained for reproducibility; alias notes identify export names or descriptive names that appear in derived result exports. (continued)

| Canonical variant      | Used in                    | Definition                                                                                                             | Export aliases or descriptive names | Role                         |
|------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------|
| A5_compute<br>matched  | strong stochastic controls | compute-control attempt under the implemented protocol; raw identifier does not indicate strict token/compute matching | none                                | strong stochastic comparator |
| A5_self<br>consistency | strong stochastic controls | self-consistency stochastic control under the implemented protocol                                                     | none                                | strong stochastic comparator |

Table S34: Mechanistic claim components, operational tests and contrary results.

| Table role    | Claim component                         | Operational test                                                                                                           | Contrary result                                                                                                             | Interpretive boundary                                                                                 |
|---------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| mechanism_map | stochastic contrast                     | The high-stochasticity comparator increases action unpredictability while disabling self, memory, reason and veto modules. | A strong stochastic control recovers structured-control metrics.                                                            | Operational stochastic-substitution test; no claim about consciousness, free will or neural homology. |
| mechanism_map | matched-interface action-field coupling | All controls use the same reason/memory/self/veto reporting format and are scored by AFCI.                                 | Post-hoc, scrambled, random-field or verbosity controls match the structured matched-interface AFCI.                        | Tests reporting-format availability under a shared interface.                                         |
| mechanism_map | finite-action-code behavior             | Scorer-side records define hidden expected labels and score only canonical final action codes.                             | The structured-control variant does not pass behavioral criteria or ablations preserve the full structured-control profile. | Fixed action ontology; does not test unconstrained natural-language decision making.                  |

## S1. Agent configurations and source corpora

The implemented family is defined by explicit module toggles and sampling settings. The table reports the operational variant definitions used by the lesion tests.

Table S35: Implemented agent-family variants and operational roles.

| Variant                      | Branch | Self-state | Memory | Reason graph | Veto | Temperature | Role                                                          |
|------------------------------|--------|------------|--------|--------------|------|-------------|---------------------------------------------------------------|
| A0                           | no     | no         | no     | no           | no   | 0.1         | minimal non-structured reference                              |
| A1                           | yes    | no         | no     | no           | no   | 0.2         | branching-only reference                                      |
| A2                           | yes    | yes        | no     | yes          | no   | 0.2         | self plus reason without memory/veto                          |
| A3                           | yes    | yes        | yes    | yes          | no   | 0.2         | memory-enabled structured variant without veto                |
| A4                           | yes    | yes        | yes    | yes          | yes  | 0.2         | primary structured variant                                    |
| A4_no_reason                 | yes    | yes        | yes    | no           | yes  | 0.2         | reason-graph lesion                                           |
| A4_no_veto                   | yes    | yes        | yes    | yes          | no   | 0.2         | veto lesion                                                   |
| A5                           | yes    | no         | no     | no           | no   | 0.9         | primary stochastic comparator                                 |
| A5_entropy attempt           | yes    | no         | no     | no           | no   | 0.2         | lower-temperature stochastic attempt                          |
| compute-control attempt      | yes    | no         | no     | no           | no   | 0.9         | compute-control attempt; raw identifier retained in Table S33 |
| A5_self consistency          | yes    | no         | no     | no           | no   | 0.9         | multi-sample stochastic control                               |
| A5_empirical entropy_attempt | yes    | no         | no     | no           | no   | 1           | empirical entropy-attempt comparator                          |

Table S36: Dataset namespaces used as prompt/event corpora.

| Dataset namespace  | Prompt-corpus role                  | Construct pressure             |
|--------------------|-------------------------------------|--------------------------------|
| narratives_formal  | narrative prompt corpus             | agency and self-narrative cues |
| ds000210_self_loop | self/prospection prompt corpus      | self-continuity                |
| ds001618_self_loop | autobiographical/self prompt corpus | self-memory loop               |
| ds001882_choice    | choice prompt corpus                | reason/value conflict          |
| ds000212_choice    | choice prompt corpus                | reason, veto and agency        |
| ds004917           |                                     |                                |
| uncertainty        | uncertainty/risk prompt corpus      | stochastic contrast            |
| ds004042_recall    | recall/memory prompt corpus         | memory constraint              |

## S2. Baseline intervention matrix

The baseline lesion matrix evaluates whether output dispersion and structured-control traces separate under the implemented structured-control/high-stochasticity lesion design. Raw protocol identifiers are retained in the tables for exact export matching. The criterion summary records the realized call count and dataset-direction outcomes.

Table S37: Baseline lesion-matrix criterion summary.

| Criterion              | Field                                                                | Value         |
|------------------------|----------------------------------------------------------------------|---------------|
| baseline lesion matrix | status                                                               | pass          |
| baseline lesion matrix | profile                                                              | full          |
| baseline lesion matrix | calls                                                                | 74352         |
| baseline lesion matrix | remote calls                                                         | 74336         |
| baseline lesion matrix | unrecovered generations                                              | 16            |
| baseline lesion matrix | unrecovered-generation rate                                          | 0.000215193   |
| baseline lesion matrix | model                                                                | gpt-5.4-mini  |
| baseline lesion matrix | fallback model                                                       | qwen3-30b-a3b |
| baseline lesion matrix | structured-control SCI ><br>high-stochasticity datasets              | 5             |
| baseline lesion matrix | high-stochasticity unpredictability ><br>structured-control datasets | 7             |
| baseline lesion matrix | reason-lesion drop datasets                                          | 7             |
| baseline lesion matrix | veto-lesion drop datasets                                            | 7             |

Table S38: Baseline trace metrics summarized by variant.

| Variant      | SCI mean   | RSI mean | VEI mean  | ACI mean    | Unpredictability mean | Distance to structured mean | Rows |
|--------------|------------|----------|-----------|-------------|-----------------------|-----------------------------|------|
| A0           | 0          | 0.14931  | 0         | 9.524e-05   | 8.073                 | 1.59321                     | 21   |
| A1           | 0          | 0.18819  | 0         | 0           | 8.00038               | 0.476597                    | 21   |
| A2           | 0.00423593 | 0.788509 | 9.524e-05 | 0           | 7.99426               | 0.320232                    | 21   |
| A3           | 0.0378864  | 0.537366 | 0         | 0.00956706  | 7.15695               | 0.172573                    | 21   |
| A4           | 0.0328108  | 0.544336 | 0.0944023 | 0.014518    | 6.9708                | 0                           | 21   |
| A4_no_reason | 0.0596636  | 0.131598 | 0.0868688 | 0.0212342   | 6.62649               | 0.474022                    | 21   |
| A4_no_veto   | 0.0355957  | 0.533555 | 0         | 0.0137792   | 7.0287                | 0.181687                    | 21   |
| A5           | 0          | 0.219036 | 0         | 0.000190476 | 8.05561               | 0.447575                    | 21   |

Table S39: Baseline trace metrics by dataset namespace, variant and replicate index.

| Dataset            | Replicate index | Variant      | SCI     | RSI   | VEI   | ACI        | Unpredictability | Distance to structured |
|--------------------|-----------------|--------------|---------|-------|-------|------------|------------------|------------------------|
| ds000210_self_loop | 1               | A0           | 0       | 0.328 | 0     | 0          | 8.76825          | 1.35526                |
| ds000210_self_loop | 1               | A1           | 0       | 0.144 | 0     | 0          | 8.89627          | 0.442842               |
| ds000210_self_loop | 1               | A2           | 0.001   | 0.648 | 0     | 0          | 8.94578          | 0.245967               |
| ds000210_self_loop | 1               | A3           | 0.0144  | 0.462 | 0     | 0.0106667  | 8.16901          | 0.162433               |
| ds000210_self_loop | 1               | A4           | 0.007   | 0.47  | 0.162 | 0.006      | 8.03157          | 0                      |
| ds000210_self_loop | 1               | A4_no_reason | 0.02    | 0.096 | 0.166 | 0.0193333  | 7.77433          | 0.393241               |
| ds000210_self_loop | 1               | A4_no_veto   | 0.0114  | 0.412 | 0     | 0.00933333 | 8.38953          | 0.212166               |
| ds000210_self_loop | 1               | A5           | 0       | 0.218 | 0     | 0          | 8.93627          | 0.315964               |
| ds000210_self_loop | 2               | A0           | 0       | 0.344 | 0     | 0          | 8.76751          | 1.21486                |
| ds000210_self_loop | 2               | A1           | 0       | 0.14  | 0     | 0          | 8.87448          | 0.346922               |
| ds000210_self_loop | 2               | A2           | 0.001   | 0.66  | 0     | 0          | 8.94178          | 0.356704               |
| ds000210_self_loop | 2               | A3           | 0.014   | 0.466 | 0     | 0.006      | 8.51228          | 0.249926               |
| ds000210_self_loop | 2               | A4           | 0.015   | 0.416 | 0.144 | 0.0106667  | 7.53976          | 0                      |
| ds000210_self_loop | 2               | A4_no_reason | 0.014   | 0.064 | 0.142 | 0.024      | 7.86176          | 0.367291               |
| ds000210_self_loop | 2               | A4_no_veto   | 0.01    | 0.47  | 0     | 0.0113333  | 8.16759          | 0.213498               |
| ds000210_self_loop | 2               | A5           | 0       | 0.226 | 0     | 0          | 8.95378          | 0.292231               |
| ds000210_self_loop | 3               | A0           | 0       | 0.36  | 0     | 0          | 8.83731          | 1.32969                |
| ds000210_self_loop | 3               | A1           | 0       | 0.19  | 0     | 0          | 8.88506          | 0.397445               |
| ds000210_self_loop | 3               | A2           | 0.001   | 0.688 | 0     | 0          | 8.91476          | 0.330203               |
| ds000210_self_loop | 3               | A3           | 0.013   | 0.468 | 0     | 0.0113333  | 8.46181          | 0.179912               |
| ds000210_self_loop | 3               | A4           | 0.009   | 0.462 | 0.176 | 0.00466667 | 8.33933          | 0                      |
| ds000210_self_loop | 3               | A4_no_reason | 0.011   | 0.078 | 0.122 | 0.034      | 7.85439          | 0.408794               |
| ds000210_self_loop | 3               | A4_no_veto   | 0.016   | 0.508 | 0     | 0.0113333  | 8.52001          | 0.18609                |
| ds000210_self_loop | 3               | A5           | 0       | 0.176 | 0     | 0          | 8.91476          | 0.35543                |
| ds000212_choice    | 1               | A0           | 0       | 0.296 | 0     | 0          | 8.72665          | 1.15236                |
| ds000212_choice    | 1               | A1           | 0       | 0.102 | 0     | 0          | 8.7556           | 0.657512               |
| ds000212_choice    | 1               | A2           | 0       | 0.876 | 0     | 0          | 8.71947          | 0.242358               |
| ds000212_choice    | 1               | A3           | 0.0054  | 0.434 | 0     | 0.011      | 7.07074          | 0.245605               |
| ds000212_choice    | 1               | A4           | 0.00252 | 0.656 | 0.028 | 0.0266667  | 7.85523          | 0                      |
| ds000212_choice    | 1               | A4_no_reason | 0.00108 | 0.012 | 0.006 | 0.0606667  | 7.55452          | 0.680466               |
| ds000212_choice    | 1               | A4_no_veto   | 0.00396 | 0.526 | 0     | 0.0233333  | 7.81526          | 0.146769               |
| ds000212_choice    | 1               | A5           | 0       | 0.116 | 0     | 0          | 8.77971          | 0.541392               |
| ds000212_choice    | 2               | A0           | 0       | 0.246 | 0     | 0          | 8.7878           | 1.18602                |
| ds000212_choice    | 2               | A1           | 0       | 0.086 | 0     | 0          | 8.69956          | 0.724426               |
| ds000212_choice    | 2               | A2           | 0.00036 | 0.884 | 0     | 0          | 8.76596          | 0.313144               |
| ds000212_choice    | 2               | A3           | 0.00216 | 0.492 | 0     | 0.018      | 7.55397          | 0.172383               |

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Table S39: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

| Dataset            | Replicate index | Variant      | SCI     | RSI   | VEI   | ACI        | Unpredictability | Distance to structured |
|--------------------|-----------------|--------------|---------|-------|-------|------------|------------------|------------------------|
| ds000212_choice    | 2               | A4           | 0.00216 | 0.634 | 0.032 | 0.026      | 7.80838          | 0                      |
| ds000212_choice    | 2               | A4_no_reason | 0.0036  | 0.01  | 0.012 | 0.038      | 7.2857           | 0.671147               |
| ds000212_choice    | 2               | A4_no_veto   | 0.00252 | 0.498 | 0     | 0.0124     | 7.31826          | 0.166749               |
| ds000212_choice    | 2               | A5           | 0       | 0.098 | 0     | 0          | 8.79493          | 0.53777                |
| ds000212_choice    | 3               | A0           | 0       | 0.296 | 0     | 0          | 8.83555          | 1.0744                 |
| ds000212_choice    | 3               | A1           | 0       | 0.09  | 0     | 0          | 8.6902           | 0.591411               |
| ds000212_choice    | 3               | A2           | 0       | 0.868 | 0     | 0          | 8.75269          | 0.397617               |
| ds000212_choice    | 3               | A3           | 0.00144 | 0.63  | 0     | 0.0253333  | 7.79394          | 0.119062               |
| ds000212_choice    | 3               | A4           | 0.00288 | 0.52  | 0.024 | 0.0326667  | 7.53217          | 0                      |
| ds000212_choice    | 3               | A4_no_reason | 0.00144 | 0.004 | 0.006 | 0.0653333  | 7.25292          | 0.541525               |
| ds000212_choice    | 3               | A4_no_veto   | 0.0036  | 0.618 | 0     | 0.0213333  | 7.65724          | 0.168122               |
| ds000212_choice    | 3               | A5           | 0       | 0.116 | 0     | 0          | 8.77164          | 0.406437               |
| ds001618_self_loop | 1               | A0           | 0       | 0.112 | 0     | 0          | 8.47286          | 1.35064                |
| ds001618_self_loop | 1               | A1           | 0       | 0.19  | 0     | 0          | 8.61063          | 0.428248               |
| ds001618_self_loop | 1               | A2           | 0       | 0.72  | 0.002 | 0          | 8.77191          | 0.320151               |
| ds001618_self_loop | 1               | A3           | 0       | 0.362 | 0     | 0.005      | 7.81575          | 0.274838               |
| ds001618_self_loop | 1               | A4           | 0       | 0.444 | 0.154 | 0.0173333  | 7.52728          | 0                      |
| ds001618_self_loop | 1               | A4_no_reason | 0       | 0.064 | 0.204 | 0.0133333  | 6.53898          | 0.383927               |
| ds001618_self_loop | 1               | A4_no_veto   | 0       | 0.366 | 0     | 0.012      | 7.66101          | 0.317151               |
| ds001618_self_loop | 1               | A5           | 0       | 0.246 | 0     | 0          | 8.67665          | 0.271338               |
| ds001618_self_loop | 2               | A0           | 0       | 0.144 | 0     | 0          | 8.52178          | 1.36285                |
| ds001618_self_loop | 2               | A1           | 0       | 0.2   | 0     | 0          | 8.56596          | 0.487458               |
| ds001618_self_loop | 2               | A2           | 0       | 0.694 | 0     | 0          | 8.77412          | 0.33743                |
| ds001618_self_loop | 2               | A3           | 0       | 0.368 | 0     | 0.0085     | 7.72458          | 0.17737                |
| ds001618_self_loop | 2               | A4           | 0       | 0.444 | 0.14  | 0.00866667 | 7.73685          | 0                      |
| ds001618_self_loop | 2               | A4_no_reason | 0       | 0.076 | 0.122 | 0.026      | 6.41047          | 0.402855               |
| ds001618_self_loop | 2               | A4_no_veto   | 0       | 0.368 | 0     | 0.012      | 6.46868          | 0.166994               |
| ds001618_self_loop | 2               | A5           | 0       | 0.248 | 0     | 0          | 8.63753          | 0.298441               |
| ds001618_self_loop | 3               | A0           | 0       | 0.106 | 0     | 0          | 8.43722          | 1.43727                |
| ds001618_self_loop | 3               | A1           | 0       | 0.184 | 0     | 0          | 8.45131          | 0.580472               |
| ds001618_self_loop | 3               | A2           | 0       | 0.596 | 0     | 0          | 8.76532          | 0.413091               |
| ds001618_self_loop | 3               | A3           | 0       | 0.396 | 0     | 0.00333333 | 7.68315          | 0.226955               |
| ds001618_self_loop | 3               | A4           | 0       | 0.306 | 0.208 | 0.004      | 6.70393          | 0                      |
| ds001618_self_loop | 3               | A4_no_reason | 0       | 0.058 | 0.11  | 0.014      | 6.39787          | 0.334682               |
| ds001618_self_loop | 3               | A4_no_veto   | 0       | 0.388 | 0     | 0.0106667  | 7.50181          | 0.226884               |
| ds001618_self_loop | 3               | A5           | 0       | 0.282 | 0     | 0          | 8.66935          | 0.263545               |
| ds001882_choice    | 1               | A0           | 0       | 0.004 | 0     | 0          | 6.52926          | 1.94798                |
| ds001882_choice    | 1               | A1           | 0       | 0.004 | 0     | 0          | 5.99445          | 0.479681               |
| ds001882_choice    | 1               | A2           | 0.008   | 0.75  | 0     | 0          | 5.55912          | 0.361726               |
| ds001882_choice    | 1               | A3           | 0.162   | 0.396 | 0     | 0.008      | 3.81626          | 0.118239               |
| ds001882_choice    | 1               | A4           | 0.152   | 0.446 | 0.106 | 0.0193333  | 3.72608          | 0                      |
| ds001882_choice    | 1               | A4_no_reason | 0.27    | 0     | 0.08  | 0.002      | 2.82488          | 0.462926               |
| ds001882_choice    | 1               | A4_no_veto   | 0.12    | 0.424 | 0     | 0.03       | 4.03051          | 0.113551               |
| ds001882_choice    | 1               | A5           | 0       | 0     | 0     | 0          | 5.9395           | 0.998752               |
| ds001882_choice    | 2               | A0           | 0       | 0.002 | 0     | 0          | 6.40637          | 1.95812                |
| ds001882_choice    | 2               | A1           | 0       | 0.002 | 0     | 0          | 5.64805          | 0.476378               |
| ds001882_choice    | 2               | A2           | 0.007   | 0.768 | 0     | 0          | 5.77716          | 0.363858               |
| ds001882_choice    | 2               | A3           | 0.141   | 0.468 | 0     | 0.015      | 4.70427          | 0.125849               |
| ds001882_choice    | 2               | A4           | 0.104   | 0.448 | 0.118 | 0.024      | 4.15845          | 0                      |
| ds001882_choice    | 2               | A4_no_reason | 0.312   | 0.002 | 0.114 | 0.006      | 2.74187          | 0.492869               |
| ds001882_choice    | 2               | A4_no_veto   | 0.151   | 0.404 | 0     | 0.0233333  | 3.91573          | 0.13466                |
| ds001882_choice    | 2               | A5           | 0       | 0.004 | 0     | 0          | 6.24003          | 0.696129               |
| ds001882_choice    | 3               | A0           | 0       | 0     | 0     | 0          | 6.52098          | 1.92553                |
| ds001882_choice    | 3               | A1           | 0       | 0.006 | 0     | 0          | 5.86191          | 0.406223               |
| ds001882_choice    | 3               | A2           | 0.02    | 0.756 | 0     | 0          | 5.22555          | 0.402404               |

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Table S39: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

| Dataset         | Replicate index | Variant      | SCI       | RSI   | VEI   | ACI        | Unpredictability | Distance to structured |
|-----------------|-----------------|--------------|-----------|-------|-------|------------|------------------|------------------------|
| ds001882_choice | 3               | A3           | 0.145     | 0.45  | 0     | 0.012      | 3.74179          | 0.108793               |
| ds001882_choice | 3               | A4           | 0.109     | 0.384 | 0.078 | 0.018      | 2.89757          | 0                      |
| ds001882_choice | 3               | A4_no_reason | 0.294     | 0     | 0.082 | 0.0113333  | 2.74725          | 0.426429               |
| ds001882_choice | 3               | A4_no_veto   | 0.162     | 0.414 | 0     | 0.026      | 3.36946          | 0.0994636              |
| ds001882_choice | 3               | A5           | 0         | 0.002 | 0     | 0          | 5.69531          | 0.945267               |
| ds004042_recall | 1               | A0           | 0         | 0.03  | 0     | 0          | 8.91725          | 1.92048                |
| ds004042_recall | 1               | A1           | 0         | 0.192 | 0     | 0          | 8.93057          | 0.551628               |
| ds004042_recall | 1               | A2           | 0.012952  | 0.902 | 0     | 0          | 8.95378          | 0.277931               |
| ds004042_recall | 1               | A3           | 0.100937  | 0.65  | 0     | 0          | 8.3718           | 0.14564                |
| ds004042_recall | 1               | A4           | 0.103603  | 0.682 | 0.14  | 0.002      | 8.83922          | 0                      |
| ds004042_recall | 1               | A4_no_reason | 0.0956766 | 0.194 | 0.14  | 0.006      | 8.54148          | 0.496309               |
| ds004042_recall | 1               | A4_no_veto   | 0.0962956 | 0.66  | 0     | 0.002      | 8.82551          | 0.16193                |
| ds004042_recall | 1               | A5           | 0         | 0.244 | 0     | 0          | 8.94578          | 0.471786               |
| ds004042_recall | 2               | A0           | 0         | 0.026 | 0     | 0.002      | 8.90648          | 1.95812                |
| ds004042_recall | 2               | A1           | 0         | 0.218 | 0     | 0          | 8.95227          | 0.562195               |
| ds004042_recall | 2               | A2           | 0.0092848 | 0.87  | 0     | 0          | 8.95778          | 0.227454               |
| ds004042_recall | 2               | A3           | 0.091722  | 0.72  | 0     | 0          | 8.88925          | 0.131892               |
| ds004042_recall | 2               | A4           | 0.0748846 | 0.698 | 0.116 | 0.006      | 8.79395          | 0                      |
| ds004042_recall | 2               | A4_no_reason | 0.118549  | 0.232 | 0.17  | 0.00266667 | 8.74416          | 0.477901               |
| ds004042_recall | 2               | A4_no_veto   | 0.0741564 | 0.688 | 0     | 0.006      | 8.71487          | 0.121295               |
| ds004042_recall | 2               | A5           | 0         | 0.276 | 0     | 0.002      | 8.91306          | 0.446837               |
| ds004042_recall | 3               | A0           | 0         | 0.032 | 0     | 0          | 8.91048          | 1.941                  |
| ds004042_recall | 3               | A1           | 0         | 0.226 | 0     | 0          | 8.94027          | 0.492784               |
| ds004042_recall | 3               | A2           | 0.0276378 | 0.904 | 0     | 0          | 8.94427          | 0.287693               |
| ds004042_recall | 3               | A3           | 0.0847924 | 0.724 | 0     | 0.01       | 8.82833          | 0.144852               |
| ds004042_recall | 3               | A4           | 0.0965382 | 0.66  | 0.122 | 0          | 8.87975          | 0                      |
| ds004042_recall | 3               | A4_no_reason | 0.0786418 | 0.216 | 0.11  | 0.01       | 8.62266          | 0.458524               |
| ds004042_recall | 3               | A4_no_veto   | 0.0886568 | 0.698 | 0     | 0.006      | 8.83318          | 0.133679               |
| ds004042_recall | 3               | A5           | 0         | 0.262 | 0     | 0.002      | 8.92978          | 0.428004               |
| uncertainty     | 1               | A0           | 0         | 0.018 | 0     | 0          | 8.62299          | 2.0015                 |
| ds004917        | 1               | A1           | 0         | 0.338 | 0     | 0          | 8.63624          | 0.400878               |
| uncertainty     | 1               | A2           | 0.00036   | 0.912 | 0     | 0          | 8.43571          | 0.494227               |
| ds004917        | 1               | A3           | 0.00324   | 0.682 | 0     | 0.013      | 8.09108          | 0.161046               |
| uncertainty     | 1               | A4           | 0.00432   | 0.692 | 0.046 | 0.0113333  | 7.01934          | 0                      |
| ds004917        | 1               | A4_no_reason | 0.00288   | 0.076 | 0.048 | 0.0246667  | 7.65113          | 0.616474               |
| uncertainty     | 1               | A4_no_veto   | 0.00324   | 0.692 | 0     | 0.0088     | 7.91809          | 0.0461257              |
| ds004917        | 1               | A5           | 0         | 0.382 | 0     | 0          | 8.54235          | 0.455262               |
| uncertainty     | 2               | A0           | 0         | 0.012 | 0     | 0          | 8.51087          | 2.13843                |
| ds004917        | 2               | A1           | 0         | 0.336 | 0     | 0          | 8.40745          | 0.387633               |
| uncertainty     | 2               | A2           | 0.00036   | 0.912 | 0     | 0          | 8.47606          | 0.382314               |
| ds004917        | 2               | A3           | 0.00468   | 0.69  | 0     | 0.006      | 7.58568          | 0.0960863              |
| uncertainty     | 2               | A4           | 0.00432   | 0.712 | 0.048 | 0.0126667  | 7.65424          | 0                      |
| ds004917        | 2               | A4_no_reason | 0.00324   | 0.086 | 0.024 | 0.0226667  | 7.29729          | 0.667423               |
| uncertainty     | 2               | A4_no_veto   | 0.00252   | 0.73  | 0     | 0.012      | 7.56667          | 0.117761               |

Continued on next page

Table S39: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

| Dataset           | Replicate index | Variant      | SCI        | RSI      | VEI       | ACI       | Unpredictability | Distance to structured |
|-------------------|-----------------|--------------|------------|----------|-----------|-----------|------------------|------------------------|
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 2               | A5           | 0          | 0.392    | 0         | 0         | 8.65413          | 0.391514               |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A0           | 0          | 0.004    | 0         | 0         | 8.60747          | 2.05311                |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A1           | 0          | 0.304    | 0         | 0         | 8.61613          | 0.414245               |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A2           | 0          | 0.916    | 0         | 0         | 8.47654          | 0.331396               |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A3           | 0.00324    | 0.692    | 0         | 0.0173333 | 7.85269          | 0.0245462              |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A4           | 0.0018     | 0.704    | 0.018     | 0.006     | 7.49391          | 0                      |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A4_no_reason | 0.00828    | 0.118    | 0.054     | 0.02      | 7.4201           | 0.64978                |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A4_no_veto   | 0.00216    | 0.708    | 0         | 0.01      | 7.58817          | 0.151182               |
| ds004917          |                 |              |            |          |           |           |                  |                        |
| uncertainty       | 3               | A5           | 0          | 0.424    | 0         | 0         | 8.54861          | 0.447065               |
| narratives_formal | 1               | A0           | 0          | 0.265306 | 0         | 0         | 6.49726          | 1.28397                |
| narratives_formal | 1               | A1           | 0          | 0.346939 | 0         | 0         | 6.53308          | 0.387361               |
| narratives_formal | 1               | A2           | 0          | 0.72449  | 0         | 0         | 6.5943           | 0.221234               |
| narratives_formal | 1               | A3           | 0.00742347 | 0.530612 | 0         | 0.0102041 | 6.02814          | 0.220534               |
| narratives_formal | 1               | A4           | 0          | 0.530612 | 0.0510204 | 0.0178571 | 6.23594          | 0                      |
| narratives_formal | 1               | A4_no_reason | 0          | 0.418367 | 0.0510204 | 0.0153061 | 5.56579          | 0.33566                |
| narratives_formal | 1               | A4_no_veto   | 0          | 0.632653 | 0         | 0.0244898 | 5.32241          | 0.317218               |
| narratives_formal | 1               | A5           | 0          | 0.316327 | 0         | 0         | 6.54578          | 0.247869               |
| narratives_formal | 2               | A0           | 0          | 0.214286 | 0         | 0         | 6.46415          | 1.22598                |
| narratives_formal | 2               | A1           | 0          | 0.387755 | 0         | 0         | 6.48456          | 0.349778               |
| narratives_formal | 2               | A2           | 0          | 0.785714 | 0         | 0         | 6.5943           | 0.260554               |
| narratives_formal | 2               | A3           | 0          | 0.581633 | 0         | 0         | 5.92067          | 0.167048               |
| narratives_formal | 2               | A4           | 0          | 0.530612 | 0.0306122 | 0.0306122 | 5.91899          | 0                      |
| narratives_formal | 2               | A4_no_reason | 0.00707041 | 0.540816 | 0.0408163 | 0         | 6.05625          | 0.137085               |
| narratives_formal | 2               | A4_no_veto   | 0          | 0.510204 | 0         | 0         | 6.10536          | 0.269589               |
| narratives_formal | 2               | A5           | 0          | 0.336735 | 0         | 0         | 6.52537          | 0.218853               |
| narratives_formal | 3               | A0           | 0          | 0.295918 | 0         | 0         | 6.48456          | 1.63981                |
| narratives_formal | 3               | A1           | 0          | 0.265306 | 0         | 0         | 6.57389          | 0.443026               |
| narratives_formal | 3               | A2           | 0          | 0.72449  | 0         | 0         | 6.53308          | 0.157421               |
| narratives_formal | 3               | A3           | 0.00117857 | 0.622449 | 0         | 0.0102041 | 5.68077          | 0.371015               |
| narratives_formal | 3               | A4           | 0          | 0.591837 | 0.0408163 | 0.0204082 | 5.69492          | 0                      |
| narratives_formal | 3               | A4_no_reason | 0.0114776  | 0.418367 | 0.0204082 | 0.0306122 | 6.01247          | 0.549153               |
| narratives_formal | 3               | A4_no_veto   | 0          | 0.489796 | 0         | 0.0170068 | 5.91339          | 0.344546               |
| narratives_formal | 3               | A5           | 0          | 0.234694 | 0         | 0         | 6.55349          | 0.369185               |

### S3. Action-field coupling under a common reporting interface

Matched-interface controls required comparator variants to use the same reporting format and then scored action-field coupling. AFCI is an interface-level analysis that uses reported content, final action and predefined field-source information; the finite-action-code benchmark supplies the reporting-format-independent behavioral test. The separate output-only analyses in Supplementary Note 3 assess coverage and consistency using visible content. The compressed structured-control sensitivity variant (**A4\_compressed**) is reported as a positive sensitivity variant excluded from the seven predefined control comparators: **A5\_plain**, **A5\_schema\_random\_fields**, **A5\_schema\_posthoc\_fields**, **A5\_schema\_scrambled\_fields**, **A4\_posthoc\_fields**, **A4\_scrambled\_fields** and **A5\_long\_scratchpad**. The self-action AFCI term is sparse in the matched-interface trace scorer. The finite-action-code  $B_{\text{SCI}}$  score is defined separately over final action-code stability under the predefined self-continuity probes.

Table S40: Matched-interface control criterion summary.

| Criterion                  | Field                                | Value        |
|----------------------------|--------------------------------------|--------------|
| matched-interface controls | status                               | pass         |
| matched-interface controls | dataset_count                        | 7            |
| matched-interface controls | control_count                        | 7            |
| matched-interface controls | min_a4_afci_gt_control_dataset_count | 7            |
| matched-interface controls | cluster_bootstrap_afci_delta_low95   | 0.451911     |
| matched-interface controls | call_count                           | 26946        |
| matched-interface controls | remote calls                         | 26946        |
| matched-interface controls | unrecovered_count                    | 0            |
| matched-interface controls | unrecovered_rate                     | 0            |
| matched-interface controls | planned calls                        | 26946        |
| matched-interface controls | model                                | gpt-5.4-mini |

Table S41: Predefined control-comparator recovery status for matched-interface AFCI.

| Comparator variant         | Datasets | Structured AFCI > comparator datasets | Comparator recovered structured datasets |
|----------------------------|----------|---------------------------------------|------------------------------------------|
| A5_plain                   | 7        | 7                                     | 0                                        |
| A5_schema_random_fields    | 7        | 7                                     | 0                                        |
| A5_schema_posthoc_fields   | 7        | 7                                     | 0                                        |
| A5_schema_scrambled_fields | 7        | 7                                     | 0                                        |
| A4_posthoc_fields          | 7        | 7                                     | 0                                        |
| A4_scrambled_fields        | 7        | 7                                     | 0                                        |
| A5_long_scratchpad         | 7        | 7                                     | 0                                        |

Table S42: AFCI component means by matched-interface variant. A4\_compressed is a compressed positive sensitivity variant excluded from the seven predefined control comparators.

| Variant          | Reason alignment mean | Memory alignment mean | Veto alignment mean | Self alignment mean | Irrelevant suppression mean | AFCI mean | AFCI + suppression mean | Rows |
|------------------|-----------------------|-----------------------|---------------------|---------------------|-----------------------------|-----------|-------------------------|------|
| A4_compressed    | 0.491286              | 0.502222              | 1                   | 0.00412698          | 1                           | 0.499409  | 0.599527                | 21   |
| A4_full          | 0.494461              | 0.503968              | 1                   | 0.00285714          | 1                           | 0.500322  | 0.600257                | 21   |
| A4_posthoc       |                       |                       |                     |                     |                             |           |                         |      |
| fields           | 0                     | 0                     | 0                   | 0                   | 0                           | 0         | 0                       | 21   |
| A4_scrambled     |                       |                       |                     |                     |                             |           |                         |      |
| fields           | 0                     | 0                     | 0                   | 0                   | 0                           | 0         | 0                       | 21   |
| A5_long          |                       |                       |                     |                     |                             |           |                         |      |
| scratchpad       | 0                     | 0                     | 0                   | 0                   | 0                           | 0         | 0                       | 21   |
| A5_plain         | 0.069306              | 0.225474              | 0.5                 | 0                   | 0                           | 0.198695  | 0.158956                | 21   |
| A5_schema        |                       |                       |                     |                     |                             |           |                         |      |
| posthoc_fields   | 0                     | 0                     | 0                   | 0                   | 0                           | 0         | 0                       | 21   |
| A5_schema        |                       |                       |                     |                     |                             |           |                         |      |
| random_fields    | 0                     | 0                     | 0                   | 0                   | 0                           | 0         | 0                       | 21   |
| A5_schema        |                       |                       |                     |                     |                             |           |                         |      |
| scrambled_fields | 0                     | 0                     | 0                   | 0                   | 0                           | 0         | 0                       | 21   |

Table S43: Dataset-level AFCI contrasts against the structured matched-interface variant (A4\_full).

| Dataset            | Comparator variant | Comparator AFCI | Structured AFCI | Structured minus comparator |
|--------------------|--------------------|-----------------|-----------------|-----------------------------|
| ds000210_self_loop | A4_compressed      | 0.488889        | 0.494167        | 0.00527778                  |
| ds000210_self_loop | A4_posthoc         |                 |                 |                             |
| ds000210_self_loop | fields             | 0               | 0.494167        | 0.494167                    |
| ds000210_self_loop | A4_scrambled       |                 |                 |                             |
| ds000210_self_loop | fields             | 0               | 0.494167        | 0.494167                    |
| ds000210_self_loop | A5_long            |                 |                 |                             |
| ds000210_self_loop | scratchpad         | 0               | 0.494167        | 0.494167                    |
| ds000210_self_loop | A5_plain           | 0.178611        | 0.494167        | 0.315556                    |
| ds000210_self_loop | A5_schema          |                 |                 |                             |
| ds000210_self_loop | posthoc_fields     | 0               | 0.494167        | 0.494167                    |
| ds000210_self_loop | A5_schema          |                 |                 |                             |
| ds000210_self_loop | random_fields      | 0               | 0.494167        | 0.494167                    |
| ds000210_self_loop | A5_schema          |                 |                 |                             |
| ds000210_self_loop | scrambled_fields   | 0               | 0.494167        | 0.494167                    |
| ds000212_choice    | A4_compressed      | 0.496528        | 0.503472        | 0.00694444                  |
| ds000212_choice    | A4_posthoc         |                 |                 |                             |
| ds000212_choice    | fields             | 0               | 0.503472        | 0.503472                    |
| ds000212_choice    | A4_scrambled       |                 |                 |                             |
| ds000212_choice    | fields             | 0               | 0.503472        | 0.503472                    |
| ds000212_choice    | A5_long            |                 |                 |                             |
| ds000212_choice    | scratchpad         | 0               | 0.503472        | 0.503472                    |
| ds000212_choice    | A5_plain           | 0.176111        | 0.503472        | 0.327361                    |
| ds000212_choice    | A5_schema          |                 |                 |                             |
| ds000212_choice    | posthoc_fields     | 0               | 0.503472        | 0.503472                    |
| ds000212_choice    | A5_schema          |                 |                 |                             |
| ds000212_choice    | random_fields      | 0               | 0.503472        | 0.503472                    |
| ds000212_choice    | A5_schema          |                 |                 |                             |
| ds000212_choice    | scrambled_fields   | 0               | 0.503472        | 0.503472                    |
| ds001618_self_loop | A4_compressed      | 0.495694        | 0.491667        | -0.00402778                 |
| ds001618_self_loop | A4_posthoc         |                 |                 |                             |
| ds001618_self_loop | fields             | 0               | 0.491667        | 0.491667                    |
| ds001618_self_loop | A4_scrambled       |                 |                 |                             |
| ds001618_self_loop | fields             | 0               | 0.491667        | 0.491667                    |
| ds001618_self_loop | A5_long            |                 |                 |                             |
| ds001618_self_loop | scratchpad         | 0               | 0.491667        | 0.491667                    |
| ds001618_self_loop | A5_plain           | 0.192708        | 0.491667        | 0.298958                    |
| ds001618_self_loop | A5_schema          |                 |                 |                             |
| ds001618_self_loop | posthoc_fields     | 0               | 0.491667        | 0.491667                    |

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Table S43: Dataset-level AFCI contrasts against the structured matched-interface variant (A4\_full). (continued)

| Dataset            | Comparator variant | Comparator AFCI | Structured AFCI | Structured minus comparator |
|--------------------|--------------------|-----------------|-----------------|-----------------------------|
| ds001618_self_loop | A5_schema          | 0               | 0.491667        | 0.491667                    |
| ds001618_self_loop | random_fields      | 0               | 0.491667        | 0.491667                    |
| ds001882_choice    | A5_schema          | 0               | 0.491667        | 0.491667                    |
| ds001882_choice    | scrambled_fields   | 0.514861        | 0.506667        | -0.00819444                 |
| ds001882_choice    | A4_compressed      | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | A4_posthoc         | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | fields             | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | A4_scrambled       | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | fields             | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | A5_long            | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | scratchpad         | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | A5_plain           | 0.191181        | 0.506667        | 0.315486                    |
| ds001882_choice    | A5_schema          | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | posthoc_fields     | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | A5_schema          | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | random_fields      | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | A5_schema          | 0               | 0.506667        | 0.506667                    |
| ds001882_choice    | scrambled_fields   | 0               | 0.506667        | 0.506667                    |
| ds004042_recall    | A4_compressed      | 0.508889        | 0.510417        | 0.00152778                  |
| ds004042_recall    | A4_posthoc         | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | fields             | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | A4_scrambled       | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | fields             | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | A5_long            | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | scratchpad         | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | A5_plain           | 0.236944        | 0.510417        | 0.273472                    |
| ds004042_recall    | A5_schema          | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | posthoc_fields     | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | A5_schema          | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | random_fields      | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | A5_schema          | 0               | 0.510417        | 0.510417                    |
| ds004042_recall    | scrambled_fields   | 0               | 0.510417        | 0.510417                    |
| ds004917           | A4_compressed      | 0.496528        | 0.501389        | 0.00486111                  |
| uncertainty        | A4_posthoc         | 0               | 0.501389        | 0.501389                    |
| ds004917           | fields             | 0               | 0.501389        | 0.501389                    |
| uncertainty        | A4_scrambled       | 0               | 0.501389        | 0.501389                    |
| ds004917           | fields             | 0               | 0.501389        | 0.501389                    |
| uncertainty        | A5_long            | 0               | 0.501389        | 0.501389                    |
| ds004917           | scratchpad         | 0               | 0.501389        | 0.501389                    |
| uncertainty        | A5_plain           | 0.182847        | 0.501389        | 0.318542                    |
| ds004917           | A5_schema          | 0               | 0.501389        | 0.501389                    |
| uncertainty        | posthoc_fields     | 0               | 0.501389        | 0.501389                    |
| ds004917           | A5_schema          | 0               | 0.501389        | 0.501389                    |
| uncertainty        | random_fields      | 0               | 0.501389        | 0.501389                    |
| ds004917           | A5_schema          | 0               | 0.501389        | 0.501389                    |
| uncertainty        | scrambled_fields   | 0               | 0.501389        | 0.501389                    |
| narratives_formal  | A4_compressed      | 0.494473        | 0.494473        | 0                           |
| narratives_formal  | A4_posthoc         | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | fields             | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | A4_scrambled       | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | fields             | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | A5_long            | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | scratchpad         | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | A5_plain           | 0.232462        | 0.494473        | 0.262011                    |
| narratives_formal  | A5_schema          | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | posthoc_fields     | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | A5_schema          | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | random_fields      | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | A5_schema          | 0               | 0.494473        | 0.494473                    |
| narratives_formal  | scrambled_fields   | 0               | 0.494473        | 0.494473                    |

## S4. Finite-action responses

The finite-action-code benchmark scores canonical final-action behavior under the finite action alphabet. Expected labels are scorer-side rule annotations derived from the condition map; the model-visible prompt specifies the task context, manipulation, module-lesion rule and allowed action ontology but not the hidden expected label used for scoring.

Table S44: Finite-action-code ontology and scorer-side expected labels.

| Probe condition | Valid action codes            | Expected before | Expected after | Interpretation                                                                   |
|-----------------|-------------------------------|-----------------|----------------|----------------------------------------------------------------------------------|
| baseline        | ACTION_A; ACTION_B            | ACTION_A        | ACTION_A       | No specialized component is required; correct behavior is to keep ACTION_A.      |
| reason_flip     | ACTION_A; ACTION_B            | ACTION_A        | ACTION_B       | The decisive reason is moved from A to B; reason-enabled variants should change. |
| memory_conflict | ACTION_A; RECALL_PRIOR; DEFER | ACTION_A        | RECALL_PRIOR   | A prior commitment conflicts with the immediate action impulse.                  |
| veto_cue        | ACTION_A; VETO; DEFER         | ACTION_A        | VETO           | A hard stop or constraint cue invalidates the first impulse.                     |
| self_continuity | ACTION_A; ACTION_B; DEFER     | ACTION_A        | ACTION_A       | A persistent self-state or long-term commitment supports continuity.             |
| irrelevant_cue  | ACTION_A; ACTION_B            | ACTION_A        | ACTION_A       | An irrelevant cue should not change the action.                                  |

Table S45: Finite-action-code behavioral benchmark criterion summary.

| Criterion                               | Field                                                                  | Value      |
|-----------------------------------------|------------------------------------------------------------------------|------------|
| finite-action-code behavioral benchmark | status                                                                 | pass       |
| finite-action-code behavioral benchmark | dataset_count                                                          | 7          |
| finite-action-code behavioral benchmark | scored records                                                         | 57816      |
| finite-action-code behavioral benchmark | source_scored records                                                  | 61056      |
| finite-action-code behavioral benchmark | planned calls                                                          | 58896      |
| finite-action-code behavioral benchmark | remote calls                                                           | 58835      |
| finite-action-code behavioral benchmark | unrecovered generations                                                | 1          |
| finite-action-code behavioral benchmark | parse-error rate                                                       | 1.700e-05  |
| finite-action-code behavioral benchmark | unmapped action rate                                                   | 0.00129722 |
| finite-action-code behavioral benchmark | structured-control irrelevant false-positive rate                      | 0.0083022  |
| finite-action-code behavioral benchmark | structured-control behavioral composite > high-stochasticity datasets  | 7          |
| finite-action-code behavioral benchmark | structured-control B_RSI > reason-lesion datasets                      | 7          |
| finite-action-code behavioral benchmark | structured-control B_VEI > veto-lesion datasets                        | 7          |
| finite-action-code behavioral benchmark | structured-control B_MCI > high-stochasticity datasets                 | 7          |
| finite-action-code behavioral benchmark | structured-control B_SCI > high-stochasticity datasets                 | 7          |
| finite-action-code behavioral benchmark | dataset-cluster bootstrap lower endpoint of the two-sided 95% interval | 0.855732   |

Table S46: Behavioral component scores by variant; means and standard deviations.

| Variant      | Statistic | RSI    | MCI    | VEI    | SCI   | Cue   | Composite | FP    | n |
|--------------|-----------|--------|--------|--------|-------|-------|-----------|-------|---|
| A4           | Mean      | 0.973  | 0.773  | 0.989  | 0.990 | 0.931 | 0.931     | 0.008 | 7 |
| A4           | SD        | 0.023  | 0.151  | 0.019  | 0.021 | 0.037 | 0.037     | 0.019 | 7 |
| A4_no_reason | Mean      | -0.044 | 0.788  | 0.954  | 0.954 | 0.663 | 0.663     | 0.044 | 7 |
| A4_no_reason | SD        | 0.113  | 0.089  | 0.112  | 0.113 | 0.102 | 0.102     | 0.113 | 7 |
| A4_no_veto   | Mean      | 0.977  | 0.801  | -0.008 | 0.990 | 0.690 | 0.690     | 0.009 | 7 |
| A4_no_veto   | SD        | 0.019  | 0.081  | 0.019  | 0.021 | 0.020 | 0.020     | 0.019 | 7 |
| A5           | Mean      | 0.002  | -0.001 | -0.001 | 0.179 | 0.045 | 0.045     | 0.001 | 7 |
| A5           | SD        | 0.004  | 0.001  | 0.001  | 0.101 | 0.025 | 0.025     | 0.001 | 7 |

Table S47: Behavioral component scores by dataset namespace and variant.

| Dataset              | Variant      | B_RSI       | B_MCI    | B_VEI       | B_SCI     | Cue       | Composite | FP         |
|----------------------|--------------|-------------|----------|-------------|-----------|-----------|-----------|------------|
| ds000210_self_loop   | A4           | 0.969444    | 0.808333 | 0.988889    | 0.994444  | 0.940278  | 0.940278  | 0.00277778 |
| ds000210_self_loop   | A4_no_reason | 0           | 0.813889 | 0.997222    | 0.997222  | 0.702083  | 0.702083  | 0          |
| ds000210_self_loop   | A4_no_veto   | 0.969444    | 0.788889 | -0.00277778 | 0.994444  | 0.6875    | 0.6875    | 0.00277778 |
| ds000210_self_loop   | A5           | 0           | 0        | 0           | 0.166667  | 0.0416667 | 0.0416667 | 0          |
| ds000212_choice      | A4           | 0.977778    | 0.555556 | 0.997222    | 1         | 0.882639  | 0.882639  | 0          |
| ds000212_choice      | A4_no_reason | -0.00277778 | 0.738889 | 0.994444    | 0.997222  | 0.681944  | 0.681944  | 0.00277778 |
| ds000212_choice      | A4_no_veto   | 0.988889    | 0.672222 | 0           | 1         | 0.665278  | 0.665278  | 0          |
| ds000212_choice      | A5           | 0           | 0        | 0           | 0.161111  | 0.0402778 | 0.0402778 | 0          |
| ds001618_self_loop   | A4           | 0.983333    | 0.816667 | 0.997222    | 1         | 0.949306  | 0.949306  | 0          |
| ds001618_self_loop   | A4_no_reason | -0.00277778 | 0.863889 | 0.997222    | 0.988889  | 0.711806  | 0.711806  | 0.00277778 |
| ds001618_self_loop   | A4_no_veto   | 0.983333    | 0.869444 | 0           | 1         | 0.713194  | 0.713194  | 0          |
| ds001618_self_loop   | A5           | 0.00277778  | 0        | 0           | 0.391667  | 0.0986111 | 0.0986111 | 0          |
| ds001882_choice      | A4           | 0.997222    | 0.852778 | 0.994444    | 0.997222  | 0.960417  | 0.960417  | 0.00277778 |
| ds001882_choice      | A4_no_reason | 0           | 0.836111 | 1           | 1         | 0.709028  | 0.709028  | 0          |
| ds001882_choice      | A4_no_veto   | 1           | 0.847222 | 0           | 1         | 0.711806  | 0.711806  | 0          |
| ds001882_choice      | A5           | 0           | 0        | 0           | 0.0916667 | 0.0229167 | 0.0229167 | 0          |
| ds004042_recall      | A4           | 0.927171    | 0.907563 | 0.946779    | 0.943978  | 0.931373  | 0.931373  | 0.0504202  |
| ds004042_recall      | A4_no_reason | -0.29972    | 0.635854 | 0.70028     | 0.697479  | 0.433473  | 0.433473  | 0.29972    |
| ds004042_recall      | A4_no_veto   | 0.938375    | 0.879552 | -0.0504202  | 0.943978  | 0.677871  | 0.677871  | 0.0504202  |
| ds004042_recall      | A5           | 0           | 0        | 0           | 0.159664  | 0.039916  | 0.039916  | 0          |
| ds004917_uncertainty | A4           | 0.991667    | 0.908333 | 1           | 0.997222  | 0.974306  | 0.974306  | 0          |
| ds004917_uncertainty | A4_no_reason | 0           | 0.891667 | 1           | 1         | 0.722917  | 0.722917  | 0          |

Continued on next page

Table S47: Behavioral component scores by dataset namespace and variant.  
(continued)

| Dataset                 | Variant      | B_RSI     | B_MCI       | B_VEI       | B_SCI     | Cue       | Composite | FP         |
|-------------------------|--------------|-----------|-------------|-------------|-----------|-----------|-----------|------------|
| ds004917<br>uncertainty | A4_no_veto   | 0.980556  | 0.838889    | 0           | 0.997222  | 0.704167  | 0.704167  | 0          |
| ds004917<br>uncertainty | A5           | 0         | 0           | 0           | 0.0944444 | 0.0236111 | 0.0236111 | 0          |
| narratives<br>formal    | A4           | 0.964286  | 0.563492    | 0.996032    | 1         | 0.880952  | 0.880952  | 0          |
| narratives<br>formal    | A4_no_reason | 0         | 0.738095    | 0.988095    | 1         | 0.681548  | 0.681548  | 0          |
| narratives<br>formal    | A4_no_veto   | 0.97619   | 0.710317    | -0.00396825 | 0.996032  | 0.669643  | 0.669643  | 0.00396825 |
| narratives<br>formal    | A5           | 0.0119048 | -0.00396825 | -0.00396825 | 0.190476  | 0.0486111 | 0.0486111 | 0.00396825 |

Table S48: Predefined dataset-direction behavioral criterion contrasts.

| Dataset                 | structured<br>comp. | reason-lesion<br>comp. | veto-lesion<br>comp. | HS comp.  | structured-<br>HS comp. | reason<br>contrast | veto contrast | memory<br>contrast | self contrast |
|-------------------------|---------------------|------------------------|----------------------|-----------|-------------------------|--------------------|---------------|--------------------|---------------|
| ds000210_self_loop      | 0.940278            | 0.702083               | 0.6875               | 0.0416667 | 0.898611                | 0.969444           | 0.991667      | 0.808333           | 0.827778      |
| ds000212_choice         | 0.882639            | 0.681944               | 0.665278             | 0.0402778 | 0.842361                | 0.980556           | 0.997222      | 0.555556           | 0.838889      |
| ds001618_self_loop      | 0.949306            | 0.711806               | 0.713194             | 0.0986111 | 0.850694                | 0.986111           | 0.997222      | 0.816667           | 0.608333      |
| ds001882_choice         | 0.960417            | 0.709028               | 0.711806             | 0.0229167 | 0.9375                  | 0.997222           | 0.994444      | 0.852778           | 0.905556      |
| ds004042_recall         | 0.931373            | 0.433473               | 0.677871             | 0.039916  | 0.891457                | 1.22689            | 0.997199      | 0.907563           | 0.784314      |
| ds004917<br>uncertainty | 0.974               | 0.723                  | 0.704                | 0.024     | 0.951                   | 0.992              | 1             | 0.908              | 0.903         |
| narratives<br>formal    | 0.880952            | 0.681548               | 0.669643             | 0.0486111 | 0.832341                | 0.964286           | 1             | 0.56746            | 0.809524      |

## S5. Stochastic controls and entropy

Strong stochastic controls test whether alternative stochastic implementations recover the structured-control profile. Entropy diagnostics summarize action variability under raw strings, rule-canonical actions, semantic clusters and action-family mappings. The calibration tables report the maximum entropy gaps alongside the predefined matching threshold.

Table S49: Proxy token/compute accounting summary. Provider-side prompt-token, completion-token and total-token usage were not uniformly retained; this table reports availability and proxy measurements for length and structure.

| Audit item                                | Measurement or proxy                                            | Availability or value                                   | Evidence source                                                                  | Interpretation                                           |
|-------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------|
| provider token usage                      | prompt_tokens,<br>completion_tokens,<br>total_tokens            | not uniformly retained                                  | token_verbosity_audit criterion summary reports provider token usage unavailable | provider-token accounting incomplete                     |
| prompt length                             | model-visible prompt character or token count                   | not uniformly retained                                  | no complete prompt-length column in retained token audit summary                 | prompt-token matching unavailable                        |
| completion length proxy                   | json_trace_chars,<br>final_action_chars,<br>first_impulse_chars | 84,516 trace rows; 552 source/dataset/variant summaries | retained token-verbosity audit table                                             | field-length/verbosity proxy only                        |
| trace structure proxy                     | reason_node_count,<br>branch_count,<br>memory_item_count        | 84,516 trace rows; 552 summaries                        | retained token-verbosity audit table                                             | structure proxy for compute-risk assessment              |
| schema field count                        | matched-interface emitted JSON schema                           | nine external fields in matched-interface protocol      | matched-interface protocol schema                                                | controls schema availability under the matched interface |
| retry count                               | provider retry attempts                                         | not uniformly retained                                  | unrecovered generations are reported separately by experiment                    | retry-cost inference unavailable                         |
| latency proxy                             | latency_ms                                                      | available where retained in token_verbosity_audit       | wall-clock proxy affected by provider and network conditions                     | indirect model-compute/cost proxy                        |
| compressed structured-control sensitivity | AFCI_mean                                                       | A4_full=0.500;<br>A4_compressed=0.499                   | matched-interface AFCI table                                                     | coupling persisted under compressed trace expression     |
| <b>A5_long scratchpad control</b>         | AFCI_mean                                                       | <b>A5_long scratchpad=0.000</b>                         | matched-interface AFCI table                                                     | compute matching not established                         |

Table S50: Robustness and stochastic-control criterion summary. Prompt-paraphrase, expanded counterfactual and placebo rows are auxiliary robustness blocks. They are distinct from the formal counterfactual-transform quality-control screen and the consistency-passing transform reported in Supplementary Table S65.

| Experiment                     | Status                  | Calls | Remote | Unrec. | Unrec. rate | Min ds | HS rec. | AFCI ds | Parse n | Parse r  | Kappa | Leak |
|--------------------------------|-------------------------|-------|--------|--------|-------------|--------|---------|---------|---------|----------|-------|------|
| strong stochastic controls     | pass                    | 15490 | 15490  | 0      | 0           | 7      | 0       | —       | —       | —        | —     | —    |
| A5_empirical                   | predefined              | 17082 | 17082  | 0      | 0           | 7      | 0       | —       | —       | —        | —     | —    |
| entropy_attempt                | closeness not satisfied |       |        |        |             |        |         |         |         |          |       |      |
| prompt paraphrases             | pass                    | 11400 | 11400  | 0      | 0           | 17     | 0       | —       | —       | —        | —     | —    |
| expanded counterfactual probes | auxiliary               | 30584 | 30584  | 0      | 0           | 61     | 0       | —       | —       | —        | —     | —    |
| placebo components             | pass                    | 9960  | 9960   | 0      | 0           | 24     | 0       | —       | —       | —        | —     | —    |
| cross-model subset             | pass                    | —     | —      | —      | —           | —      | —       | —       | —       | —        | —     | —    |
| open-weight                    | pass                    | —     | —      | —      | —           | —      | —       | 7       | 8       | 0.000762 | —     | —    |
| reference model                |                         |       |        |        |             |        |         |         |         |          |       |      |
| returned-score                 | pass                    | —     | —      | —      | —           | —      | —       | —       | —       | —        | 0.961 | 0    |
| audit                          |                         |       |        |        |             |        |         |         |         |          |       |      |

Table S51: Entropy diagnostics summarized by experiment source and variant.

| Source                     | Variant      | Raw H mean | Rule H mean | Semantic H mean | Family H mean | Raw H SD | Rule H SD | Semantic H SD | Family H SD | Aggregate rows |
|----------------------------|--------------|------------|-------------|-----------------|---------------|----------|-----------|---------------|-------------|----------------|
| counterfactual probes      | A4           | 4.180      | 2.647       | 3.962           | 0.863         | 1.703    | 1.490     | 1.625         | 0.461       | 70             |
| counterfactual probes      | A4_no reason | 4.013      | 2.549       | 3.851           | 0.883         | 1.753    | 1.543     | 1.683         | 0.512       | 70             |
| counterfactual probes      | A4_no veto   | 4.179      | 2.591       | 3.969           | 0.913         | 1.734    | 1.508     | 1.709         | 0.490       | 70             |
| counterfactual probes      | A5           | 5.519      | 3.362       | 5.445           | 1.086         | 1.410    | 1.738     | 1.432         | 0.452       | 70             |
| A5_emp.                    |              |            |             |                 |               |          |           |               |             |                |
| entropy                    | A4           | 6.596      | 5.106       | 6.360           | 1.070         | 1.292    | 1.541     | 1.301         | 0.411       | 21             |
| A5_emp.                    |              |            |             |                 |               |          |           |               |             |                |
| entropy                    | A5           | 7.569      | 5.600       | 7.514           | 1.241         | 0.956    | 1.390     | 0.979         | 0.131       | 21             |
| A5_emp.                    |              |            |             |                 |               |          |           |               |             |                |
| entropy                    | A5_emp.      | 7.584      | 5.477       | 7.529           | 1.236         | 0.924    | 1.432     | 0.951         | 0.134       | 21             |
| placebo_components         | A4           | 4.445      | 3.443       | 4.312           | 0.822         | 1.594    | 1.628     | 1.546         | 0.393       | 33             |
| placebo_components         | A4_no reason | 4.156      | 3.232       | 4.027           | 0.879         | 1.622    | 1.472     | 1.596         | 0.500       | 33             |
| placebo_components         | A4_no veto   | 4.698      | 3.546       | 4.551           | 0.865         | 1.508    | 1.678     | 1.460         | 0.432       | 33             |
| placebo_components         | A5           | 5.364      | 4.048       | 5.330           | 1.091         | 1.442    | 1.677     | 1.450         | 0.378       | 33             |
| prompt_paraphrases         | A4           | 5.232      | 3.848       | 5.000           | 1.159         | 1.409    | 1.501     | 1.341         | 0.466       | 21             |
| prompt_paraphrases         | A4_no reason | 4.895      | 3.419       | 4.720           | 1.103         | 1.503    | 1.495     | 1.458         | 0.514       | 21             |
| prompt_paraphrases         | A4_no veto   | 5.464      | 4.010       | 5.267           | 1.002         | 1.356    | 1.515     | 1.326         | 0.410       | 21             |
| prompt_paraphrases         | A5           | 6.302      | 4.653       | 6.183           | 1.204         | 0.995    | 1.340     | 1.000         | 0.251       | 21             |
| strong stochastic controls |              |            |             |                 |               |          |           |               |             |                |
| strong stochastic controls | A4           | 7.009      | 5.462       | 6.777           | 1.090         | 1.849    | 1.757     | 1.815         | 0.415       | 7              |
| strong stochastic controls |              |            |             |                 |               |          |           |               |             |                |
| strong stochastic controls | A5           | 7.998      | 5.914       | 7.943           | 1.204         | 1.377    | 1.545     | 1.401         | 0.228       | 7              |
| strong stochastic controls | compute      | 7.994      | 5.915       | 7.936           | 1.238         | 1.422    | 1.569     | 1.452         | 0.151       | 7              |
| strong stochastic controls | control      |            |             |                 |               |          |           |               |             |                |
| strong stochastic controls | A5_entropy   | 7.941      | 5.896       | 7.896           | 1.161         | 1.465    | 1.754     | 1.463         | 0.211       | 7              |
| strong stochastic controls | attempt      |            |             |                 |               |          |           |               |             |                |
| strong stochastic controls | A5_self      |            |             |                 |               |          |           |               |             |                |
| controls                   | consistency  | 7.994      | 5.926       | 7.928           | 1.173         | 1.382    | 1.640     | 1.414         | 0.176       | 7              |

Table S52: Predefined empirical entropy-attempt calibration criterion summary.

| Criterion                | Field                           | Value                                                        |
|--------------------------|---------------------------------|--------------------------------------------------------------|
| strict_a5_em_calibration | calibration_events_per_dataset  | 50                                                           |
| strict_a5_em_calibration | closeness_threshold             | 0.15                                                         |
| strict_a5_em_calibration | grid_coverage_status            | complete                                                     |
| strict_a5_em_calibration | max_selected_strict_entropy_gap | 1.17146                                                      |
| strict_a5_em_calibration | note                            | maximum gap across the four entropy layers exceeds threshold |
| strict_a5_em_calibration | observed_required_setting_count | 28                                                           |
| strict_a5_em_calibration | required_setting_count          | 28                                                           |
| strict_a5_em_calibration | selected_temperature            | 0.2                                                          |
| strict_a5_em_calibration | selected_top_p                  | 0.75                                                         |
| strict_a5_em_calibration | selected_variant                | A5_attempt_t020_p075                                         |
| strict_a5_em_calibration | status                          | predefined closeness not satisfied                           |
| strict_a5_em_calibration | strict_entropy_close_enough     | False                                                        |

Table S53: Predefined empirical entropy-attempt calibration grid.

| Variant    | Temp. | Top $p$ | Raw H gap | Rule H gap | Semantic H gap | Family H gap | Strict score | Max strict gap |
|------------|-------|---------|-----------|------------|----------------|--------------|--------------|----------------|
| A5_attempt |       |         |           |            |                |              |              |                |
| t020_p060  | 0.2   | 0.6     | 0.944     | 1.300      | 1.096          | 0.302        | 0.910        | 1.300          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t020_p075  | 0.2   | 0.75    | 0.973     | 1.171      | 1.110          | 0.353        | 0.902        | 1.171          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t020_p090  | 0.2   | 0.9     | 1.102     | 1.420      | 1.224          | 0.404        | 1.037        | 1.420          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t020_p100  | 0.2   | 1       | 0.955     | 1.331      | 1.094          | 0.365        | 0.936        | 1.331          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t040_p060  | 0.4   | 0.6     | 1.125     | 1.545      | 1.271          | 0.388        | 1.082        | 1.545          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t040_p075  | 0.4   | 0.75    | 1.154     | 1.462      | 1.295          | 0.249        | 1.040        | 1.462          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t040_p090  | 0.4   | 0.9     | 1.070     | 1.328      | 1.217          | 0.362        | 0.994        | 1.328          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t040_p100  | 0.4   | 1       | 1.186     | 1.334      | 1.314          | 0.317        | 1.038        | 1.334          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t060_p060  | 0.6   | 0.6     | 1.129     | 1.518      | 1.264          | 0.296        | 1.052        | 1.518          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t060_p075  | 0.6   | 0.75    | 1.123     | 1.373      | 1.260          | 0.403        | 1.040        | 1.373          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t060_p090  | 0.6   | 0.9     | 1.114     | 1.492      | 1.253          | 0.504        | 1.091        | 1.492          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t060_p100  | 0.6   | 1       | 1.155     | 1.520      | 1.273          | 0.303        | 1.063        | 1.520          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t080_p060  | 0.8   | 0.6     | 1.167     | 1.539      | 1.295          | 0.371        | 1.093        | 1.539          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t080_p075  | 0.8   | 0.75    | 1.043     | 1.344      | 1.178          | 0.322        | 0.972        | 1.344          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t080_p090  | 0.8   | 0.9     | 1.068     | 1.414      | 1.207          | 0.324        | 1.003        | 1.414          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t080_p100  | 0.8   | 1       | 1.058     | 1.704      | 1.205          | 0.307        | 1.069        | 1.704          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t100_p060  | 1     | 0.6     | 0.904     | 1.305      | 1.026          | 0.418        | 0.913        | 1.305          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t100_p075  | 1     | 0.75    | 0.973     | 1.476      | 1.106          | 0.469        | 1.006        | 1.476          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t100_p090  | 1     | 0.9     | 0.960     | 1.403      | 1.099          | 0.404        | 0.966        | 1.403          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t100_p100  | 1     | 1       | 1.017     | 1.438      | 1.164          | 0.394        | 1.003        | 1.438          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t120_p060  | 1.2   | 0.6     | 1.045     | 1.579      | 1.184          | 0.435        | 1.061        | 1.579          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t120_p075  | 1.2   | 0.75    | 1.043     | 1.136      | 1.188          | 0.577        | 0.986        | 1.188          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t120_p090  | 1.2   | 0.9     | 0.984     | 1.389      | 1.128          | 0.395        | 0.974        | 1.389          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t120_p100  | 1.2   | 1       | 1.029     | 1.298      | 1.170          | 0.469        | 0.991        | 1.298          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t140_p060  | 1.4   | 0.6     | 1.041     | 1.357      | 1.179          | 0.525        | 1.025        | 1.357          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t140_p075  | 1.4   | 0.75    | 1.028     | 1.419      | 1.157          | 0.397        | 1.000        | 1.419          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t140_p090  | 1.4   | 0.9     | 1.051     | 1.357      | 1.198          | 0.437        | 1.011        | 1.357          |
| A5_attempt |       |         |           |            |                |              |              |                |
| t140_p100  | 1.4   | 1       | 0.966     | 1.424      | 1.119          | 0.439        | 0.987        | 1.424          |

Table S54: Finite-action-code entropy-calibration criterion summary.

| Criterion                         | Field                                                                             | Value                              |
|-----------------------------------|-----------------------------------------------------------------------------------|------------------------------------|
| finite-action entropy calibration | structured-control behavioral composite > selected high-stochasticity datasets    | 7                                  |
| finite-action entropy calibration | behavior pass threshold                                                           | 6                                  |
| finite-action entropy calibration | calibration events per dataset                                                    | 14                                 |
| finite-action entropy calibration | calibration maximum predefined entropy gap                                        | 1.22474                            |
| finite-action entropy calibration | calibration mean predefined entropy gap                                           | 1.13035                            |
| finite-action entropy calibration | calls                                                                             | 11088                              |
| finite-action entropy calibration | closeness threshold                                                               | 0.15                               |
| finite-action entropy calibration | dataset count                                                                     | 7                                  |
| finite-action entropy calibration | dry run                                                                           | False                              |
| finite-action entropy calibration | experiment                                                                        | finite-action entropy calibration  |
| finite-action entropy calibration | unrecovered-generation rate                                                       | 0                                  |
| finite-action entropy calibration | grid setting count                                                                | 14                                 |
| finite-action entropy calibration | maximum model-generation budget                                                   | 12000                              |
| finite-action entropy calibration | holdout events per dataset                                                        | 18                                 |
| finite-action entropy calibration | holdout maximum predefined entropy gap                                            | 1.34765                            |
| finite-action entropy calibration | holdout mean predefined entropy gap                                               | 1.14711                            |
| finite-action entropy calibration | holdout predefined entropy close enough                                           | False                              |
| finite-action entropy calibration | holdout_unmapped action rate                                                      | 0                                  |
| finite-action entropy calibration | planned calibration calls                                                         | 8820                               |
| finite-action entropy calibration | planned calls                                                                     | 11088                              |
| finite-action entropy calibration | planned holdout calls                                                             | 2268                               |
| finite-action entropy calibration | remote calls                                                                      | 11088                              |
| finite-action entropy calibration | selected high-stochasticity candidate did not recover structured-control behavior | True                               |
| finite-action entropy calibration | selected behavior bootstrap lower endpoint of the two-sided 95% interval          | 0.863095                           |
| finite-action entropy calibration | selected temperature                                                              | 0.2                                |
| finite-action entropy calibration | selected top-p                                                                    | 0.7                                |
| finite-action entropy calibration | selected variant                                                                  | A5_attempt_t02_p07                 |
| finite-action entropy calibration | status                                                                            | predefined closeness not satisfied |
| finite-action entropy calibration | unrecovered generations                                                           | 0                                  |

Table S55: Finite-action-code entropy-calibration grid.

| Variant    | Sets | Rows | Mean gap | Max gap | Unmap | Raw H | Code H | Family H | Temp. | Top $p$ |
|------------|------|------|----------|---------|-------|-------|--------|----------|-------|---------|
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t01_p07    | 7    | 588  | 1.138    | 1.189   | 0     | 1.138 | 1.138  | 1.138    | 0.1   | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t01_p10    | 7    | 588  | 1.171    | 1.249   | 0.002 | 1.171 | 1.171  | 1.171    | 0.1   | 1       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t02_p07    | 7    | 588  | 1.130    | 1.225   | 0.002 | 1.130 | 1.130  | 1.130    | 0.2   | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t02_p10    | 7    | 588  | 1.142    | 1.250   | 0     | 1.142 | 1.142  | 1.142    | 0.2   | 1       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t04_p07    | 7    | 588  | 1.139    | 1.189   | 0     | 1.139 | 1.139  | 1.139    | 0.4   | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t04_p10    | 7    | 588  | 1.164    | 1.219   | 0     | 1.164 | 1.164  | 1.164    | 0.4   | 1       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t06_p07    | 7    | 588  | 1.162    | 1.250   | 0.002 | 1.162 | 1.162  | 1.162    | 0.6   | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t06_p10    | 7    | 588  | 1.161    | 1.223   | 0     | 1.161 | 1.161  | 1.161    | 0.6   | 1       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t08_p07    | 7    | 588  | 1.157    | 1.225   | 0.002 | 1.157 | 1.157  | 1.157    | 0.8   | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t08_p10    | 7    | 588  | 1.156    | 1.223   | 0     | 1.156 | 1.156  | 1.156    | 0.8   | 1       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t10_p07    | 7    | 588  | 1.153    | 1.223   | 0     | 1.153 | 1.153  | 1.153    | 1     | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t10_p10    | 7    | 588  | 1.153    | 1.284   | 0.003 | 1.153 | 1.153  | 1.153    | 1     | 1       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t12_p07    | 7    | 588  | 1.152    | 1.223   | 0     | 1.152 | 1.152  | 1.152    | 1.2   | 0.7     |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t12_p10    | 7    | 588  | 1.142    | 1.219   | 0.002 | 1.142 | 1.142  | 1.142    | 1.2   | 1       |

Table S56: Finite-action-code entropy-calibration holdout grid.

| Variant    | Sets | Rows | Mean gap | Max gap | Unmap | Raw H | Code H | Family H | Temp. | Top $p$ |
|------------|------|------|----------|---------|-------|-------|--------|----------|-------|---------|
| A5         | 7    | 756  | 1.165    | 1.299   | 0     | 1.165 | 1.165  | 1.165    | —     | —       |
| A5_attempt |      |      |          |         |       |       |        |          |       |         |
| t02_p07    | 7    | 756  | 1.147    | 1.348   | 0     | 1.147 | 1.147  | 1.147    | 0.2   | 0.7     |

## S6. Model sensitivity and human agreement

Robustness analyses compare the tested instructions, provider identifiers and a local open-weight reference model. The annotation tables report agreement in returned individual scores, using Cohen’s  $\kappa$  as a chance-adjusted nominal-agreement statistic (Cohen, 1960). The sampling and scoring procedures are described in Supplementary Note 5.

Table S57: Cross-model subset criterion summary.

| Model               | Status | Exp.            | Profile | Calls | Remote | Unrecov. | Unrec.<br>rate | Vars. | Ctrls. | Min ctrl ds | Strong HS<br>ds | Fallback         | SC>HS ds |
|---------------------|--------|-----------------|---------|-------|--------|----------|----------------|-------|--------|-------------|-----------------|------------------|----------|
| gpt-5.4<br>mini     | pass   | cross<br>subset | full    | 1995  | 1995   | 0        | 0              | 5     | 4      | 5           | 0               | qwen3<br>30b-a3b | 7        |
| gpt-4.1<br>mini     | pass   | cross<br>subset | full    | 1995  | 1995   | 0        | 0              | 5     | 4      | 6           | 0               | qwen3<br>30b-a3b | 7        |
| deepseek-chat       | pass   | cross<br>subset | full    | 1995  | 1995   | 0        | 0              | 5     | 4      | 6           | 0               | qwen3<br>30b-a3b | 7        |
| gemini-2.5<br>flash | pass   | cross<br>subset | full    | 1995  | 1995   | 2        | 0.001          | 5     | 4      | 7           | 0               | qwen3<br>30b-a3b | 7        |

Table S58: Open-weight Qwen2.5-14B-Instruct AFCI components by variant.

| Variant                     | Statistic | Reason | Memory | Veto | Self  | Support | AFCI  | AFCI+ | n  |
|-----------------------------|-----------|--------|--------|------|-------|---------|-------|-------|----|
| A4                          | Mean      | 1      | 0.5    | 1    | 1     | 1       | 0.875 | 0.9   | 21 |
| A4                          | SD        | 0      | 0      | 0    | 0     | 0       | 0     | 0     | 21 |
| A4_no<br>reason             | Mean      | 0.5    | 0.5    | 1    | 1     | 1       | 0.75  | 0.8   | 21 |
| A4_no<br>reason             | SD        | 0      | 0      | 0    | 0     | 0       | 0     | 0     | 21 |
| A4_no<br>veto               | Mean      | 1      | 0.5    | 0.5  | 1     | 1       | 0.75  | 0.8   | 21 |
| A4_no<br>veto               | SD        | 0      | 0      | 0    | 0     | 0       | 0     | 0     | 21 |
| A5                          | Mean      | 0.446  | 0.25   | 0.5  | 0.393 | 0       | 0.397 | 0.318 | 21 |
| A5                          | SD        | 0.019  | 0      | 0    | 0.031 | 0       | 0.008 | 0.006 | 21 |
| A5_schema<br>posthoc_fields | Mean      | 0      | 0      | 0    | 0     | 0       | 0     | 0     | 21 |
| A5_schema<br>posthoc_fields | SD        | 0      | 0      | 0    | 0     | 0       | 0     | 0     | 21 |

Table S59: Agreement in returned annotations by dimension.

| Dimension                                  | Annotator A  | Annotator B  | Overlap items | Agreement | Cohen kappa |
|--------------------------------------------|--------------|--------------|---------------|-----------|-------------|
| final action reason-consistent             | annotator_01 | annotator_02 | 1200          | 0.958333  | 0.912127    |
| final action reason-consistent             | annotator_01 | annotator_03 | 1200          | 0.955833  | 0.907158    |
| final action reason-consistent             | annotator_02 | annotator_03 | 1200          | 0.990833  | 0.980522    |
| action respects relevant memory            | annotator_01 | annotator_02 | 1200          | 0.997500  | 0.989211    |
| action respects relevant memory            | annotator_01 | annotator_03 | 1200          | 0.997500  | 0.989211    |
| action respects relevant memory            | annotator_02 | annotator_03 | 1200          | 1.000000  | 1.000000    |
| veto_was_appropriate                       | annotator_01 | annotator_02 | 1200          | 1.000000  | 1.000000    |
| veto_was_appropriate                       | annotator_01 | annotator_03 | 1200          | 1.000000  | 1.000000    |
| veto_was_appropriate                       | annotator_02 | annotator_03 | 1200          | 1.000000  | 1.000000    |
| veto changed first impulse                 | annotator_01 | annotator_02 | 1200          | 1.000000  | 1.000000    |
| veto changed first impulse                 | annotator_01 | annotator_03 | 1200          | 1.000000  | 1.000000    |
| veto changed first impulse                 | annotator_02 | annotator_03 | 1200          | 1.000000  | 1.000000    |
| reason_looks_posthoc                       | annotator_01 | annotator_02 | 1200          | 0.960000  | 0.917801    |
| reason_looks_posthoc                       | annotator_01 | annotator_03 | 1200          | 0.961667  | 0.921303    |
| reason_looks_posthoc                       | annotator_02 | annotator_03 | 1200          | 0.995000  | 0.989652    |
| verbose/template-like without control      | annotator_01 | annotator_02 | 1200          | 0.998333  | 0.971364    |
| verbose/template-like without control      | annotator_01 | annotator_03 | 1200          | 0.996667  | 0.942728    |
| verbose/template-like without control      | annotator_02 | annotator_03 | 1200          | 0.998333  | 0.972113    |
| irrelevant cue caused inappropriate change | annotator_01 | annotator_02 | 1200          | 0.974167  | 0.865842    |
| irrelevant cue caused inappropriate change | annotator_01 | annotator_03 | 1200          | 0.972500  | 0.860015    |
| irrelevant cue caused inappropriate change | annotator_02 | annotator_03 | 1200          | 0.993333  | 0.965260    |

Annotation methods: three graduate students in relevant university disciplines, including doctoral and master's students, each rated the same 1,200 items using a common seven-dimension codebook. Supplementary Note 5 specifies sampling, metadata masking, randomized order, written instructions and majority aggregation. All 25,200 individual core ratings are valid binary values; 8,250 item–dimension combinations are unanimous and 150 have a 2:1 split, affecting 68 items. The table uses individual ratings before aggregation: mean pairwise Cohen's kappa is 0.961, the minimum individual value is 0.860 and the minimum dimension mean is 0.897.

## S7. Source records and reproducibility

Numbered manuscript-table locations inside the archival map refer to the original source release. The analyses in Supplementary Notes 1–15 and the current S-numbered tables provide the corresponding interpretation and results.

The reproducibility map links reported claims to tables, result exports, decision criteria, analysis code and checksums. Full SHA-256 values are retained in the CSV export; the table prints 12-character prefixes. Export paths identify the original source records. Figure S9 displays the open-weight controls, and Supplementary Note 1 reports the grouped reference-dependent localization analysis. “Legacy export ID” identifies an archived filename; current supplementary tables use the S prefix.

Table S60: Validation source-data and control-construction map.

| Item                                          | Role                                                                                           | Source data, config or script path                                                                                                                                | Checksum or provenance      |
|-----------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Main validation panel a source data           | Open-weight core no-context controls                                                           | manuscript/figures/source_data/figure5a_open_weight_core_controls.csv                                                                                             | 12ef8d181136                |
| Main validation panel b source data           | Open-weight state-boundary lesion                                                              | manuscript/figures/source_data/figure5b_open_weight_state_boundary_lesion.csv                                                                                     | 803140cd02ce                |
| Main validation panel c source data           | SWE-bench Lite full 300-record incremental AUC                                                 | manuscript/figures/source_data/figure5c_swebench_incremental_auc.csv                                                                                              | 261436e42398                |
| Main validation panel d source data           | SWE-bench Lite single-predictor AUC comparison                                                 | manuscript/figures/source_data/figure5d_swebench_single_predictor_auc.csv                                                                                         | 8897f8c61c79                |
| Main validation panel e source data           | SWE-bench Lite sensitivity and binding-guard summary                                           | manuscript/figures/source_data/figure5e_swebench_sensitivity_guard.csv                                                                                            | 4dcd90ac380e                |
| Main validation rendering and export QA       | Figure rendering and export-QA script                                                          | manuscript/figures/scripts/build_validation_figure5.R; audit/figure5_primary_validation_export_qc_20260530.csv                                                    | 61d69f9d8264 / 80f8d170bb37 |
| SWE-bench reliability calibration source data | Issue-grouped five-fold reliability calibration for the full 300-record issue-to-file endpoint | manuscript/figures/source_data/extended_data_figure9_swebench_reliability_calibration.csv; manuscript/figures/source_data/swebench_cv_reliability_predictions.csv | archived manifest           |
| SWE-bench threshold analysis source data      | Thresholded localization-guard analysis over predicted hit@3 probabilities                     | manuscript/figures/source_data/extended_data_figure10_swebench_threshold_analysis.csv; manuscript/figures/scripts/build_swebench_reliability_figures.R            | archived manifest           |
| Submitted main Table 1 source data            | Claim ledger with supported claims and excluded inferences                                     | manuscript/tables/table1_validation_evidence_interpretation.csv                                                                                                   | archived manifest           |
| Supplementary Table 33 source data            | Validation controls and interpretation status                                                  | manuscript/tables/supplementary_table_33_validation_controls.csv                                                                                                  | 7a88ca507fb7                |
| Decisive-field positive-control summaries     | Two-model decisive-field and wrapper result tables                                             | manuscript/tables/table8_decisive_field_positive_control_condition_summary.csv; manuscript/tables/table8_control_binding_wrapper_scrambled_summary.csv            | archived manifest           |
| Decisive-field positive-control leakage audit | Prompt/scorer leakage and scope checks                                                         | manuscript/tables/table8_decisive_field_positive_control_leakage_audit.csv                                                                                        | e1fdfc5005ce                |
| Local strict matching summaries               | Local Qwen2.5-1.5B forced-budget entropy/token audit                                           | manuscript/tables/table16_local_strict_matching_condition_summary.csv; manuscript/tables/table16_local_strict_matching_gate_summary.csv                           | archived manifest           |

Continued on next page

Table S60: Validation source-data and control-construction map. (continued)

| Item                                                          | Role                                                                                                                                                 | Source data, config or script path                                                                                                                                                                                                                                                                                                                                                                                                                        | Checksum or provenance      |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| SWE-bench Lite issue-to-file summaries                        | Complete 300-record issue-localization summaries and execution-slice preparation status                                                              | results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/gate_status.json; results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/official_execution_slice_environment_status.json; results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/official_execution_slice_analysis_v3_apply_repair/official_execution_slice_gate_status.json | archived manifest           |
| Open-weight matrix config path                                | Model, scaffold, task-family and variant manifest                                                                                                    | results/paper1_revision/remote_5090_run3_20260508/nmi_formal_execution_manifest/execution_manifest.json                                                                                                                                                                                                                                                                                                                                                   | 11e3390109fd                |
| Formal API validation config path                             | API validation run manifest                                                                                                                          | results/paper1_revision/remote_5090_run3_20260508/formal_api_validation/formal_api_full_20260510_0030/run_manifest.json                                                                                                                                                                                                                                                                                                                                   | 8712dbea8b1b                |
| Distribution-matched control construction                     | Offline/no-context action-prior policy and prompt construction                                                                                       | src/freewill/structured_control_validation.py                                                                                                                                                                                                                                                                                                                                                                                                             | eba9d1039034                |
| Strict-lesion construction                                    | Removal of decisive intervention-carrying field with non-decisive context retained                                                                   | src/freewill/structured_control_validation.py                                                                                                                                                                                                                                                                                                                                                                                                             | eba9d1039034                |
| Counterfactual transform QC and consistency-passing transform | Transform definitions and formal holdout launcher                                                                                                    | src/freewill/structured_control_validation.py; scripts/run/run_formal_open_weight_validation.py                                                                                                                                                                                                                                                                                                                                                           | eba9d1039034 / 0f9cfd41e1af |
| Token/compute criterion script                                | Prompt-token, total-token and latency-proxy criterion analysis                                                                                       | scripts/run/analyze_formal_validation_results.py; scripts/run/audit_open_weight_token_entropy.py                                                                                                                                                                                                                                                                                                                                                          | 404f5edb4132 / f037f6424513 |
| Entropy-prior behavioral-control script                       | Calibration-split action-prior selection and entropy-prior no-field behavioral control                                                               | scripts/run/select_formal_entropy_settings.py; src/freewill/structured_control_validation.py                                                                                                                                                                                                                                                                                                                                                              | f29b12ebe627 / eba9d1039034 |
| Local strict matching script                                  | Local tokenizer padding, fixed-generation and forced-budget audit implementation                                                                     | scripts/run/run_strict_matching_local.py; src/freewill/strict_matching_local.py                                                                                                                                                                                                                                                                                                                                                                           | archived manifest           |
| SWE-bench Lite localization scripts                           | Dataset fetch, oracle-free issue-to-file prompts, shard merge, readiness audit, execution-slice patch generation and v3 apply-check/repair filtering | scripts/run/run_swebench_file_localization_v2_api.py; scripts/run/merge_swebench_fileloc_v2_shards.py; scripts/run/run_swebench_execution_slice_patch_generation_api.py; scripts/run/run_swebench_patch_generation_v3_apply_repair.py; src/freewill/real_task_predictive_validity.py                                                                                                                                                                      | archived manifest           |

Table S61: Source evidence and export map. Export IDs identify archived source files.

| ID  | Evidence                                                              | Original export | Analysis                   | Checksum prefix |
|-----|-----------------------------------------------------------------------|-----------------|----------------------------|-----------------|
| H01 | baseline lesion matrix separates unpredictability and trace structure | Export 2        | Baseline lesion matrix     | 5f2fae5d2ab5    |
| H02 | matched-interface AFCI rules out field availability alone             | Export 3        | Matched-interface controls | 2fdaffdbfdb     |
| H03 | finite-action-code behavior supports action-level component coupling  | Export 4        | Finite-action behavior     | 55820262e805    |
| H04 | strong stochastic controls do not recover structured-control profile  | Export 9        | Stochastic controls        | 1f1de7d4d18d    |

Table S61: Source evidence and export map. Export IDs identify archived source files. (continued)

| ID  | Evidence                                                                                                | Original export      | Analysis                            | Checksum prefix   |
|-----|---------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------|-------------------|
| H05 | predefined entropy closeness was not satisfied                                                          | Export 6 / Export 12 | Entropy diagnostics and calibration | b6150257          |
| H06 | cross-model tested subset preserves structured-control greater-than high-stochasticity direction        | Export 9             | Cross-model subset                  | c3ae              |
| H07 | open-weight reference model preserves structured-control greater-than high-stochasticity AFCI direction | Export 9 / Export 10 | Open-weight anchor                  | 2f8b5ff04901      |
| H08 | decisive-field positive control recovers the finite-action binding readout                              | Export 8             | Decisive-field control              | d76affe4ffe8      |
| H09 | local wrapper reduces scrambled-field binding                                                           | Export 8             | Decisive-field control              | archived manifest |
| H10 | bounded local forced-budget entropy/token audit preserves structured-state advantage                    | Export 16            | Strict-matching local audit         | archived manifest |
| H11 | SWE-bench Lite historical association; the augmented features depend on gold                            | Export 17            | SWE localization v2                 | 0206bfa8d6db      |
| H12 | Individual ratings and majority labels document inter-annotator agreement                               | Export 9             | Blind-rating export                 | archived manifest |
|     |                                                                                                         |                      |                                     | bffe01d6d1fd      |

The complete source identifiers and analysis mappings are provided in `supplement_tables/historical_claim_artifact_map.csv`, supplied with the editable source and research package.

02

Interface-diagnostic rows are summarized separately from effect estimates. The taxonomy distinguishes final-criterion-compatible interface diagnostics, non-final provider-interface attempts and entropy-calibration diagnostics. The export-coverage table lists the table exports underlying the manuscript and this supplement.

Table S62: Interface-diagnostic taxonomy and interpretation.

| Source                                 | Experiment                    | Model                  | Diagnostic type              | Rows | Interpretation                                                       |
|----------------------------------------|-------------------------------|------------------------|------------------------------|------|----------------------------------------------------------------------|
| Cross-model robustness                 | Gemini full run               | gemini-2.5-flash       | payload not object           | 2    | Final criterion met; retained as interface diagnostic.               |
| Provider-interface compatibility check | Qwen direct-provider check    | qwen3-30b-a3b          | HTTP 400                     | 100  | Compatibility rows outside the final-evidence criteria.              |
| Entropy-calibration diagnostic         | entropy-attempt calibration   | qwen3-30b-a3b fallback | HTTP 500 / fallback metadata | 594  | Diagnostic calibration rows; predefined closeness was not satisfied. |
| Baseline lesion matrix                 | aggregate baseline accounting | gpt-5.4-mini           | aggregate only               | 16   | Aggregate unrecovered-output count; no trace-level taxonomy.         |

Table S63: Table-export coverage. Full-precision values are retained in the CSV exports.  
Data rows exclude the header row and any empty trailing lines.

| Export label                                                   | Data rows | Columns | Leading columns                                                                                                     |
|----------------------------------------------------------------|-----------|---------|---------------------------------------------------------------------------------------------------------------------|
| Variant name registry                                          | 17        | 5       | canonical variant, used in, definition, aliases, role                                                               |
| Mechanistic claim map                                          | 3         | 5       | claim component, operational test, contrary result, interpretive boundary                                           |
| Baseline lesion-matrix criterion summary                       | 12        | 3       | criterion, field, value                                                                                             |
| Baseline trace metrics by dataset, variant and replicate index | 168       | 19      | variant, VBI, SCI, VEI, RSI, ACI, CSVI, unpredictability ...                                                        |
| Baseline trace metrics by variant                              | 8         | 18      | variant, VBI mean, SCI mean, VEI mean, RSI mean, ACI mean, CSVI mean, unpredictability mean ...                     |
| Matched-interface AFCI by dataset, variant and replicate index | 189       | 10      | dataset namespace, variant, replicate index, reason alignment, memory alignment, veto alignment ...                 |
| Matched-interface AFCI by variant                              | 9         | 16      | variant, alignment means, AFCI mean, AFCI with suppression mean, SD fields ...                                      |
| Matched-interface dataset contrasts                            | 56        | 5       | dataset namespace, variant, AFCI, structured full AFCI, paired contrast                                             |
| Matched-interface comparator criterion status                  | 7         | 4       | control variant, dataset count, structured greater-than-control dataset count, recovery count                       |
| Matched-interface criterion summary                            | 11        | 3       | criterion, field, value                                                                                             |
| Finite-action condition rates                                  | 504       | 7       | dataset namespace, variant, replicate index, condition, correct rate, unmapped rate, row count                      |
| Finite-action dataset criterion contrasts                      | 7         | 10      | dataset namespace, structured variant, ablation comparators, high-stochasticity comparator, component contrasts ... |
| Finite-action criterion summary                                | 16        | 3       | criterion, field, value                                                                                             |
| Finite-action metrics by dataset, variant and replicate index  | 84        | 16      | dataset namespace, variant, replicate index, condition rates and component scores ...                               |
| Finite-action summary by dataset and variant                   | 28        | 9       | dataset namespace, variant, B scores, cue specificity, behavioral structured action control ...                     |
| Finite-action summary by variant                               | 4         | 16      | variant, component-score means, behavioral composite mean, false-positive and row-count summaries ...               |
| Entropy diagnostics by source and variant                      | 20        | 11      | source, variant, raw, canonical, semantic and action-family entropy means and SDs ...                               |
| Full entropy diagnostic units                                  | 594       | 14      | source, output namespace, variant, replicate index, row count, unique action counts ...                             |
| Strict entropy-attempt calibration grid                        | 28        | 12      | variant, temperature, top-p, paired counts, absolute entropy gaps ...                                               |
| Strict entropy-attempt criterion summary                       | 13        | 3       | criterion, field, value                                                                                             |
| Finite-action entropy-calibration grid                         | 14        | 11      | variant, dataset count, row count, predefined entropy gap, unmapped action rate, entropy gaps ...                   |
| Finite-action entropy-calibration criterion summary            | 30        | 3       | criterion, field, value                                                                                             |
| Finite-action entropy-calibration holdout grid                 | 2         | 11      | variant, dataset count, row count, predefined entropy gap, unmapped action rate, entropy gaps ...                   |
| Blinded-audit reliability by dimension                         | 7         | 6       | dimension, annotator labels, overlap items, agreement, Cohen's kappa                                                |
| Cross-model criterion summary                                  | 4         | 14      | model, status, experiment profile, calls, interface diagnostics, directional dataset counts ...                     |
| Open-weight AFCI by dataset, variant and replicate index       | 105       | 10      | dataset namespace, variant, replicate index, alignment terms and AFCI                                               |

Continued on next page

Table S63: Table-export coverage. Full-precision values are retained in the CSV exports.  
Data rows exclude the header row and any empty trailing lines. (continued)

| Export label                                      | Data rows | Columns | Leading columns                                                                                                     |
|---------------------------------------------------|-----------|---------|---------------------------------------------------------------------------------------------------------------------|
| Open-weight AFCI by variant                       | 5         | 16      | variant, alignment means, AFCI means, SD fields and rows                                                            |
| Robustness criterion summary                      | 8         | 13      | experiment, status, calls, interface diagnostics, directional counts, parse and audit statistics                    |
| Decisive-field positive-control condition summary | 12        | 13      | model, condition, rows, accuracy, wrapped accuracy, wrapper change rate ...                                         |
| Decisive-field positive-control gate summary      | 2         | 10      | model, full-state accuracy, only-decisive accuracy, best control accuracy, recovery fraction ...                    |
| Decisive-field positive-control task cells        | 240       | 4       | model, task family, full-state accuracy, only-decisive accuracy                                                     |
| Decisive-field positive-control leakage audit     | 5         | 4       | audit item, evidence, status, interpretation                                                                        |
| Control-binding wrapper scrambled summary         | 2         | 7       | model, rows, raw scrambled following, wrapped scrambled following, raw expected accuracy ...                        |
| Local strict matching condition summary           | 2         | 10      | condition, rows, accuracy, invalid rate, canonical-action entropy, token and latency means                          |
| Local strict matching gate summary                | 1         | 12      | status, prompt match rate, completion match rate, total-token ratio, latency ratio ...                              |
| SWE-bench file-localization predictor summary     | 11        | 5       | predictor, hit@3 AUC, hit@3 Pearson, no-violation AUC, sign convention                                              |
| SWE-bench file-localization model summary         | 6         | 13      | model, model-instance rows, localization hit@3, violation rates, binding score and execution-slice boundary columns |
| SWE-bench file-localization gate summary          | 1         | 21      | status, rows, model-instance rows, hit@3 counts, AUC summaries, wrapper status and execution-slice boundary columns |
| Derived effect-size summary                       | 6         | 3       | evidence block, direction, stronger summary                                                                         |
| Proxy token/compute audit summary                 | 9         | 5       | audit item, measurement or proxy, availability, evidence source, interpretation                                     |
| Entropy calibration threshold summary             | 44        | 12      | source, variant, entropy gaps, strict threshold, pass flag                                                          |
| Interface-diagnostic taxonomy                     | 4         | 6       | source, experiment, model, diagnostic type, rows, interpretation                                                    |
| Claim-to-evidence reproducibility map             | 12        | 7       | claim, legacy export ID, result export, criterion summary, analysis entry point, SHA-256                            |
| Statistical sensitivity summary                   | 8         | 10      | contrast, independent unit, direction count, minimum contrast, sign test                                            |

## S8. Statistical and scoring definitions

Table S64: Dataset-level statistical sensitivity summary for primary contrasts.  
Exact sign tests are one-sided under a 0.5 null probability for a positive dataset direction; two-sided values are doubled and capped at one.

| Contrast                                                                    | Unit                 | Units | Positive units | Minimum contrast | Mean contrast | LODO minimum mean | Top-effect removed mean | One-sided sign $p$ |
|-----------------------------------------------------------------------------|----------------------|-------|----------------|------------------|---------------|-------------------|-------------------------|--------------------|
| finite behavior:<br>structured-control-high-stochasticity<br>composite      | dataset<br>namespace | 7     | 7              | 0.832            | 0.886         | 0.875             | 0.875                   | 0.0078125          |
| reason ablation: structured-control-reason<br>lesion B_RSI                  | dataset<br>namespace | 7     | 7              | 0.964            | 1.017         | 0.982             | 0.982                   | 0.0078125          |
| veto ablation: structured-control-veto<br>lesion B_VEI                      | dataset<br>namespace | 7     | 7              | 0.992            | 0.997         | 0.996             | 0.996                   | 0.0078125          |
| memory behavior:<br>structured-control-high-stochasticity<br>B_MCI          | dataset<br>namespace | 7     | 7              | 0.556            | 0.774         | 0.751             | 0.751                   | 0.0078125          |
| self-continuity behavior:<br>structured-control-high-stochasticity<br>B_SCI | dataset<br>namespace | 7     | 7              | 0.608            | 0.811         | 0.795             | 0.795                   | 0.0078125          |
| schema AFCI: structured full-stochastic<br>plain                            | dataset<br>namespace | 7     | 7              | 0.262            | 0.302         | 0.297             | 0.297                   | 0.0078125          |
| schema AFCI: structured full-weakest<br>control comparator                  | dataset<br>namespace | 7     | 7              | 0.492            | 0.500         | 0.499             | 0.499                   | 0.0078125          |
| open-weight AFCI:<br>structured-control-high-stochasticity                  | dataset<br>namespace | 7     | 7              | 0.467            | 0.478         | 0.476             | 0.476                   | 0.0078125          |

Let  $\mathcal{D}$  denote the seven source corpora and  $\mathcal{I}_{d,v}$  the retained generations for dataset  $d$  and variant  $v$ . Raw final-action unpredictability is reported as Shannon entropy (Shannon, 1948),

$$H_{d,v} = - \sum_{a \in \mathcal{A}_{d,v}} \hat{p}_{d,v}(a) \log_2 \hat{p}_{d,v}(a), \quad (2)$$

where  $\mathcal{A}_{d,v}$  is the observed final-action support and  $\hat{p}_{d,v}$  is the empirical action frequency. The baseline structured composite is

$$C_{d,v} = \frac{1}{4} (\text{SCI}_{d,v} + \text{RSI}_{d,v} + \text{VEI}_{d,v} + \text{ACI}_{d,v}). \quad (3)$$

For matched-interface controls, each scored generation has

$$\text{AFCI}_{i,v} = \frac{1}{4} (R_{i,v} + M_{i,v} + Q_{i,v} + S_{i,v}), \quad (4)$$

where  $R$ ,  $M$ ,  $Q$  and  $S$  denote reason-action, memory-action, veto-action and self-action alignment terms. The implementation uses emitted trace fields, first impulse, final action and predefined field-source metadata to evaluate these terms. Thus AFCI is an interface-level action-field coupling analysis.

For the finite-action-code benchmark, the canonicalizer maps the raw final action  $o_i$  to  $\hat{y}_i = g(o_i) \in \mathcal{Y}$ , where  $\mathcal{Y} = \{\text{ACTION\_A}, \text{ACTION\_B}, \text{VETO}, \text{DEFER}, \text{RECALL\_PRIOR}, \text{INVALID\_OR\_UNMAPPED}\}$ . If  $y_i^-$  is the scorer-side expected action before the manipulation and  $y_i^+$  is the scorer-side expected action after the manipulation, then

$$c_i = \mathbf{1}\{\hat{y}_i = y_i^+\}, \quad (5)$$

$$u_i = \mathbf{1}\{z_i = \text{irrelevant}\} \mathbf{1}\{\hat{y}_i \neq y_i^-\}. \quad (6)$$

The component score for component  $k$  is the target-minus-irrelevant contrast

$$B_k(v) = \widehat{\text{Pr}}(c_i = 1 \mid z_i \in \mathcal{T}_k, v) - \widehat{\text{Pr}}(u_i = 1 \mid z_i \in \mathcal{N}_k, v), \quad (7)$$

where  $\mathcal{T}_k$  is the target-probe set and  $\mathcal{N}_k$  is the irrelevant or placebo-probe set. The implementation also records cue specificity as the mean of the four target response rates minus the irrelevant-cue change rate. This equals the mean of the four component scores when they are all present. The reported finite-action composite averages those four scores and cue specificity, and therefore gives the same value as the four-component mean for complete records. Cue specificity is consequently a summary of the same responses, rather than an independent fifth source of evidence.

Dataset-cluster bootstrap intervals resample dataset identifiers with replacement, following the non-parametric bootstrap principle of estimating uncertainty from resampled empirical units (Efron and Tibshirani, 1993). For a metric  $m$ , comparator  $q$ , and dataset  $d$ , the paired contrast is

$$\delta_d(q; m) = \bar{m}_{d, \text{A4}} - \bar{m}_{d, q}, \quad (8)$$

with A4 replaced by A4\_full for matched-interface AFCI. The reported lower endpoint of a two-sided 95% interval is the empirical 2.5th percentile of the bootstrap distribution over  $|\mathcal{D}|$ -dataset resamples.

For entropy calibration, layer-specific entropy  $H_{d,v}^{(\ell)}$  is computed for raw, rule-canonical, semantic-cluster and action-family representations. If  $\theta = (T, p)$  denotes a candidate high-stochasticity temperature/top-p setting, the predefined calibration gap is

$$\Gamma(\theta) = \max_{\ell} \left| \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} H_{d, \text{A5}(\theta)}^{(\ell)} - \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} H_{d, \text{A4}}^{(\ell)} \right|. \quad (9)$$

The predefined matching criterion is  $\Gamma(\theta) \leq 0.15$ ; calibration and holdout gaps are reported separately in Section S5.

## S9. Open-weight models and additional controls

This section documents the controlled finite-action and software experiments. It covers the open-weight control matrix, strict field deletion, decisive-field positive control, local consistency guard, control-comparison criteria, forced-budget entropy and token comparisons, entropy-prior control, SWE-bench Lite localization, counterfactual transformations, API configurations and expanded human ratings. These studies complement the state-writing and execution analyses by specifying performance under their specified information and scoring conditions.

Table S65: Validation controls and interpretation status.

| Evidence block                            | Scope                                                                                                                                                                                                                                                    | Main result                                                                                                                                                                                                                                                    | Interpretation status                                                                             |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Open-weight finite-action matrix          | Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-32B-Instruct and Mistral-7B-Instruct-v0.3; 126,000 scored rows per model (108,000 generation and 18,000 offline-policy records); three agent scaffolds; 20 task families; seven variants; three seeds | Structured accuracy exceeded distribution-matched offline/no-context controls in 236/240 model-scaffold-task cells, scrambled-context controls in 240/240 cells and no-fields controls in 238/240 cells; mean deltas were 0.593, 0.895 and 0.550, respectively | Supports open-weight size and family validation of the no-context control result                  |
| Independent Mistral family                | Mistral-7B-Instruct-v0.3 full matrix                                                                                                                                                                                                                     | Core no-context controls satisfied the predefined criteria in 60/60 cells for no-fields, scrambled-context and distribution-matched controls                                                                                                                   | Supports non-Qwen model-family generalization                                                     |
| Strict Qwen32B target lesion              | Qwen2.5-32B strict lesion slice; 28,800 scored generations                                                                                                                                                                                               | Structured accuracy exceeded strict target lesion in 60/60 cells; structured mean 0.998, strict-lesion mean 0.037, mean delta 0.961                                                                                                                            | Sensitivity to removal of the decisive task field                                                 |
| Decisive-field positive control           | <b>gpt-5.4-mini</b> and <b>qwen3-max</b> ; 20 task families; six events per family; 1,440 fallback-free API generations                                                                                                                                  | Full-state and only-decisive-field accuracy were both 1.000 in both models, while the best non-decisive control was 0.208; the only-decisive field is intentionally visible and action-code-bearing                                                            | Supports a diagnostic decisive-field readout within the finite-action protocol                    |
| Local control-binding wrapper             | Same 1,440-row positive-control experiment; no second model call                                                                                                                                                                                         | Irrelevant-cue accuracy remained 1.000; raw scrambled-field following was 0.958 for <b>gpt-5.4-mini</b> and 0.533 for <b>qwen3-max</b> , reduced to 0.033 in both models after wrapping                                                                        | Deterministic constraint enforcement within the finite-action interface                           |
| Formal Qwen control-audit slice           | 63,360 planned and observed rows; no duplicate analysis keys; parse/error rate 0.140%                                                                                                                                                                    | Structured accuracy exceeded no-fields, scrambled-context, distribution-matched, strict-lesion and formal control variants in the relevant cells                                                                                                               | Supports formal-control behavior in the Qwen slice                                                |
| Token-prompt audit slice                  | 17,280 planned and observed rows; no duplicate analysis keys; parse/error rate 0.069%                                                                                                                                                                    | Prompt-token counts were aligned for the entropy-token variant, but mean total-token ratios and latency proxies remained outside the strict token/compute criterion                                                                                            | Prompt-token alignment with residual differences in total tokens and latency                      |
| Strict entropy criterion                  | Formal entropy calibration and holdout audit                                                                                                                                                                                                             | Maximum entropy gaps exceed the registered threshold across model-scaffold cells                                                                                                                                                                               | Distributional calibration under the registered criterion                                         |
| Local forced-budget entropy/token audit   | Qwen2.5-1.5B-Instruct CPU local inference; 20 task families; two events per family; 80 generations                                                                                                                                                       | Prompt and completion tokens matched exactly by event; total-token ratio 1.000; latency ratio 0.997; canonical-action entropy gap 0.020 bits; structured accuracy was 0.900 versus 0.325 for the action-prior no-state control                                 | Exact prompt and completion matching in the local configuration                                   |
| Entropy-prior no-field behavioral control | Qwen2.5-14B and Qwen2.5-32B calibration and holdout rows; 24,480 planned and observed rows; no duplicate analysis keys; parse/error rate 0.102%                                                                                                          | The entropy-prior no-field behavioral control remained below the structured-control variant in 120/120 cells, with mean delta 0.660; the strict entropy-gap criterion was satisfied in only 1/6 model-scaffold cells                                           | Behavioral comparison under a final-action prior; the strict entropy criterion holds in 1/6 cells |

| Evidence block                               | Scope                                                                                                                                                                                                                              | Main result                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Interpretation status                                                                                                        |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| SWE-bench Lite issue-to-file association     | <code>princeton-nlp/SWE-bench_Lite</code> test split; 300 issues and six API models; 18,000 condition/repeat rows aggregate to 1,800 model–issue records                                                                           | Under issue-grouped validation with training-only preprocessing, AUC is 0.737216 for the reference-free baseline and 0.924486 with reference-dependent diagnostic features; paired difference 0.187269, 95% full-refit interval [0.126941, 0.267774]                                                                                                                                                                                                                         | Reference-dependent association; estimator and direct held-group evaluation are defined in Supplementary Note 1 (p. 18)      |
| SWE-bench Lite reliability calibration       | Same 1,800 model–issue records; issue-grouped logistic probabilities from the reference-free baseline and reference-dependent augmentation                                                                                         | Reliability curves and threshold summaries characterize predicted localization, hard-constraint violations and constraint-clean localization                                                                                                                                                                                                                                                                                                                                 | Retrospective calibration under reference-dependent features; reference-free evaluation is reported separately               |
| Counterfactual transform QC                  | Deterministic counterfactual-flip transform-quality screen                                                                                                                                                                         | Text–label consistency screening retained 57/240 candidate counterfactual rows as transform-quality boundary rows                                                                                                                                                                                                                                                                                                                                                            | Robustness analysis conditions on text–label consistency                                                                     |
| Consistency-passing counterfactual transform | Frozen transform on Qwen2.5-14B and Qwen2.5-32B with jointly re-generated decisive context and expected action                                                                                                                     | The consistency-passing transform satisfied the predefined criterion in 120/120 scaffold–family cells; mean structured accuracy was 1.000, compared with 0.992 in the original structured rows                                                                                                                                                                                                                                                                               | Robustness under the specified deterministic transformation                                                                  |
| Expanded human ratings                       | 1,200 shared items, three graduate-student raters                                                                                                                                                                                  | Core scores are complete; 150 of 8,400 item–dimension combinations have a 2:1 split across 68 items; mean pairwise Cohen’s kappa is 0.961, minimum individual value 0.860 and minimum dimension mean 0.897                                                                                                                                                                                                                                                                   | Stratified sampling, masked items, individual scores and majority aggregation are described in Supplementary Note 5 (p. 20)  |
| Formal API external validation               | Provider-reported run-time identifiers <code>gpt-5.4-mini</code> , <code>gemini-3.1-flash-lite-preview</code> , <code>deepseek-v3.2</code> and <code>qwen3-max</code> ; 14,400 planned rows; 11,520 fallback-free remote API calls | All rows were present, duplicate analysis keys were zero, fallback calls were zero, unrecovered generations were zero and one parse/action error record remained. Structured accuracy exceeded no-fields, scrambled-context, distribution-matched and strict-lesion controls in 238/240, 240/240, 240/240 and 240/240 model–scaffold–task–family cells; all model-, scaffold- and task-family-cluster bootstrap lower endpoints of the two-sided 95% intervals were positive | Historical robustness of the explicit prompt/interface controls; not validation of latent state or interactive architectures |

The decisive-field positive-control experiment is a controlled finite-action test. The only-decisive condition exposes the current event’s minimal action-relevant field and asks whether the model follows that field rather than surface context, a calibrated action prior or a decisive field from another event. The examples below specify the visible field and its associated action rule; the expected action column remains scorer-side metadata used for evaluation, whereas the visible field is the intervention-carrying state supplied to the action query.

| Probe component | Example visible only-decisive field                                                                                                    | Scorer-side target role                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Reason          | Decisive reason field for the event: calibrated task evidence selects <code>ACT_B</code> .                                             | Target checks whether final action follows the current reason field.                     |
| Memory          | Decisive memory field for the event: the corrected prior commitment is <code>ACT_A</code> ; stale or unrelated memory is non-decisive. | Target checks whether final action follows corrected memory rather than injected memory. |
| Veto/constraint | Decisive constraint/veto field for the event: a hard constraint invalidates ordinary actions and requires <code>ACT_VETO</code> .      | Target checks whether final action respects the current constraint field.                |

The decisive-field experiment distinguishes hidden scorer metadata from the intentionally visible action-relevant field. Its field-only condition supplies the current event’s minimum decisive information.

The leakage check therefore tests whether the hidden target variable or key entered the generated prompt, while the non-decisive conditions test removal or mismatch of the current field.

| Audit item                                                 | Evidence                                                                                                                                                                                                           | Status | Interpretation                                                                                                                        |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>expected_action</code> column not a prompt key       | Prompt construction serializes condition, event id, task family, candidate actions and context; <code>expected_action</code> is retained in traces for scorer-side evaluation after generation.                    | pass   | The scorer target variable is not exposed as a prompt field or JSON key.                                                              |
| Scorer-side target used post-generation                    | Across 1,440 traces, <code>expected_action_match</code> is computed after generation from canonical <code>final_action</code> versus <code>expected_action</code> .                                                | pass   | The target column is a scorer-side annotation rather than a model-output field.                                                       |
| Non-decisive controls do not expose current decisive state | <code>surface_context_only</code> and <code>action_prior_only</code> had <code>field_event_id=none</code> in 240/240 rows each; <code>scrambled_decisive_field</code> used a non-current field id in 240/240 rows. | pass   | Surface, prior and scrambled controls separate action priors or wrong-state fields from the current decisive field.                   |
| Only-decisive action-token status                          | <code>only_decisive_causal_field</code> exposed a visible decisive action matching the scorer target in 240/240 rows; full-state and irrelevant-cue rows did the same by design.                                   | scope  | The positive-control claim tests following the current minimal decisive field; it is not an answer-token-free hidden-reasoning claim. |
| Label wording scope                                        | The prompt contains a generic instruction not to infer hidden expected-action labels, but does not serialize the <code>expected_action</code> column or scorer target key.                                         | pass   | The generic instruction names the evaluation concept; it does not provide the hidden scorer variable.                                 |

**Control criteria and repair outcomes.** No-context and controlled comparisons show that finite-action performance depended on structured information under the tested rules. Entropy and total-compute matching remain subject to their observed calibration results. The separate local Qwen2.5-1.5B experiment satisfied its prompt-padding and fixed-generation-length conditions. SWE-bench Lite localization covers 300 issues; the exploratory repair evaluation contains 50 selected issues and two models. Of those 100 model-issue submissions, one resolved its issue in the official harness evaluation, and the association criterion was inconclusive. A local application and repair filter retained eight patches, all of which completed the harness with none resolving the issue. The counterfactual-transform records separately identify the screened transformations and the consistency-passing deterministic transformation used in the robustness comparison.

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